

Erdős Problem 848: Exact Hall Closure by Square-Sieve Descent and Prefix Colouring

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Abstract

For $N \geq 1$, let $A \subseteq [1, N]$ have the property that $ab + 1$ is nonsquarefree for every $a, b \in A$. We prove that

$$|A| \leq |\{n \leq N : n \equiv 7 \pmod{25}\}|,$$

with equality attained by the progression $7 \pmod{25}$. The proof first rewrites the assertion as an exact Hall inequality. An exact prefix colouring proves the inequality through $1.5 \cdot 10^6$; two finite square-sieve arguments cover the interval up to $5 \cdot 10^6$; and a valuation-cell-fibre descent covers the four intervals $[5, 10)$, $[10, 20)$, $[20, 40)$, and $[40, \infty)$ million. All finite enumerations used in the proof employ integer or rational arithmetic and are reproduced by the accompanying source files.

1 Statement and Hall formulation

Write

$$C_N = \{n \in [1, N] : n \equiv 7 \pmod{25}\}.$$

We prove

$$|A| \leq |C_N| \tag{1}$$

for every N and every $A \subseteq [1, N]$ such that $ab + 1$ is nonsquarefree for all $a, b \in A$. Sawhney proved the assertion for all sufficiently large N by a density sieve and structural reduction [1]; the argument below supplies an explicit all- N proof.

Proposition 1.1 (Sharpness). *The set C_N is admissible: if $a, b \in C_N$, then $ab + 1$ is not squarefree.*

Proof. Since $a \equiv b \equiv 7 \pmod{25}$,

$$ab + 1 \equiv 7^2 + 1 = 50 \equiv 0 \pmod{25}.$$

Thus $5^2 \mid ab + 1$. □

2 The direct Hall cut

Join $b \in [1, N] \setminus C_N$ to $a \in C_N$ when $ab + 1$ is squarefree. For $B \subseteq [1, N] \setminus C_N$, write

$$\Gamma_N(B) = \{a \in C_N : ab + 1 \text{ is squarefree for some } b \in B\}$$

and

$$T_N(B) = C_N \setminus \Gamma_N(B).$$

Thus $a \in T_N(B)$ precisely when $ab + 1$ is non-squarefree for every $b \in B$.

For every set $B \subseteq [1, N] \setminus C_N$ such that $bc + 1$ is nonsquarefree for all $b, c \in B$, the Hall inequality to be proved is

$$|B| + |T_N(B)| \leq |C_N|, \quad (2)$$

equivalently $|\Gamma_N(B)| \geq |B|$.

Theorem 2.1 (Hall reduction). *If the Hall inequality (2) holds, then the exact bound (1) holds for every N .*

Proof. Let A be admissible and put $B = A \setminus C_N$. Every subset $D \subseteq B$ is again a compatible outside set, so $|\Gamma_N(D)| \geq |D|$ by the Hall inequality (2). Hall's marriage theorem gives an injection from B into $\Gamma_N(B)$. Admissibility gives

$$\Gamma_N(B) \cap (A \cap C_N) = \emptyset,$$

because every vertex of $\Gamma_N(B)$ has a squarefree edge to some element of $B \subseteq A$. Hence

$$|B| \leq |\Gamma_N(B)| \leq |C_N \setminus A|,$$

and therefore

$$|A| = |A \cap C_N| + |B| \leq |A \cap C_N| + |C_N \setminus A| = |C_N|.$$

□

Theorem 2.2 (Hall equivalence). *the Hall inequality (2) is equivalent to the exact Erdős 848 upper bound (1).*

Proof. The forward implication is Theorem 2.1. Conversely, assume (1) and let B satisfy the hypotheses of the Hall inequality (2). Put

$$A = B \cup T_N(B).$$

Products internal to B are non-squarefree by hypothesis. Products internal to $T_N(B)$ are divisible by 25, because all its elements are $7 \pmod{25}$. Finally, every cross-product between B and $T_N(B)$ is non-squarefree by the definition of $T_N(B)$. Hence A is admissible. Moreover B and $T_N(B)$ are disjoint, so (1) gives

$$|B| + |T_N(B)| = |A| \leq |C_N|,$$

which is (2). □

Thus the Hall inequality is the original theorem written at the exact exchange boundary. All subsequent estimates prove (2).

3 An exact finite prefix close

The whole finite remainder admits a single exact colouring certificate.

Theorem 3.1 (Two-matching prefix colouring). *For every $1 \leq N \leq 1.5 \cdot 10^6$, let*

$$\mathcal{D}_N = \{x \leq N : x^2 + 1 \text{ is nonsquarefree}\},$$

and join two distinct members $x, y \in \mathcal{D}_N$ when $xy + 1$ is nonsquarefree. This graph has a proper colouring with exactly $|\mathcal{C}_N|$ colours. Consequently (1) and the Hall inequality (2) hold throughout this interval.

Proof. Every member of an admissible set belongs to $\mathcal{D}_N$, and every two distinct members are joined. Thus it is enough to construct the asserted proper colouring.

Put $L = 1.5 \cdot 10^6$. The diagonal set $\mathcal{D}_L$ is enumerated without factoring L separate integers. If $p^2 \mid x^2 + 1$, then p is odd, $p \equiv 1 \pmod{4}$, and $p \leq x \leq L$. Conversely every solution of $z^2 \equiv -1 \pmod{p^2}$ marks an arithmetic progression of diagonal vertices. For each prime $p \leq L$, sieve primality, find the two roots modulo p , lift them uniquely modulo p^2 , verify the lifted congruence, and mark both progressions. This is an exact equivalence.

Process the marked vertices in increasing order. At a given prefix write

$$U = \{u : u \equiv 7 \pmod{25}\}, \quad V = \{v : v \equiv 18 \pmod{25}\}, \quad E = \mathcal{D}_N \setminus (U \cup V).$$

Maintain two injections into the colour set U . The first is $\mu : V \rightarrow U$ with

$$v\mu(v) + 1 \text{ squarefree.}$$

The second is $\nu : E \rightarrow U$, and it satisfies

$$e\nu(e) + 1 \text{ squarefree}, \quad e\mu^{-1}(\nu(e)) + 1 \text{ squarefree}$$

whenever the second factor exists. Then for every $u \in U$ the class

$$\{u\} \cup \mu^{-1}(u) \cup \nu^{-1}(u)$$

has at most three vertices, and every product plus one inside it is squarefree. These are therefore $|U| = |\mathcal{C}_N|$ independent colour classes.

The two injections are maintained prefix by prefix, using no future vertex. When a new 7-class vertex appears, add its singleton colour. When a new 18-class vertex appears, either put it in the unique colour not yet used by μ , or exchange it with one resident 18-class vertex; both new products in the exchange are checked. On the complete interval there are 39290 direct insertions and 20710 one-exchange insertions, and no longer augmenting path is used. An exceptional vertex is placed in a colour not yet used by ν after checking its one or two required products. If an exchange in μ invalidates a resident exceptional vertex, that vertex is immediately reinserted. There are 40692 such insertions or repairs: 40688 use a free colour directly, while only the initial vertices

$$41, \quad 70, \quad 99, \quad 117$$

use augmenting paths, of maximum depth four. Hence the two injections, and therefore the proper colouring, exist after every marked vertex is processed. Between marked vertices the

graph is unchanged, while a new colour appears exactly at a new member of C_N . The certificate consequently covers every prefix, not merely the upper endpoint.

The terminal exact counts are

quantity	value	quantity	value
primes through L	114155	lifted roots inspected	114044
$ \mathcal{D}_L $	157733	$ U = V $	60000
$ E $	37733	pure edge tests	125789
exceptional edge tests	88407	final verification pairs	135466.

All 270784 construction-oracle product queries are factored exactly. The terminal verifier then passes all 135466 class pairs through a fresh factorization oracle; in this run their products are distinct. Deterministic 64-bit Miller–Rabin verifies every factor as prime, their product is reconstructed, and repeated factors are rejected; 128-bit modular multiplication prevents overflow. A final pass checks one diagonal witness for every marked vertex, checks that no unmarked vertex was coloured, and refactors every pair within every terminal colour class. Compilation fails if assertions are disabled. The complete producer and verifier are `scripts/systematic_finite_prefix_coloring.cpp`.

Thus every admissible set is a clique in a graph properly coloured with $|C_N|$ colours, and so has at most $|C_N|$ members. Hall equivalence, Theorem 2.2, gives the last assertion. $\square$

4 An aggregate explicit large- N close

We now make the known sufficiently-large- N part numerically explicit and sharpen its threshold. We retain Sothanaphan’s explicit progression estimate [2], but aggregate the 23 non-base residue classes before estimating the diagonal Pell tail. This makes the large-square tail a single set of integers rather than 23 separately charged copies.

Let

$$C_{2,5} = 1 - \frac{25}{3\pi^2} < 0.155656803, \quad C_5 = 1 - \frac{25}{4\pi^2} < 0.366742603,$$

and

$$C_{\text{quad}} = 1 - \prod_{\substack{p \equiv 1 \pmod{4} \\ p \geq 13}} \left(1 - \frac{2}{p^2}\right) < 0.027346508.$$

Exact rational multiplication through 10^7 gives

$$1 - \prod_{\substack{13 \leq p \leq 10^7 \\ p \equiv 1 \pmod{4}}} \left(1 - \frac{2}{p^2}\right) < 0.027346490.$$

For the omitted factors,

$$1 - \prod_{p > 10^7} \left(1 - \frac{2}{p^2}\right) \leq 2 \sum_{p > 10^7} \frac{1}{p^2} < 0.000000017856.$$

The last inequality retains the negative boundary term in partial summation. The calculation is given in Lemma 6.65. Adding the two strict bounds proves the displayed constant. Fix

$q_0 \in \{25, 100\}$ and $b \leq N$. For every residue class t such that $p^2 \nmid bt + 1$ for every prime $p \mid q_0$, the explicit off-diagonal progression estimate gives

$$\#\{a \leq N : a \equiv t \pmod{q_0}, \mu(ab + 1) = 0\} \leq \frac{N}{q_0} \left(1 - \prod_{(p, q_0 b) = 1} \left(1 - \frac{1}{p^2} \right) \right) + \Phi_{q_0} N^{3/4}, \quad (3)$$

where $\Phi_{25} = 4.7$ and $\Phi_{100} = 5.3$. For completeness, the source's normalized error function is

$$\begin{aligned} E_q(N) = & \frac{2.367}{N^{3/4}} + \alpha_q + \frac{1}{N^{3/4}} + \frac{N/q + 1}{\alpha_q N^{3/2}} + \frac{1}{\alpha_q \sqrt{q} N^{3/2}} \\ & + \frac{2\lambda q^{1/4}(N/q + 1)}{\alpha_q N^{5/4}} + \frac{2\lambda}{\alpha_q q^{3/8} N^{3/2}} + \frac{\lambda q^{5/4}(N/q + 1)}{\alpha_q^2 N} + \frac{\lambda q^{1/4}}{\alpha_q^2 N^2}, \end{aligned}$$

where

$$\alpha_{25} = \frac{25}{8}, \quad \alpha_{100} = \frac{7}{2}, \quad \lambda = \frac{128}{(8 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13)^{1/4}} < 6.876.$$

It bounds the coefficient of $N^{3/4}$ in (3). Every nonconstant term decreases with N , and direct substitution at $N = 1.6 \cdot 10^{15}$ gives

$$E_{25}(1.6 \cdot 10^{15}) < 4.69949, \quad E_{100}(1.6 \cdot 10^{15}) < 5.27503.$$

Thus (3) with the displayed rounded constants is valid throughout the range used below, not only at the older published threshold.

The root-count factor in this estimate can be made much smaller in the present application. Let $\rho(m)$ be the maximum number of roots of $y^2 \equiv u \pmod{m}$ over units u , and let ℓ_j be the j -th odd prime, $P_k = \prod_{j \leq k} \ell_j$. Put

$$\begin{aligned} Q_* &= 8P_{12} = 1217001054108840, \\ N_{100} &= Q_*/20 = 60850052705442, \quad N_{25} = Q_*/5 = 243400210821768. \end{aligned}$$

For $7.67769 \cdot 10^9 \leq N \leq 1.6 \cdot 10^{15}$ define

$$H_{25}(N) = \begin{cases} 8192, & N < N_{25}, \\ 16384, & N \geq N_{25}, \end{cases} \quad H_{100}(N) = \begin{cases} 8192, & N < N_{100}, \\ 16384, & N \geq N_{100}. \end{cases}$$

For $q_0 \in \{25, 100\}$, $H = H_{q_0}(N)$, set

$$d_H = (2H)^{1/3} N^{2/3}$$

and define the normalized error envelope

$$\begin{aligned} \mathcal{E}_{q_0, H}(N) = & \frac{1}{N} \left(2.367 + d_H + 1 + \frac{(1 + 2H)(N/q_0 + 1)}{d_H} + \frac{1 + 2H}{\sqrt{q_0} d_H} \right. \\ & \left. + \frac{H q_0 N (N/q_0 + 1)}{d_H^2} + \frac{H}{d_H^2} \right). \end{aligned} \quad (4)$$

Lemma 4.1 (Primorial progression error). *In (3), throughout $7.67769 \cdot 10^9 \leq N \leq 1.6 \cdot 10^{15}$, the term $\Phi_{q_0} N^{3/4}$ may be replaced by*

$$N \mathcal{E}_{q_0, H_{q_0}(N)}(N).$$

On every interval on which the corresponding H_{q_0} is constant, the error envelope is decreasing.

Proof. In the source reduction the relevant modulus is $d_1 = q_0 b/g$, where g divides the radical of q_0 . Removing one redundant factor 5 when its exponent is at least two shows that there is an $m \leq q_0 N/5$ with $\rho(m) = \rho(d_1)$. If $m = 2^\nu m_0$ with m_0 odd, then

$$\rho(m) \leq c_\nu 2^{\omega(m_0)}, \quad c_0 = c_1 = 1, \quad c_2 = 2, \quad c_\nu = 4 \quad (\nu \geq 3).$$

The least modulus with root count greater than 8192 is $8P_{12} = Q_*$. The least one with root count greater than 16384 is $8P_{13} = 52331045326680120$, which exceeds $20(1.6 \cdot 10^{15})$. This proves the two step-function bounds for $\rho(d_1)$.

Repeat the Möbius split underlying (3), but keep $\rho(d_1) \leq H$ instead of replacing it by $6.876d_1^{1/4}$. With $X \leq N/q_0 + 1$, $\sqrt{q_0} \leq d_1 \leq q_0 N$, and $D = \lceil d_H \rceil$, the total additive error is at most

$$2.367 + D + \frac{(1 + 2H)X}{D} + \frac{1 + 2H}{\sqrt{q_0}D} + \frac{Hq_0NX}{D^2} + \frac{H}{D^2}.$$

Use $d_H \leq D \leq d_H + 1$, increasing the positive D term and decreasing all denominators, to obtain (4). Every normalized term is decreasing while H is fixed. $\square$

For the diagonal part put

$$A_7 = \{a \in A : a \equiv 7 \pmod{25}\}, \quad A_{18} = \{a \in A : a \equiv 18 \pmod{25}\}, \quad A^* = A \setminus (A_7 \cup A_{18}),$$

and define

$$U_{25} = \{x \leq N : x \not\equiv 7, 18 \pmod{25}\}, \quad U_{50} = \{x \in U_{25} : x \text{ is odd}\}.$$

For $r \in \{0, 1, 2, 3\}$ also put

$$U_{100,r} = \{x \in U_{25} : x \equiv r \pmod{4}\};$$

this is a union of 23 classes modulo 100. Two coprimality moduli will be used. Put

i	M_i	$\eta_i = \varphi(M_i)/M_i$	$h_i > \sum_{\substack{1 \leq a \leq M_i \\ (a, M_i) = 1}} 1/a$
0	2310	16/77	2.189
1	43890	288/1463	2.685

and $e_0 = 64$, $e_1 = 128$. For $R \geq M_i^2$, put

$$K_i(R) = 2\eta_i R \left(h_i + \eta_i \left(1 + \log \frac{\sqrt{R}}{M_i} \right) \right) + e_i \sqrt{R}.$$

For $q \in \{25, 50, 100\}$ write $\theta_q = 23/q$, put

$$r_N = \frac{12}{5} N^{2/3},$$

and set, whenever $r_N \geq M_i^2$,

$$\begin{aligned} \Delta_q^{(i)}(N) &= \frac{23}{N} + \frac{3K_i(r_N + 1)}{N} + \frac{46(\log(r_N + 1) + 2)}{qr_N} \\ &\quad + \frac{N^2 + 1}{Nr_N^2} \left(1 + \log_{2+\sqrt{3}} N \right). \end{aligned} \tag{5}$$

Lemma 4.2 (Aggregate diagonal estimate). *For $q = 25$ and $U_q = U_{25}$, for $q = 50$ and $U_q = U_{50}$, or for $q = 100$ and $U_q = U_{100,r}$ with any fixed $r \pmod{4}$,*

$$\#\{x \in U_q : \mu(x^2 + 1) = 0\} \leq N(\theta_q C_{\text{quad}} + \Delta_q^{(i)}(N)) \quad (i = 0, 1), \quad (6)$$

provided $r_N \geq M_i^2$. The $i = 0$ envelopes decrease for $N \geq 7.8 \cdot 10^{10}$, and at that endpoint

$$\Delta_{25}^{(0)}(N) < 0.002663263, \quad \Delta_{50}^{(0)}(N) < 0.002662851, \quad \Delta_{100}^{(0)}(N) < 0.002662646.$$

The $i = 1$ envelopes decrease for $N \geq 3 \cdot 10^{13}$, and at that endpoint

$$\Delta_{25}^{(1)}(N) < 0.000402450, \quad \Delta_{50}^{(1)}(N) < 0.000402441.$$

Proof. Let $R = \lceil r_N \rceil$. In the Möbius expansion of the indicator of $x^2 + 1$ being squarefree, only squarefree s composed of primes $p \equiv 1 \pmod{4}$, $p \geq 13$, contribute. For such an s , the congruence $x^2 \equiv -1 \pmod{s^2}$ has exactly $2^{\omega(s)}$ roots. Counting complete blocks in the 23 allowed residue classes gives

$$\left| A_s - \theta_q N \frac{2^{\omega(s)}}{s^2} \right| \leq 3 \cdot 2^{\omega(s)},$$

where A_s counts $x \in U_q$ with $s^2 \mid x^2 + 1$.

Every such s is coprime to both M_i . If $C_{M_i}(y) = \#\{n \leq y : (n, M_i) = 1\}$, inclusion–exclusion gives $C_{M_i}(y) \leq \eta_i y + 2^{\omega(M_i)}$. The hyperbola decomposition therefore gives

$$\begin{aligned} \sum_{\substack{s \leq R \\ (s, M_i) = 1}} d(s) &\leq 2 \sum_{\substack{a \leq \sqrt{R} \\ (a, M_i) = 1}} C_{M_i}(R/a) \\ &\leq 2\eta_i R \sum_{\substack{a \leq \sqrt{R} \\ (a, M_i) = 1}} \frac{1}{a} + e_i \sqrt{R} \leq K_i(R). \end{aligned}$$

The two first-block harmonic bounds are finite rational computations. The last inequality follows by separating the first block modulo M_i and bounding each later block by η_i/k . Hence the small- s rounding loss is at most $3K_i(R)$, while

$$\sum_{s > R} \frac{d(s)}{s^2} \leq \frac{2(\log R + 2)}{R}.$$

The remaining large-square tail is charged only once. Writing $x^2 + 1 = ds^2$, the negative Pell equation has at most $1 + \log_{2+\sqrt{3}} N$ positive solutions for each fixed d , so

$$\sum_{s > R} A_s \leq \frac{N^2 + 1}{R^2} \left(1 + \log_{2+\sqrt{3}} N \right).$$

Finally $|U_q| \leq \theta_q N + 23$. Combining the bounds and using $r_N \leq R \leq r_N + 1$ proves (6). Notice that $r_N \geq M_0^2$ already for $N \geq 7.8 \cdot 10^{10}$, whereas $r_N \geq M_1^2$ for $N \geq 2.274 \cdot 10^{13}$. Elementary differentiation proves the asserted monotonicity on the displayed ranges; direct endpoint evaluation gives the constants. $\square$

For the last refinement define

$$V_8 = \{x \leq N : x \not\equiv 7, 18 \pmod{25}, \quad (x \text{ odd or } 8 \mid x)\}$$

and, whenever $r_N \geq M_i^2$,

$$\begin{aligned} \Delta_8^{(i)}(N) &= \frac{115}{N} + \frac{6K_i(r_N + 1)}{N} + \frac{23(\log(r_N + 1) + 2)}{20r_N} \\ &\quad + \frac{N^2 + 1}{Nr_N^2} \left(1 + \log_{2+\sqrt{3}} N\right). \end{aligned} \quad (7)$$

Lemma 4.3 (Two-adic split of the even non-base branch). *Let $3 \cdot 10^{13} \leq N \leq 6 \cdot 10^{13}$ and let b be even. In the two $q_0 = 25$ progression estimates obtained from b , one may use*

$$H = 2048 \quad \text{if } \nu_2(b) = 1, \quad H = 4096 \quad \text{if } \nu_2(b) = 2, \quad H = 8192 \quad \text{if } \nu_2(b) \geq 3.$$

Moreover

$$\#\{x \in V_8 : \mu(x^2 + 1) = 0\} \leq N \left(\frac{23}{40} C_{\text{quad}} + \Delta_8^{(1)}(N) \right),$$

and at the left endpoint $\Delta_8^{(1)}(N) < 0.000667629$.

Proof. After removal of the redundant factor 5, the effective root-count modulus is $5b$. On the stated interval it is at most $3 \cdot 10^{14}$. If $\nu_2(b) = 1$, a root count greater than 2048 would require a modulus at least $2P_{12} = 304250263527210$; if $\nu_2(b) = 2$, a count greater than 4096 would require at least $4P_{12} = 608500527054420$. If $8 \mid b$, the existing 8192 bound applies. This proves the progression assertions.

For the diagonal assertion, the odd non-base classes have density $23/50$ and the classes divisible by 8 have density $23/200$. For each root modulo s^2 , counting the two parts successively modulo 2, 8, and 25 gives total discrepancy at most 6. Their union has 115 classes modulo 200, so the proof of Lemma 4.2, with small-modulus loss $6K_1(R)$ and the Pell tail still paid once, gives (7) with $i = 1$. Its terms decrease, and direct substitution gives the stated endpoint value. $\square$

Corollary 4.4 (Low-range two-adic diagonal). *For $N \geq 7.8 \cdot 10^{10}$,*

$$\#\{x \in V_8 : \mu(x^2 + 1) = 0\} \leq N \left(\frac{23}{40} C_{\text{quad}} + \Delta_8^{(0)}(N) \right).$$

The envelope decreases, and at the left endpoint

$$\Delta_8^{(0)}(N) < 0.004510951.$$

Proof. Repeat the diagonal part of Lemma 4.3 with M_0 in place of M_1 . Here $r_N \geq M_0^2$ throughout the stated range, so the same 115 residue-class endpoint loss, $6K_0(R)$ small-square loss, and single Pell tail give (7) with $i = 0$. Elementary differentiation shows that every term decreases; direct substitution at $N = 7.8 \cdot 10^{10}$ gives the displayed value. $\square$

For $\varepsilon \in \{0, 1, 2, 3\}$ put

$$W_\varepsilon = \{x \in U_{25} : x \not\equiv \varepsilon \pmod{4}\}$$

and define

$$\begin{aligned} \Delta_W(N) &= \frac{69}{N} + \frac{6K_0(r_N + 1)}{N} + \frac{69(\log(r_N + 1) + 2)}{50r_N} \\ &\quad + \frac{N^2 + 1}{Nr_N^2} \left(1 + \log_{2+\sqrt{3}} N\right). \end{aligned} \quad (8)$$

Lemma 4.5 (Three-mod-4-class diagonal). *For $N \geq 7.8 \cdot 10^{10}$ and every choice of ε ,*

$$\#\{x \in W_\varepsilon : \mu(x^2 + 1) = 0\} \leq N \left(\frac{69}{100} C_{\text{quad}} + \Delta_W(N) \right).$$

The envelope decreases, and at the left endpoint

$$\Delta_W(N) < 0.004511053.$$

Proof. For every squarefree small-square modulus s in the proof of Lemma 4.2, its root count on W_ε is the difference of its root counts on U_{25} and $U_{100,\varepsilon}$. Subtracting their main terms gives density $69/100$, while the two discrepancies give at most $6 \cdot 2^{\omega(s)}$. Also W_ε is a union of 69 classes modulo 100. Thus the same hyperbola calculation has endpoint loss 69, small-square loss $6K_0(R)$, harmonic-tail coefficient $69/50$, and one common Pell tail. These are exactly the four terms in (8). Their monotonicity and the displayed endpoint value follow by differentiation and direct substitution. $\square$

The final low-range refinement uses the stronger coprimality modulus before a complete residue block is available. Put

$$M_* = 43890, \quad \eta_* = \frac{288}{1463}, \quad h_* := \sum_{\substack{1 \leq a \leq 7194 \\ (a, M_*)=1}} \frac{1}{a} < 2.329039$$

and

$$K_*(R) = 2\eta_* R h_* + 128\sqrt{R}.$$

The displayed bound for h_* is an exact finite rational summation. On $6.7 \cdot 10^{10} \leq N \leq 1.001 \cdot 10^{11}$ define

$$\begin{aligned} \tilde{\Delta}_q(N) &= \Delta_q^{(0)}(N) - \frac{3K_0(r_N + 1)}{N} + \frac{3K_*(r_N + 1)}{N} \quad (q = 25, 50, 100), \\ \tilde{\Delta}_W(N) &= \Delta_W(N) - \frac{6K_0(r_N + 1)}{N} + \frac{6K_*(r_N + 1)}{N}, \\ \tilde{\Delta}_8(N) &= \Delta_8^{(0)}(N) - \frac{6K_0(r_N + 1)}{N} + \frac{6K_*(r_N + 1)}{N}. \end{aligned}$$

Lemma 4.6 (Short-block diagonal refinement). *Throughout this interval the conclusions of Lemma 4.2, Corollary 4.4, and Lemma 4.5 hold with their respective error terms replaced by $\tilde{\Delta}_q$, $\tilde{\Delta}_8$, and $\tilde{\Delta}_W$. These envelopes decrease, and at $N = 6.7 \cdot 10^{10}$,*

$$\begin{aligned} \tilde{\Delta}_{25} &< 0.002514345, \quad \tilde{\Delta}_{50} < 0.002513892, \quad \tilde{\Delta}_{100} < 0.002513665, \\ \tilde{\Delta}_W &< 0.004175714, \quad \tilde{\Delta}_8 < 0.004175602. \end{aligned}$$

Proof. For $R = r_N + 1$ on the stated interval, direct endpoint evaluation gives $\sqrt{R} < 7194$. Inclusion–exclusion gives

$$C_{M_*}(y) = \#\{n \leq y : (n, M_*) = 1\} \leq \eta_* y + 64.$$

The hyperbola step in Lemma 4.2 therefore yields, without requiring $R \geq M_*^2$,

$$\sum_{\substack{s \leq R \\ (s, M_*)=1}} d(s) \leq 2\eta_* R \sum_{\substack{a \leq \sqrt{R} \\ (a, M_*)=1}} \frac{1}{a} + 128\sqrt{R} \leq K_*(R).$$

Every squarefree modulus s occurring in the three diagonal arguments is coprime to M_* , because its prime factors are 1 (mod 4), whereas the added prime is 19. Thus only the displayed small-square rounding term changes; all main terms, harmonic tails, and the single Pell tail remain unchanged. This gives the three tilded formulas. Each normalized term decreases by elementary differentiation, and direct substitution gives the endpoint bounds. $\square$

Theorem 4.7 (Explicit large range). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 3 \cdot 10^{13}$.*

Proof. If A is contained in A_7 or A_{18} , the result is immediate. Otherwise the exhaustive parity decomposition gives the following four normalized upper bounds, in order: A^* has an even element; A^* is odd and the base part is odd; A^* is odd and a base class has an even element; and $A^* = \emptyset$ with both base classes nonempty:

$$\begin{aligned} \frac{23}{25}C_{\text{quad}} + \frac{2}{25}C_{2,5} &< 0.0376115, \\ \frac{23}{50}C_{\text{quad}} + \frac{1}{50} + \frac{1}{50}C_{2,5} &< 0.0356927, \\ \frac{23}{50}C_{\text{quad}} + \frac{1}{25}C_5 + \frac{1}{25}C_{2,5} &< 0.0334755, \\ \frac{2}{25}C_5 &< 0.0293395. \end{aligned}$$

First let $3 \cdot 10^{13} \leq N < 6 \cdot 10^{13}$. In the last three cases, using $H = 8192$ in the progression envelope gives at the left endpoint the respective bounds

$$0.038548, \quad 0.036331, \quad 0.031792,$$

and all three decrease thereafter. In the first case choose an even $b \in A^*$. If $\nu_2(b) = 1$, Lemma 4.3 with the unrestricted diagonal estimate gives an endpoint bound below 0.039559; if $\nu_2(b) = 2$, it gives a bound below 0.039961. If neither type of element exists, every even element of A^* is divisible by 8, so $A^* \subseteq V_8$. The restricted diagonal estimate and $H = 8192$ then give

$$\frac{23}{40}C_{\text{quad}} + \frac{2}{25}C_{2,5} + \Delta_8^{(1)}(N) + 2\mathcal{E}_{25,8192}(N) < 0.031297$$

at the left endpoint, again with decreasing right side. Thus the entire interval $3 \cdot 10^{13} \leq N < 6 \cdot 10^{13}$ is closed.

It remains to prove the result for $N \geq 6 \cdot 10^{13}$. Indeed, in the first case an even member of A^* sieves both base classes. In the second, the four odd base progressions modulo 100 split into two on which $4 \mid ab + 1$ automatically and two sieved progressions. In the third, an even base member sieves the opposite class and an odd member of A^* sieves its own base class. In the fourth, one member of each base class sieves the other. Lemmas 4.1 and 4.2 therefore give

$$\begin{aligned} U_1(N) &= 0.0376115 + \Delta_{25}^{(1)}(N) + 2\mathcal{E}_{25,H_{25}(N)}(N), \\ U_2(N) &= 0.0356927 + \Delta_{50}^{(1)}(N) + 2\mathcal{E}_{100,H_{100}(N)}(N) + 2/N, \\ U_3(N) &= 0.0334755 + \Delta_{50}^{(1)}(N) + 2\mathcal{E}_{25,H_{25}(N)}(N), \\ U_4(N) &= 0.0293395 + 2\mathcal{E}_{25,H_{25}(N)}(N). \end{aligned}$$

The only increases can occur at N_{100} and N_{25} . Evaluating the right-hand envelopes at the initial point and immediately after each jump gives

N	U_1	U_2	U_3	U_4
$6 \cdot 10^{13}$	0.0399044	0.0379855	0.0357684	0.0312858
N_{100}	0.0398938	0.0384785	0.0357578	0.0312767
N_{25}	0.0393740	0.0374551	0.0352380	0.0308770

All entries are strictly below $1/25$. Between rows they decrease. For $N \geq 1.6 \cdot 10^{15}$, use the original rounded progression errors $4.7N^{-1/4}$ and $5.3N^{-1/4}$ together with the decreasing aggregate diagonal bound; at the endpoint the four bounds are respectively below

$$0.039224, \quad 0.037495, \quad 0.035088, \quad 0.030827.$$

Thus every case is closed. The Hall statement follows from Theorem 2.2. $\square$

Corollary 4.8 (Parity pruning below the large close). *Let $8.858 \cdot 10^{12} \leq N < 3 \cdot 10^{13}$. If a compatible outside set B violates the Hall inequality, then $B \setminus C_N^{18}$ contains an even element.*

Proof. Put $A = B \cup T_N(B)$ as in Theorem 2.2; then $A^* = B \setminus C_N^{18}$. If $A^* = \emptyset$, the pure opposite-base matching, Corollary 5.11, closes the Hall inequality. Suppose A^* is nonempty and odd. If the base part $A_7 \cup A_{18}$ is odd, use the low-range diagonal envelope

$$0.0356927 + \Delta_{50}^{(0)}(N) + 2\mathcal{E}_{100,8192}(N) + 2/N < 0.039999984$$

at $N = 8.858 \cdot 10^{12}$; it remains decreasing throughout the stated interval. If the base part has an even element, the corresponding bound

$$0.0334755 + \Delta_{50}^{(0)}(N) + 2\mathcal{E}_{25,8192}(N) < 0.037783$$

holds at $N = 8.858 \cdot 10^{12}$ and also decreases. Either alternative would give $|A| < N/25$, contradicting the assumed Hall defect. Hence A^* , and therefore $B \setminus C_N^{18}$, contains an even element. $\square$

Corollary 4.9 (Quantitative even-pivot pruning). *Let $8.858 \cdot 10^{12} \leq N < 3 \cdot 10^{13}$, and suppose that a compatible outside set B violates the Hall inequality. If*

$$E = \{b \in B \setminus C_N^{18} : 2 \mid b\}, \quad e = |E|,$$

Define

$$U_{\text{odd}}(N) = 0.0356927 + \Delta_{50}^{(0)}(N) + 2\mathcal{E}_{100,8192}(N) + 2/N.$$

Then

$$e > \left(\frac{1}{25} - U_{\text{odd}}(N) \right) N - \frac{7}{25} > 2.$$

Thus there are always two distinct even elements of $B \setminus C_N^{18}$. If $N \geq 10^{13}$, the stronger bound

$$e > \frac{21N}{125000} - \frac{7}{25}. \tag{9}$$

holds, and there are distinct even $b, c \in B \setminus C_N^{18}$ with

$$1 \leq |b - c| \leq 5952.$$

For this pair the prime 2 cannot occur in the same-prime part and

$$|S_N(b, c)| \leq \frac{58N}{11025} + 2. \quad (10)$$

The original Hall inequality follows if

$$|T_N(B) \cap D_N(b, c)| \leq \frac{383N}{11025} - |B| - \frac{57}{25}. \quad (11)$$

If moreover $2.25 \cdot 10^{13} \leq N < 3 \cdot 10^{13}$, the pair may be chosen with $|b - c| \leq 880$. Then at most one same-prime witness occurs,

$$|S_N(b, c)| \leq \frac{N}{225} + 1, \quad (12)$$

and it is sufficient that

$$|T_N(B) \cap D_N(b, c)| \leq \frac{8N}{225} - |B| - \frac{32}{25}. \quad (13)$$

Thus every residual defect in this range is unconditionally excluded from the one-even-pivot and 2-adic same-prime branches.

Proof. Put $A = B \cup T_N(B)$ and $A^* = B \setminus C_N^{18}$, and remove the set E from A . The remaining non-base part is odd. If it is nonempty, the larger of the two low-range odd envelopes in the preceding proof is at most $U_{\text{odd}}(N)N$. If it is empty, retain any $b \in E$, whose existence follows from Corollary 4.8. The two off-diagonal progression estimates from b give instead

$$|A \setminus E| \leq N \left(\frac{2}{25} C_{2,5} + 2\mathcal{E}_{25,8192}(N) \right) < 0.016136N$$

at $N = 8.858 \cdot 10^{12}$, with a decreasing right side. Hence in all cases $|A \setminus E| \leq U_{\text{odd}}(N)N$. A Hall defect has $|A| > |C_N| \geq N/25 - 7/25$, so subtraction proves the first displayed bound. At the left endpoint $U_{\text{odd}}(N) < 0.039999984$, so its right side exceeds $1.6 \cdot 10^{-8}N - 7/25 > 2$. At $N = 10^{13}$ one has $U_{\text{odd}}(N) < 0.039832$, and monotonicity proves (9) on the remaining range.

List the e even elements increasingly. Since

$$\frac{21}{125000} - \frac{1}{5953} = \frac{13}{744125000}$$

and $N \geq 10^{13}$, equation (9) gives $e - 1 > (N - 1)/5953$. A least consecutive gap is therefore less than 5953; being even, it is at most 5952.

For an even pivot, $ab + 1$ is odd, so 2 cannot witness a square divisor. Every same-prime witness also differs from 5 and has $p^2 \mid |b - c|$. There are at most two such primes, because $(3 \cdot 7 \cdot 11)^2 = 53361 > 5952$. The two largest residue-class budgets are therefore those for $p = 3, 7$, and

$$|S_N(b, c)| \leq \frac{N}{25} \left(\frac{1}{9} + \frac{1}{49} \right) + 2 = \frac{58N}{11025} + 2.$$

Combining this with $|C_N| \geq N/25 - 7/25$ proves that (11) is sufficient for the original Hall inequality, exactly as in Proposition 6.31.

At $N = 2.25 \cdot 10^{13}$, direct substitution in the decreasing low-range odd envelope gives

$$0.04 - \left(0.0356927 + \Delta_{50}^{(0)}(N) + 2\mathcal{E}_{100,8192}(N) + 2/N \right) > \frac{1}{881} + 4 \cdot 10^{-6}.$$

The preceding argument therefore gives $e - 1 > (N - 1)/881$ throughout the final subinterval. The least even gap is less than 881, hence at most 880. Two distinct same-prime witnesses would force $(3 \cdot 7)^2 = 441$ to divide this even gap, and consequently would force the gap to be at least 882, a contradiction. Thus only $p = 3$ need be charged, which proves (12). Subtracting its bound from $|C_N| \geq N/25 - 7/25$ gives (13). $\square$

Lemma 4.10 (Finite two-pivot sieve on the base classes). *Let $8.858 \cdot 10^{12} \leq N < 3 \cdot 10^{13}$, and let $b, c \leq N$ be distinct even integers, neither congruent to 7 or 18 (mod 25). Then*

$$\#\{a \leq N : a \equiv 7 \text{ or } 18 \pmod{25}, \mu(ab + 1) = \mu(ac + 1) = 0\} < 0.013992N.$$

If in addition $N \geq 10^{13}$ and $|b - c| \leq 5952$, the right side improves to $0.012132N$.

Proof. Let

$$\mathcal{P} = \{3, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}.$$

For $p \in \mathcal{P}$, each of $p^2 \mid ab + 1$ and $p^2 \mid ac + 1$ specifies at most one residue of a modulo p^2 . When both residues exist, they coincide exactly when $p^2 \mid b - c$.

Fix one of the two classes $a \equiv 7, 18 \pmod{25}$. Inclusion–exclusion for the event that some $p \in \mathcal{P}$ divides $ab + 1$ to second order, the analogous event for $ac + 1$, and their union gives the local intersection density

$$I_J = 1 - 2 \prod_{p \in \mathcal{P}} \left(1 - \frac{1}{p^2}\right) + \prod_{p \in J} \left(1 - \frac{1}{p^2}\right) \prod_{p \in \mathcal{P} \setminus J} \left(1 - \frac{2}{p^2}\right),$$

where $J = \{p \in \mathcal{P} : p^2 \mid b - c\}$. In general it is largest when $J = \mathcal{P}$, and direct rational multiplication gives

$$I_J \leq I_{\mathcal{P}} = 1 - \prod_{p \in \mathcal{P}} \left(1 - \frac{1}{p^2}\right) < 0.152377. \quad (14)$$

If $|b - c| \leq 5952$, at most two primes lie in J , because $(3 \cdot 7 \cdot 11)^2 > 5952$. In that case the worst choice is $J = \{3, 7\}$ and $I_J < 0.129757$. Every nonempty inclusion–exclusion intersection is one CRT class after $a \pmod{25}$ is fixed. The total rounding discrepancy is at most

$$\epsilon = 2(2^{13} - 1) + (3^{13} - 1) = 1610704.$$

Thus the small-prime common failures in both base classes contribute at most

$$\frac{2I_J}{25}N + 2\epsilon. \quad (15)$$

It remains to charge a witness prime larger than 47 in at least one of the two linear forms. Put $Y = \lfloor N/333 \rfloor$. The finite computation

$$\sum_{47 < p \leq 1000} \frac{1}{p^2} + \sum_{n > 1000} \frac{1}{n^2} < 0.004751 =: \tau$$

gives $\sum_{p > 47} p^{-2} < \tau$. For $47 < p \leq Y$, a fixed prime contributes at most $N/(25p^2) + 1$ in a fixed base class. Hence one form in one base class contributes at most

$$\frac{\tau N}{25} + \pi(Y)$$

in this intermediate range.

For $p > Y$, use the tail calculation in Proposition 1 of Sothanaphan [2]. After the same removal of one redundant factor 5 as in Lemma 4.1, the root count is at most 8192 because $5N < Q_*$. If $X \leq N/25 + 1$ is the length of the base progression and d_1 is its transformed modulus, then

$$\sum_{r>Y} A_r \leq 8192 \left(\frac{2x}{d_1 Y} + \frac{x}{Y^2} \right) \leq L(N),$$

where

$$L(N) = 8192 \left(\frac{2(N/25 + 2)}{Y} + \frac{N^2 + 25N + 1}{Y^2} \right).$$

This bounds, in particular, the prime-square tail. There are two forms and two base classes, so (15) and the union bound give

$$\frac{|G_N(b, c)|}{N} \leq \frac{2I_J}{25} + \frac{2\epsilon}{N} + \frac{4\tau}{25} + \frac{4\pi(Y)}{N} + \frac{4L(N)}{N}, \quad (16)$$

where $G_N(b, c)$ denotes the set in the statement.

Rosser and Schoenfeld's explicit estimate $\pi(y) < 1.25506y/\log y$ for $y > 1$ [3] applies. Since $N/334 \leq Y \leq N/333$, all nonconstant terms in the resulting upper envelope decrease for $N \geq 8.858 \cdot 10^{12}$. At the left endpoint its five terms, using $I_J < 0.152377$, are respectively less than

$$0.012190160, \quad 0.000000364, \quad 0.000760160, \quad 0.000628127, \quad 0.000412773.$$

Their sum is less than 0.013992. Under the additional gap hypothesis, evaluate instead at $N = 10^{13}$ with $I_J < 0.129757$; the five terms are respectively less than

$$0.01038056, \quad 0.000000323, \quad 0.000760160, \quad 0.000624970, \quad 0.000365635.$$

and their sum is less than 0.012132. This proves both assertions. $\square$

Theorem 4.11 (Improved explicit large range). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 8.858 \cdot 10^{12}$.*

Proof. Theorem 4.7 handles $N \geq 3 \cdot 10^{13}$. Suppose instead that $8.858 \cdot 10^{12} \leq N < 3 \cdot 10^{13}$ and that B is a Hall defect. Put $A = B \cup T_N(B)$. Corollary 4.9 supplies distinct even $b, c \in B \setminus C_N^{18}$.

Every $x \in A^*$ has $\mu(x^2 + 1) = 0$, so the repaired low-range aggregate diagonal estimate gives

$$|A^*| \leq N \left(\frac{23}{25} C_{\text{quad}} + \Delta_{25}^{(0)}(N) \right) < 0.025784N.$$

Every member of $A_7 \cup A_{18}$ is a common non-neighbour of b, c . Lemma 4.10 therefore gives

$$|A| < (0.025784 + 0.013992)N = 0.039776N < \frac{N}{25} - \frac{7}{25} \leq |C_N|,$$

contradicting the Hall defect. Thus the middle interval is closed, and the Hall equivalence finishes the proof. $\square$

5 A direct singleton Hall close

The full large- N argument spends density on all parts of an admissible set. For a singleton Hall defect we only need one squarefree neighbour, and the same explicit progression estimate becomes effective much earlier.

We use the following all-parameter form of the Möbius splitting estimate proved in Proposition 1 of Sothanaphan [2]. If $x \geq q \geq 1$, $(u, q) = 1$, and $D \geq 1$ is an integer, then

$$S_2(x; q, u) \geq \frac{x}{q} \prod_{p \nmid q} \left(1 - \frac{1}{p^2}\right) - D - \frac{x}{qD} - \lambda q^{1/4} \left(\frac{2x}{qD} + \frac{x}{D^2}\right), \quad (17)$$

where $S_2(x; q, u)$ counts squarefree positive integers at most x in the class $u \pmod{q}$ and

$$\lambda = \frac{128}{(8 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13)^{1/4}} < 6.876.$$

For $N \geq 2 \cdot 10^9$ define

$$\begin{aligned} \mathcal{L}(N) = & 0.6079 \left(\frac{N}{25} - 2\right) - (3.2N^{3/4} + 1) - \frac{N/25 + 1}{3.2N^{3/4}} \\ & - 6.876\sqrt{5}N^{1/4} \left(\frac{2(N/25 + 1)}{3.2N^{3/4}} + \frac{N^2 + 25N}{(3.2N^{3/4})^2}\right). \end{aligned} \quad (18)$$

Theorem 5.1 (Uniform one-pivot Hall expansion). *Let $N \geq 2 \cdot 10^9$, and let $b \in [1, N] \setminus C_N$. Then there is an $a \in C_N$ for which $ab + 1$ is squarefree. More precisely,*

$$|\Gamma_N(\{b\})| \geq \mathcal{L}(N) > 0.00207N,$$

and $\mathcal{L}(2 \cdot 10^9) > 4.14 \cdot 10^6$. Consequently every nonempty compatible outside set B with $|B| \leq \mathcal{L}(N)$ satisfies the Hall inequality (2). This includes all singleton defects.

Proof. Write $a = 7 + 25n$ and put

$$X = |C_N|, \quad g = (7b + 1, 25b), \quad q = \frac{25b}{g}, \quad c = \frac{7b + 1}{g}.$$

Since $(b, 7b + 1) = 1$, one has $g \mid 25$. The exclusion $b \not\equiv 7 \pmod{25}$ gives $g \in \{1, 5\}$, $(c, q) = 1$, and $g \mid q$ but $g \nmid c$. Therefore g is squarefree and

$$(7 + 25n)b + 1 = g(qn + c)$$

is squarefree exactly when $qn + c$ is squarefree.

The X values $qn + c$, $0 \leq n < X$, are precisely the positive integers in the class $c \pmod{q}$ up to

$$x = q(X - 1) + c.$$

Here $c \leq q$, $q \leq 25N$, $x/q \leq X$, and

$$\frac{x}{q} \geq X - 1, \quad \frac{N}{25} - 1 \leq X \leq \frac{N}{25} + 1, \quad x \leq N^2 + 25N.$$

Take $D = \lceil 3.2N^{3/4} \rceil$ in (17). Since

$$\prod_{p \nmid q} \left(1 - \frac{1}{p^2}\right) \geq \prod_p \left(1 - \frac{1}{p^2}\right) = \frac{6}{\pi^2} > 0.6079$$

and $q^{1/4} \leq \sqrt{5}N^{1/4}$, the number of squarefree values is at least $\mathcal{L}(N)$. After division by N , every subtracted term is decreasing for $N \geq 2 \cdot 10^9$, while the positive main term is increasing. Using $\sqrt{5} < 2.237$, at that endpoint this lower bound exceeds $4.14 \cdot 10^6$ and its quotient by N exceeds 0.00207. Thus a squarefree neighbour exists, and the normalized lower bound remains above 0.00207 thereafter. For a nonempty B , choose $b \in B$. Then $\Gamma_N(\{b\}) \subseteq \Gamma_N(B)$, so $|\Gamma_N(B)| \geq \mathcal{L}(N) \geq |B|$ whenever the displayed size hypothesis holds. This is precisely the Hall inequality. $\square$

For the matching range we can reuse the primorial progression error instead of the coarser all-parameter singleton bound. Put

$$N_{4096} = \frac{8P_{11}}{5} = 5936590507848, \quad H_{\deg}(N) = \begin{cases} 4096, & N < N_{4096}, \\ 8192, & N \geq N_{4096}, \end{cases}$$

and set

$$\kappa_5 = \frac{12769}{20164}.$$

For $1.8263 \cdot 10^{11} \leq N < N_{25}$, put

$$\mathcal{M}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25, H_{\deg}(N)}(N).$$

Proposition 5.2 (Primorial one-pivot degree). *On this interval every outside vertex has at least $\mathcal{M}(N)$ squarefree neighbours in C_N , and every vertex of C_N has at least the same number in C_N^{18} . Moreover $\mathcal{M}(N)/N$ is increasing on each of the two intervals separated by N_{4096} , and*

$$\mathcal{M}(1.8263 \cdot 10^{11}) > \frac{N}{50} + 2.8 \cdot 10^4.$$

Proof. For an outside pivot, Corollary 1 of Sothanaphan's progression estimate gives the bad-value upper bound used in (3). The effective root modulus is at most $5N$. The least moduli with root count greater than 4096 and 8192 are $8P_{11}$ and Q_* , respectively, so Lemma 4.1 allows the displayed $H_{\deg}$. Also

$$\prod_{(p, 25b)=1} \left(1 - \frac{1}{p^2}\right) \geq \prod_{p \neq 5} \left(1 - \frac{1}{p^2}\right) = \frac{25}{4\pi^2} > \kappa_5.$$

Indeed the classical bound $\pi < 355/113$ gives

$$\frac{25}{4\pi^2} > \frac{25}{4} \left(\frac{113}{355}\right)^2 = \frac{12769}{20164} = \kappa_5.$$

Since $|C_N| \geq N/25 - 7/25$, subtraction gives the claimed lower degree. The reverse 7–18 calculation is identical. Every nonconstant term of $\mathcal{E}_{25, H_{\deg}}$ decreases while $H_{\deg}$ is fixed. Direct evaluation gives the displayed lower-end margin and $\mathcal{M}(N_{4096})/N_{4096} > 0.0222$ immediately after the sole jump. This proves both the asserted piecewise monotonicity and the strict half-class lower bound throughout the interval. $\square$

For $4.5654 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}$ define the correlated degree envelope

$$\mathcal{M}_1(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,1024}(N).$$

Proposition 5.3 (Root-product correlated degree). *Throughout this interval every outside vertex has at least $\mathcal{M}_1(N)$ squarefree neighbours in C_N , and the same reverse degree holds from C_N to C_N^{18} . Moreover $\mathcal{M}_1(N)/N$ is increasing and*

$$\mathcal{M}_1(4.5654 \cdot 10^{10}) > \frac{N}{50} + 800.$$

Proof. Let $m = 2^\nu m_0 \leq 5N$ be the effective root modulus. If $\rho(m) \leq 1024$, the proof of Proposition 5.2 applies directly with $H = 1024$. Suppose instead that $\rho(m) > 1024$. Since $5N < P_{11}$ throughout the present interval, the cases $\nu \leq 1$ are impossible: they require at least eleven odd prime factors. Removing a factor 5 did not change the 2-adic valuation, so $\nu \geq 2$ implies $4 \mid b$. The Euler product then omits $p = 2$ as well as $p = 5$, and is at least $4\kappa_5/3$; meanwhile $\rho(m) \leq 4096$, since the least modulus with more than 4096 roots is $8P_{11} > 5N$. Thus the $H = 4096$ lower degree is at least the asserted $H = 1024$ lower degree provided

$$\mathcal{E}_{25,4096}(N) - \mathcal{E}_{25,1024}(N) \leq \frac{\kappa_5}{75}.$$

The left side decreases; at $N = 4.5654 \cdot 10^{10}$ it is below 0.003132, whereas $\kappa_5/75 > 0.008443$. This proves the correlated bound in both directions. The monotonicity and endpoint margin follow directly from the formula for $\mathcal{E}_{25,1024}$. $\square$

For $2.2827 \cdot 10^{10} \leq N < 2.47 \cdot 10^{11}$ put

$$\mathcal{M}_{1/2}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,512}(N).$$

Proposition 5.4 (Second root–Euler correlated degree). *Throughout this interval every outside vertex has at least $\mathcal{M}_{1/2}(N)$ squarefree neighbours in C_N , and the same reverse degree holds from C_N to C_N^{18} . The quotient $\mathcal{M}_{1/2}(N)/N$ increases, and*

$$\mathcal{M}_{1/2}(2.2827 \cdot 10^{10}) > \frac{N}{50} + 411.$$

Proof. Let $m = 2^\nu m_0 \leq 5N$ be the effective root modulus. If $\rho(m) \leq 512$, use the progression estimate directly with $H = 512$. Suppose $\rho(m) > 512$ and $\nu \leq 1$. Then m_0 has at least ten distinct odd prime factors. At most one is the modulus prime 5, so the pivot is divisible by at least nine distinct odd primes other than 5. If 3 did not divide the pivot, then

$$b \geq 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 37 = 247357937827 > N,$$

a contradiction. Hence the Euler product omits 3 and is at least $9\kappa_5/8$. Also $\rho(m) \leq 1024$, because $5N < P_{11}$, and at the lower endpoint

$$\mathcal{E}_{25,1024} - \mathcal{E}_{25,512} < 0.001386 < \frac{\kappa_5}{200}.$$

Thus the missing-3 main-term gain pays the larger root error.

If $\nu \geq 2$, then $4 \mid b$, the Euler product omits 2, and $\rho(m) \leq 4096$. At the same endpoint

$$\mathcal{E}_{25,4096} - \mathcal{E}_{25,512} < 0.005332 < \frac{\kappa_5}{75},$$

so the missing-2 gain pays this error. Both differences decrease. The reverse progression has the identical alternatives. This proves the degree bounds; monotonicity and the endpoint margin follow by direct evaluation. $\square$

Put

$$\beta_8 := 0.00097181149.$$

For $1.382 \cdot 10^{10} \leq N \leq 2.2827 \cdot 10^{10}$ define

$$\mathcal{M}_{1/4}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,512}(N) + \beta_8 N.$$

Proposition 5.5 (Eight-prime root–Euler correlated degree). *Throughout this interval every outside vertex has at least $\mathcal{M}_{1/4}(N)$ squarefree neighbours in C_N , and the same reverse degree holds from C_N to C_N^{18} . The quotient $\mathcal{M}_{1/4}(N)/N$ increases, and*

$$\mathcal{M}_{1/4}(1.382 \cdot 10^{10}) > \frac{N}{50} + 16356.$$

Proof. If the effective root count is at most 256, use $H = 256$. Direct endpoint evaluation gives

$$\mathcal{E}_{25,512} - \mathcal{E}_{25,256} > 0.001099 > \beta_8$$

throughout the interval.

Suppose next that the root count exceeds 256 and the effective modulus has 2-adic valuation at most one. It has at least nine distinct odd prime factors. At most one is the modulus prime 5, so the pivot is divisible by at least eight distinct non-5 odd primes. Exact ordered-prime enumeration under product at most $2.2827 \cdot 10^{10}$ gives the unique least Euler gain at

$$\{7, 13, 17, 19, 23, 29, 31, 37\}, \quad \prod p = 22487085257,$$

and

$$\prod_p \left(1 - \frac{1}{p^2}\right)^{-1} - 1 > 0.038365586453.$$

Consequently its normalized main-term gain is greater than

$$\frac{\kappa_5}{25}(0.038365586453) > 0.000971811492 > \beta_8.$$

If the 2-adic valuation is at least two, the missing-2 gain $\kappa_5/75$ is larger still. Thus every root count at most 512 pays the claimed improved degree.

For the remaining higher-root strata, at the lower endpoint

$$\mathcal{E}_{25,1024} - \mathcal{E}_{25,512} + \beta_8 < 0.002610 < \frac{\kappa_5}{200},$$

and

$$\mathcal{E}_{25,4096} - \mathcal{E}_{25,512} + \beta_8 < 0.007275 < \frac{\kappa_5}{75}.$$

The forced missing-3 and missing-2 gains from the proof of Proposition 5.4 therefore continue to pay those strata. The reverse progression is identical. Monotonicity and the endpoint margin follow from the displayed formula. $\square$

Put

$$\beta'_8 := 0.00111752951, \quad \mathcal{M}_{1/8}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,512}(N) + \beta'_8 N.$$

Proposition 5.6 (Second eight-prime root–Euler degree). *For $1.29 \cdot 10^{10} \leq N \leq 1.382 \cdot 10^{10}$, every outside vertex has at least $\mathcal{M}_{1/8}(N)$ squarefree neighbours in C_N , with the same reverse degree. Moreover*

$$\mathcal{M}_{1/8}(1.29 \cdot 10^{10}) > \frac{N}{50} + 6728.$$

Proof. Repeat Proposition 5.5, but impose the current product bound $1.382 \cdot 10^{10}$. Complete ordered-prime enumeration now visits 828 nodes and is uniquely minimized by

$$\{7, 11, 13, 17, 23, 29, 31, 37\}, \quad \prod p = 13018838833.$$

Exact rational multiplication gives

$$\prod_p \left(1 - \frac{1}{p^2}\right)^{-1} - 1 = \frac{13286935886873651}{301166059585536000} > 0.044118304383,$$

and hence normalized gain greater than $0.001117529515 > \beta'_8$. The low-root saving $\mathcal{E}_{25,512} - \mathcal{E}_{25,256}$ exceeds 0.001299, while at the lower endpoint the two remaining comparisons are

$$\mathcal{E}_{25,1024} - \mathcal{E}_{25,512} + \beta'_8 < 0.002794 < \frac{\kappa_5}{200}$$

and

$$\mathcal{E}_{25,4096} - \mathcal{E}_{25,512} + \beta'_8 < 0.007568 < \frac{\kappa_5}{75}.$$

Thus every root stratum pays the asserted degree. Direct endpoint evaluation gives the margin. $\square$

Put

$$\beta''_8 := 0.00113583855, \quad \mathcal{M}_{1/16}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,512}(N) + \beta''_8 N.$$

Proposition 5.7 (Third eight-prime root–Euler degree). *For $1.279 \cdot 10^{10} \leq N \leq 1.29 \cdot 10^{10}$, every outside vertex has at least $\mathcal{M}_{1/16}(N)$ squarefree neighbours in C_N , with the same reverse degree. Moreover*

$$\mathcal{M}_{1/16}(1.279 \cdot 10^{10}) > \frac{N}{50} + 5090.$$

Proof. At product bound $1.29 \cdot 10^{10}$, complete enumeration is uniquely minimized by

$$\{7, 11, 13, 17, 19, 29, 31, 43\}, \quad \prod p = 12498697211.$$

Its exact relative Euler gain is

$$\frac{12438467965272539}{277389791723520000} > 0.044841116495,$$

which gives normalized gain $> 0.001135838556 > \beta''_8$. At the lower endpoint the low-root saving and the two higher-root comparisons remain strict:

$$\mathcal{E}_{25,512} - \mathcal{E}_{25,256} > \beta''_8,$$

$$\mathcal{E}_{25,1024} - \mathcal{E}_{25,512} + \beta_8'' < 0.002799 < \frac{\kappa_5}{200},$$

$$\mathcal{E}_{25,4096} - \mathcal{E}_{25,512} + \beta_8'' < 0.007586 < \frac{\kappa_5}{75}.$$

Thus the same root-stratified degree proof applies, and endpoint evaluation gives the margin. $\square$

Put

$$\mathcal{M}_{\text{QR2}}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,128}(N).$$

Proposition 5.8 (Quadratic-residue root hierarchy). *For $7.67769 \cdot 10^9 \leq N \leq 1.15 \cdot 10^{10}$, every outside vertex has at least $\mathcal{M}_{\text{QR2}}(N)$ squarefree neighbours in C_N , with the same reverse degree. Moreover*

$$\mathcal{M}_{\text{QR2}}(7.67769 \cdot 10^9) > \frac{N}{50} + 3853771.$$

Proof. The root count is at most 2048: more roots would require at least nine non-5 odd prime divisors, whose least product is

$$3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 = 20056049013 > N.$$

If the root count is at most 128, use Lemma 4.1 directly. For the four remaining root strata, let k be the number of non-5 odd prime divisors. Root count in $(128, 256]$, $(256, 512]$, $(512, 1024]$, or $(1024, 2048]$ forces $k \geq 5, 6, 7, 8$, respectively.

Each stratum also forces enough small prime divisors. If the asserted number were absent, the least possible radical would already exceed the upper endpoint:

H	forced small primes	excluded least product	value
256	3 primes ≤ 1000	$3 \cdot 7 \cdot 1009 \cdot 1013 \cdot 1019$	21872281683
512	4 primes ≤ 360	$3 \cdot 7 \cdot 11 \cdot 367 \cdot 373 \cdot 379$	11984670159
1024	5 primes ≤ 160	$3 \cdot 7 \cdot 11 \cdot 13 \cdot 163 \cdot 167 \cdot 173$	14141826699
2048	6 primes ≤ 60	$3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 61 \cdot 67 \cdot 71$	14813826027.

Choose $D = 20000000$. Then the complementary variable satisfies

$$m \leq \frac{N^2 + 25N + 1}{D^2} \leq 330626.$$

Intersect every possible collection of the forced quadratic-residue masks. The exact maxima are

H	mask cutoff/size	max $\#m$	H max $\#m$
256	1000/3	42697	10930432
512	360/4	21570	11043840
1024	160/5	10663	10918912
2048	60/6	4798	9826304.

Thus every higher-root tail is at most 11043840. The calculation is reproduced by `scripts/quadratic_residue_hierarchy_scan.py`.

Discarding every Euler gain, the progression proof gives at the new lower endpoint more than

$$\frac{\kappa_5(N - 43)}{25} - D - \frac{N/25 + 1}{D} - \frac{1}{5D} - 11043840 - 1 > \frac{N}{50} + 9933718.$$

This lower bound increases with N , whereas $\mathcal{M}_{\text{QR2}}(N) - N/50 < 1.277 \cdot 10^7$ throughout the interval. Hence every higher-root stratum also supplies the asserted degree. Direct evaluation gives the displayed endpoint margin. $\square$

Put

$$\mathcal{M}_{\text{QR}}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,256}(N).$$

Proposition 5.9 (Quadratic-residue screened degree). *For $1.15 \cdot 10^{10} \leq N \leq 1.2643 \cdot 10^{10}$, every outside vertex has at least $\mathcal{M}_{\text{QR}}(N)$ squarefree neighbours in C_N , with the same reverse degree. Moreover*

$$\mathcal{M}_{\text{QR}}(1.15 \cdot 10^{10}) > \frac{N}{50} + 154290.$$

Proof. If the transformed root count is at most 256, this is Lemma 4.1 with $H = 256$. We replace the crude root payment only in the stratum $256 < \rho(d_1) \leq 512$ whose 2-adic exponent is at most one. There the effective modulus contains exactly eight non-5 odd prime divisors S . In the large-square tail of Proposition 1 of Sothanaphan, the complementary variable m and a root d satisfy

$$d^2 m \equiv c_1 \pmod{d_1}.$$

For every $p \in S$ one has $c_1 \equiv 1 \pmod{p}$; hence any contributing m must be a nonzero quadratic residue modulo every $p \in S$.

Choose $D = 8000000$. On the whole interval,

$$m \leq \frac{N^2 + 25N + 1}{D^2} \leq 2497585.$$

Complete ordered-prime enumeration at the upper endpoint gives 5700 feasible supports, involving 101 possible primes. Intersecting the exact quadratic-residue masks through 2497585 leaves at most 6024 values of m for every support. Since each has at most 512 roots, the complete tail is at most

$$512 \cdot 6024 = 3084288.$$

The scan is reproduced by `scripts/quadratic_residue_support_scan.py`.

We may ignore the Euler gain from all eight primes. If X is either base progression length, then $X - 1 \geq (N - 43)/25$, $X \leq N/25 + 1$, and the proof of the progression estimate now gives at least

$$\frac{\kappa_5(N - 43)}{25} - D - \frac{N/25 + 1}{D} - \frac{1}{5D} - 3084288 - 1 > \frac{N}{50} + 50214005$$

squarefree values at the lower endpoint. The left side minus $N/50$ increases, while $\mathcal{M}_{\text{QR}}(N) - N/50$ is below $2.26 \cdot 10^6$ on the interval. Thus this root stratum more than pays the asserted degree.

For root count above 512 with 2-adic exponent at most one, the forced missing-3 gain applies and, at the lower endpoint,

$$\mathcal{E}_{25,1024} - \mathcal{E}_{25,256} < 0.003124 < \frac{\kappa_5}{200}.$$

For 2-adic exponent at least two, the missing-2 gain applies and

$$\mathcal{E}_{25,4096} - \mathcal{E}_{25,256} < 0.008083 < \frac{\kappa_5}{75}.$$

Both differences decrease with N . This covers every root stratum in both directions. Direct endpoint substitution gives the stated margin. $\square$

Put

$$\beta_8''' := 0.00116052, \quad \mathcal{M}_{1/32}(N) = \frac{\kappa_5 N}{25} - \frac{7}{25} - N\mathcal{E}_{25,512}(N) + \beta_8''' N.$$

Proposition 5.10 (Modulus-correlated eight-prime degree). *For $1.2643 \cdot 10^{10} \leq N \leq 1.279 \cdot 10^{10}$, every outside vertex has at least $\mathcal{M}_{1/32}(N)$ squarefree neighbours in C_N , with the same reverse degree. Throughout this interval the degree is strictly larger than half a base class, and at the lower endpoint*

$$\mathcal{M}_{1/32}(1.2643 \cdot 10^{10}) > \frac{N}{50} + 1442.$$

Proof. We retain the transformed modulus in the proof of Lemma 4.1. For $q_0 = 25$, root bound H , pivot b , and $X \leq N/25 + 1$, the last term before the uniform enlargement $d_1 \leq 25N$ is

$$\frac{Hd_1 X}{D^2} \leq \frac{25HbX}{D^2}.$$

Consequently, with $H = 512$ and $d_{512} = 1024^{1/3} N^{2/3}$, the actual error is at most

$$N\mathcal{E}_{25,512}(N) - \frac{512 \cdot 25(N - b)(N/25 + 1)}{d_{512}^2}. \quad (19)$$

The reciprocal d_1 term is still bounded with $d_1 \geq 5$, so this is a one-sided strengthening of the already proved envelope.

Only the root stratum $256 < \rho(d_1) \leq 512$ with odd b needs the new correlation. In this stratum b has exactly eight distinct non-5 odd prime divisors. Write their set as S and their product as $r(S)$. Every prime divisor of b lies in $S \cup \{5\}$; hence

$$b = r(S)k, \quad k \leq N/r(S), \quad p \mid k \implies p \in S \cup \{5\}.$$

For fixed S , the Euler gain is

$$G(S) = \frac{\kappa_5}{25} \left(\prod_{p \in S} (1 - p^{-2})^{-1} - 1 \right),$$

whereas (19) is worst at the largest admissible k . Complete ordered-prime enumeration through the upper endpoint visits 5831 supports and 157 support/multiplier breakpoints. Direct differentiation shows that the degree for each fixed admissible pair (S, b) increases between consecutive breakpoints. The unique global minimum is at $N = 12643000000$, with

$$S = \{7, 11, 13, 17, 19, 29, 31, 43\}, \quad r(S) = b = 12498697211,$$

and exact Euler contribution

$$G(S) = \frac{158826797448565050491}{139832194007826432000000}.$$

Adding the retained-modulus saving gives

$$G(S) + \frac{512 \cdot 25(N - b)(N/25 + 1)}{d_{512}^2 N} > 0.001160529889 > \beta_8'''.$$

This finite calculation is reproduced by `scripts/modulus_correlated_eight_prime_scan.py`.

If the root count is at most 256, then at the lower endpoint

$$\mathcal{E}_{25,512} - \mathcal{E}_{25,256} > 0.00133914 > \beta_8'''.$$

If b is even, the missing-2 gain is already larger than the asserted bonus. The remaining higher-root strata satisfy

$$\mathcal{E}_{25,1024} - \mathcal{E}_{25,512} + \beta_8''' < 0.002848 < \frac{\kappa_5}{200},$$

$$\mathcal{E}_{25,4096} - \mathcal{E}_{25,512} + \beta_8''' < 0.007653 < \frac{\kappa_5}{75},$$

so the forced missing-3 and missing-2 gains pay them exactly as before. The reverse progression has the same alternatives. Endpoint substitution gives the displayed margin; the breakpoint enumeration and piecewise monotonicity preserve it on the full interval. $\square$

Define the degree function

$$\mathcal{D}(N) = \begin{cases} \mathcal{M}_{\text{QR2}}(N), & 7.67769 \cdot 10^9 \leq N < 1.15 \cdot 10^{10}, \\ \mathcal{M}_{\text{QR}}(N), & 1.15 \cdot 10^{10} \leq N < 1.2643 \cdot 10^{10}, \\ \mathcal{M}_{1/32}(N), & 1.2643 \cdot 10^{10} \leq N < 1.279 \cdot 10^{10}, \\ \mathcal{M}_{1/16}(N), & 1.279 \cdot 10^{10} \leq N < 1.29 \cdot 10^{10}, \\ \mathcal{M}_{1/8}(N), & 1.29 \cdot 10^{10} \leq N < 1.382 \cdot 10^{10}, \\ \mathcal{M}_{1/4}(N), & 1.382 \cdot 10^{10} \leq N < 2.2827 \cdot 10^{10}, \\ \mathcal{M}_{1/2}(N), & 2.2827 \cdot 10^{10} \leq N < 2.47 \cdot 10^{11}, \\ \mathcal{M}_1(N), & 2.47 \cdot 10^{11} \leq N < 3.4375 \cdot 10^{11}, \\ \mathcal{M}(N), & 3.4375 \cdot 10^{11} \leq N < N_{25}, \\ \mathcal{L}(N), & N \geq N_{25}. \end{cases}$$

Corollary 5.11 (Global matching of the opposite base class). *Let*

$$C_N^{18} = \{b \in [1, N] : b \equiv 18 \pmod{25}\}.$$

For every $N \geq 7.67769 \cdot 10^9$, the bipartite squarefree-edge graph between C_N^{18} and C_N has a matching that saturates C_N^{18} . Consequently every outside set $B \subseteq C_N^{18}$ satisfies the Hall inequality (2).

Proof. Below $1.15 \cdot 10^{10}$ use Proposition 5.8, from there to $1.2643 \cdot 10^{10}$ use Proposition 5.9, from there to $1.279 \cdot 10^{10}$ use Proposition 5.10, from there to $1.29 \cdot 10^{10}$ use Proposition 5.7, from there to $1.382 \cdot 10^{10}$ use Proposition 5.6, from there to $2.2827 \cdot 10^{10}$ use Proposition 5.5; from there to $2.47 \cdot 10^{11}$, Proposition 5.4 gives degree at least $\mathcal{D}(N)$ in both directions; from there to $3.4375 \cdot 10^{11}$ use Proposition 5.3, and from there to N_{25} use Proposition 5.2. Above that point, Theorem 5.1 gives degree at least $\mathcal{D}(N)$ from every vertex of C_N^{18} into C_N . Applying the same proof with the roles reversed and the progression $b \equiv 18 \pmod{25}$ gives the same lower bound from every vertex of C_N into C_N^{18} : in that calculation $(18a + 1, 25a) = 1$ for $a \equiv 7 \pmod{25}$, and all the inequalities used in Theorem 5.1 remain valid.

The quotient $\mathcal{L}(N)/N$ is increasing. Direct substitution at $N = 1.409 \cdot 10^{12}$ gives

$$\mathcal{L}(N) > \frac{N}{50} + 5.8 \cdot 10^5 > \frac{1}{2} \max\{|C_N^{18}|, |C_N|\}.$$

For the lower intervals, Propositions 5.8, 5.9, 5.10, 5.7, 5.6, 5.5, 5.4, 5.3, and 5.2, together with their endpoint and switch evaluations, give the same strict half-size inequality with $\mathcal{D}(N)$. Write $L = C_N^{18}$ and $R = C_N$. If $S \subseteq L$ is nonempty and $|S| \leq \mathcal{D}(N)$, the neighbourhood of any member of S already has size at least $|S|$. If $|S| > \mathcal{D}(N)$ and $|\Gamma(S)| < |S|$, then, because $|L| \leq |R|$, there is a vertex $y \in R \setminus \Gamma(S)$. All neighbours of y lie in $L \setminus S$, so

$$\deg(y) \leq |L| - |S| < |L| - \mathcal{D}(N) < \mathcal{D}(N),$$

contradicting the reverse minimum-degree bound. Hall's theorem supplies the matching. Restricting it to any $B \subseteq C_N^{18}$ proves the last claim. $\square$

The half-minimum-degree argument is not needed below the preceding endpoint. Splitting both base classes by parity makes the square witnesses 2 and 5 identically inactive, while a Hall defect itself supplies a short triple. This gives a systemic matching down to the one-pivot scale.

Put

$$L_0 = \{x \leq N : x \equiv 18 \pmod{50}\}, \quad R_0 = \{y \leq N : y \equiv 7 \pmod{50}\},$$

and

$$L_1 = \{x \leq N : x \equiv 43 \pmod{50}\}, \quad R_1 = \{y \leq N : y \equiv 32 \pmod{50}\}.$$

Thus $C_N^{18} = L_0 \sqcup L_1$, $C_N = R_0 \sqcup R_1$, and $|L_i| \leq |R_i|$. Join L_i to R_i when the product plus one is squarefree.

Lemma 5.12 (Zero-coincidence three-form block). *Let x_1, x_2, x_3 be distinct elements of one L_i with $\max x_j - \min x_j \leq 200$. The number of $y \in R_i$ for which each of $x_1y + 1, x_2y + 1, x_3y + 1$ has a prime-square divisor at most 353 is less than*

$$0.003707 \left(\frac{N}{50} + 1 \right) + 1628952. \quad (20)$$

Proof. Let $\mathcal{P} = \{p : 3 \leq p \leq 353, p \neq 5\}$ and put $z_p = p^{-2}$. There are 69 such primes. No residue classes belonging to two different forms can coincide at a prime in $\mathcal{P}$: coincidence would give $p^2 \mid x_j - x_k$, whereas $50 \mid x_j - x_k$, so a nonzero difference would have absolute value at least $50p^2 \geq 450$.

For a list $w = (w_p)_{p \in \mathcal{P}}$ write

$$L_3(w) = 1 - e_1(w) + e_2(w) - e_3(w), \quad U_2(w) = 1 - e_1(w) + e_2(w),$$

where the e_j are elementary symmetric functions. The degree-3 lower and degree-2 upper Bonferroni inequalities applied to the survivor events give the common-bad density at most

$$I_3 := 1 - 3L_3(z) + 3U_2(2z) - L_3(3z).$$

Exact rational evaluation gives

$$\sum_{p \in \mathcal{P}} \frac{1}{p^2} < 0.161841, \quad I_3 < 0.003707. \quad (21)$$

Every nonconstant CRT term has one terminal incomplete class. The total number of such terms in the displayed Bonferroni combination is

$$3 \sum_{j=1}^3 e_j(1, \dots, 1) + 3 \sum_{j=1}^2 e_j(2, \dots, 2) + \sum_{j=1}^3 e_j(3, \dots, 3) = 1628952.$$

This proves (20). $\square$

For $N \geq 2 \cdot 10^9$ define

$$P(N) = \frac{N}{19(\log(N/20) - 1.1)}, \quad \tau = 0.000407315,$$

and let

$$H(N) = \begin{cases} 1024, & 2 \cdot 10^9 \leq N < 2587877292, \\ 2048, & 2587877292 \leq N < 7.67769 \cdot 10^9. \end{cases}$$

Put

$$E_H(N) = \left(\frac{N}{50} + 1 \right) \tau + P(N) + 400.801H(N), \quad (22)$$

$$D_H(N) = \frac{N}{50} - 1 - 0.161841 \left(\frac{N}{50} + 1 \right) - 69 - E_H(N), \quad (23)$$

$$U_H(N) = 0.003707 \left(\frac{N}{50} + 1 \right) + 1628952 + 3E_H(N). \quad (24)$$

Lemma 5.13 (Uniform medium-prime payment in the parity blocks). *On $2 \cdot 10^9 \leq N < 7.67769 \cdot 10^9$, every vertex of each graph L_i – R_i has degree at least $D_H(N)$, in both directions. Moreover, every triple in Lemma 5.12 has common non-neighbourhood in R_i of size at most $U_H(N)$, and*

$$D_H(N) > \frac{N}{100}, \quad U_H(N) < D_H(N). \quad (25)$$

Proof. For a pivot t in either block and a variable in the opposite block, 2 and 5 cannot be square witnesses. For primes through 353, the union bound and (21) give the first two subtractions in (23).

For the remaining primes take $Y = \lfloor N/19 \rfloor$. The density part is at most $(N/50 + 1)\tau$, since the rational prime-square tail calculation gives

$$\sum_{p>353} \frac{1}{p^2} < 0.000407315.$$

For $353 < p \leq Y$, pay one terminal class per prime. Dusart's estimate (D), together with $N/20 \leq Y \leq N/19$, gives

$$\pi(Y) \leq P(N).$$

For $p > Y$, write the variable as $r + 50n$. The transformed progression has modulus $50t$ and coprime initial term $rt + 1$. Removing one redundant factor 5 leaves a root modulus at most $10N$. The least modulus with more than 1024 unit square roots is

$$8P_9 = 25878772920,$$

and the least with more than 2048 is

$$8P_{10} = 802241960520.$$

This proves the displayed step bound $H(N)$. The standard complementary variable split used in Lemma 6.63 bounds the large-prime terminal classes by

$$H(N) \left(\frac{2(N/50 + 2)}{Y} + \frac{N^2 + 50N + 1}{Y^2} \right) < 400.801H(N).$$

Thus the full tail is bounded by $E_H(N)$, proving the degree assertion. For a common non-neighbour of three pivots, either all witnesses are at most 353, when Lemma 5.12 applies, or at least one form uses a larger witness; the union bound pays $3E_H(N)$. This gives (24).

With $H = 1024$, direct substitution at $N = 2 \cdot 10^9$ gives

$$\frac{D_H(N)}{N} > 0.01351113, \quad \frac{U_H(N)}{N} < 0.01064464.$$

At the sole root jump, $N = 2587877292$ and $H = 2048$, it gives

$$\frac{D_H(N)}{N} > 0.01344370, \quad \frac{U_H(N)}{N} < 0.01066191.$$

On each constant- H interval, $D_H(N)/N$ increases and $U_H(N)/N$ decreases. The two endpoint checks prove (25). $\square$

Theorem 5.14 (Systemic pure-18 parity matching). *For every $N \geq 2 \cdot 10^9$, the squarefree-edge graph between C_N^{18} and C_N has a matching that saturates C_N^{18} . Consequently every outside set $B \subseteq C_N^{18}$ satisfies the original Hall inequality (2).*

Proof. The range $N \geq 7.67769 \cdot 10^9$ is Corollary 5.11, so consider the preceding interval. Fix one block L_i – R_i and suppose $S \subseteq L_i$ is a Hall defect. Write $m = |S|$. If $m \leq D_H(N)$, the neighbourhood of one member of S already has size at least m , a contradiction. Hence

$$m > D_H(N) > \frac{N}{100}.$$

List S increasingly as $x_1 < \dots < x_m$. Since

$$\sum_{j=1}^{m-2} (x_{j+2} - x_j) \leq 2(N-1),$$

some three consecutive members have span less than 201, and hence at most 200. Lemma 5.13 bounds their common non-neighbourhood by $U_H(N) < D_H(N)$.

On the other hand let $T = R_i \setminus \Gamma(S)$. Every neighbour of a vertex of T lies in $L_i \setminus S$, so the reverse degree bound gives $m \leq |L_i| - D_H(N)$. The assumed Hall defect therefore implies

$$|T| > |R_i| - m \geq |R_i| - |L_i| + D_H(N) \geq D_H(N),$$

contradicting the common-non-neighbourhood bound. Hall's theorem supplies a matching in each parity block. Their disjoint union saturates $C_N^{18} = L_0 \sqcup L_1$. Restricting the matching to any $B \subseteq C_N^{18}$ proves the last statement. $\square$

The exact rational constants and both root-jump endpoint margins are reproduced by

`scripts/systematic_pure_matching_close.py`.

We next retain the quadratic-residue screen in the one-pivot progression itself. This is the ingredient needed to make the preceding matching robust against a genuinely mixed outside set.

Put

$$X_0(N) = \frac{501}{25} + \frac{501}{N}, \quad M_0 = 251002, \quad \eta_0 = 0.002267575.$$

For $0 \leq k \leq 8$, define s_k by

k	0	1	2	3	4	5	6	7	8
s_k	251002	251002	251002	125464	63250	32653	16510	8125	3679.

Proposition 5.15 (QR-correlated mixed degree). *For*

$$2 \cdot 10^9 \leq N < 7.67769 \cdot 10^9,$$

every outside vertex has more than

$$\frac{N}{50} + 4535153 \quad (26)$$

squarefree neighbours in C_N . The same lower bound holds in the reverse progression from C_N into C_N^{18} . More precisely, the normalized excess in (26) is greater than 0.0022675765 at the lower endpoint and increases on the interval.

Proof. Take $D = \lfloor N/500 \rfloor$. Then $D \geq N/501$, and hence

$$\frac{X}{D} \leq X_0(N), \quad \frac{N^2 + 25N + 1}{D^2} < 251002 = M_0.$$

Let k be the number of non-5 odd prime divisors of the effective root modulus. Nine such divisors are impossible, because

$$3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 \cdot 31 = 20056049013 > N.$$

If the 2-adic exponent is ν , the root count is at most

$$H_{\nu,k} = c_\nu 2^{k+1}, \quad c_0 = c_1 = 1, \quad c_2 = 2, \quad c_\nu = 4 \quad (\nu \geq 3).$$

For $k \geq 3$, product forcing supplies the following small subsets of the odd prime divisors:

k	cutoff	forced subset	max $\#\{m \leq M_0 : m \text{ survives}\}$
3	2000	1	125464
4	1400	2	63250
5	1000	3	32653
6	360	4	16510
7	160	5	8125
8	60	6	3679.

Indeed, failure of the first two assertions would force respectively the products

$$2003 \cdot 2011 \cdot 2017 = 8124542561, \quad 3 \cdot 1409 \cdot 1423 \cdot 1427 = 8583434967,$$

and the remaining four excluded products are the ones displayed in the proof of Proposition 5.8; all exceed the upper endpoint. Every contributing complementary variable m is a nonzero quadratic residue modulo each selected prime divisor. Exact intersections of those residue masks give the last column. For $k \leq 2$ we simply retain all M_0 values, giving the first three entries $s_k = M_0$.

The same Möbius split used in Proposition 5.9 therefore gives, for an odd pivot, the lower bound

$$G_{0,k}(N) = \frac{\kappa_5(N-43)}{25} - \frac{N}{500} - X_0(N) - H_{0,k} \left(s_k + \frac{X_0(N)}{2} \left(4 + \frac{s_k}{3026} \right) \right) - \frac{501}{5N} - 1. \quad (27)$$

For an even pivot the Euler product omits the factor at 2, so the first term is instead $4\kappa_5(N-43)/75$; use the corresponding c_ν in $H_{\nu,k}$. These bounds discard the modulus-5 and 2-adic residue screens and every further Euler gain.

The term $s_k/3026$ cannot be discarded. The reciprocal sum over the permitted transformed parameters through 3025 is at most 4, while each remaining survivor contributes at most $1/3026$. Therefore

$$\sum_{m \text{ surviving}} \frac{1}{m} \leq 4 + \frac{s_k}{3026}.$$

The elementary transformed-parameter count then gives the root term in (27).

There are only $4 \cdot 9$ displayed valuation/root strata. Direct exact substitution at $N = 2 \cdot 10^9$ shows that the minimum is the odd $k = 6$ stratum:

$$G_{0,6}(N) - \frac{N}{50} > 4535153.1398.$$

The next smallest odd strata retain more than $4.55 \cdot 10^6$, and every even stratum retains more than $1.50 \cdot 10^7$. In each fixed stratum the affine main term increases, while $X_0(N)$ and all normalized constant payments decrease. Feasible prime supports only disappear as N decreases. Thus the lower endpoint controls the full interval. The reverse 18 (mod 25) progression has the identical root and residue-screen calculation. This proves the proposition. $\square$

The support forcing, all exact QR-mask intersections, and all 36 endpoint comparisons are reproduced by

`scripts/systematic_mixed_degree_close.py.`

The reciprocal arithmetic is checked by

`scripts/verify_four_range_paper_arithmetic.py.`

Corollary 5.16 (A systemic mixed residual). *Let B be a Hall defect in the interval of Proposition 5.15, and put $B^* = B \setminus C_N^{18}$. Then*

$$|B^*| > 0.00453515N. \quad (28)$$

Consequently B^ , and hence $A^* = (B \cup T_N(B)) \setminus (C_N \cup C_N^{18})$, contains a mod-4 class with three members of span less than 1764.*

Proof. The pure case is already excluded by Theorem 5.14, so choose $x \in B^*$. Compatibility and the reverse degree bound give

$$|B \cap C_N^{18}| < |C_N^{18}| - \frac{N}{50} - 4535153,$$

whereas Hall defect and the forward degree give $|B| > N/50 + 4535153$. Since $|C_N^{18}| \leq N/25 + 1$,

$$|B^*| > 9070305 > 0.00453515N$$

at the lower endpoint; the normalized lower bound increases thereafter by the preceding proof.

Let S be the largest mod-4 class in B^* . Then $|S| > 0.0011337875N$. If $x_1 < \dots < x_m$ are its members, then

$$\sum_{j=1}^{m-2} (x_{j+2} - x_j) \leq 2(N-1).$$

At the lower endpoint this forces a three-term span below 1764, and the quotient decreases thereafter. Hence an integral span is at most 1763. $\square$

We now give the terminal three-pivot payment. Set

$$\tau_{131} = 0.001257623, \quad P_- = 13, \quad H_\Sigma = 384.$$

The first constant is Lemma 6.53, the second is Lemma 6.30, and the last follows from the QR hierarchy: for $Y = \lfloor N/77 \rfloor$, each pivot's high-root tail is no larger than the root-128 unscreened tail. Indeed $m \leq 5929$, and the four higher-root mask maxima are 986, 592, 315, 146; in every case

$$H \left(\#m + \frac{N/25 + 2}{2Y} \left(4 + \frac{\#m}{3026} \right) \right) \leq 128 \left(5929 + \frac{N/25 + 2}{2Y} \left(4 + \frac{5929}{3026} \right) \right).$$

The four mask maxima remain valid; their comparison is checked by `scripts/verify_four_range_paper_arithmetic.py`.

The exact restricted Möbius support in the aggregate diagonal estimate gives, uniformly on the present interval,

$$\Delta_{25}^{\text{mix}} < 0.002980825. \quad (29)$$

If three pivots lie in one mod-4 class which is globally concentrated in one mod-9 class, the allowed union of residue classes has density 161/225; its prefix discrepancy modulo 900 is less than 5, and

$$\Delta_{900}^{\text{mix}} < 0.003766884. \quad (30)$$

The maximum in (29) occurs at

$$N = 2 \cdot 10^9, \quad R = 3809763, \quad \sum_{s \leq R} 2^{\omega(s)} = 787461$$

on the permitted squarefree semigroup. The factor 13 is used in both Pell tails.

For primes through 131, apply degree-3, degree-2, and degree-3 Bonferroni bounds to the one-, two-, and three-form survivors. A prime common to all three forms has multiplicity 1; a pair coincidence has multiplicity 1 on that pair and 2 in the three-form union. Concentrating all allowed pair coincidences on one edge maximizes the sum of the three quadratic pair bounds. Direct expansion gives the following terminal table. The column θ multiplies C_{quad} ; “charge” is an explicitly discarded part of A^* .

branch	θ	common primes	charge	(I, ϵ)	$ A /N$
even generic	23/25	7, 11, 13, 17	0	(0.041080150, 45066)	< 0.038927972
two odd classes	23/50	7, 11, 13, 17	η_0	(0.115095250, 49865)	< 0.034542160
two odd, common 3	23/50	3, 7, 11, 13, 17	η_0	(0.160187242, 48180)	< 0.038147835
one odd class	23/100	2, 7, 11, 13, 17	0	(0.290282102, 48180)	< 0.039998152
one odd, common 2, 3	23/900	2, 3	0.0012	(0.344765994, 53410)	< 0.039971251
global mod 9	23/225	3, 7, 11, 13, 17	0	(0.145156038, 43495)	< 0.024889106
even concentrated	161/225	3	0	(0.113917972, 50119)	< 0.039955268
mixed parity	23/100	7, 11, 13, 17	η_0	(0.261748734, 49285)	< 0.039984162

Here every row includes

$$\frac{6\tau_{131}}{25} < 0.000301830$$

and the common intermediate-prime/root payment

$$\begin{aligned} & \frac{6}{77} \left(\frac{1}{L} + \frac{1}{L^2} + \frac{2.51}{L^3} \right) + \frac{2H_\Sigma}{N} \left(5929 + \frac{156}{25} + \frac{312}{N} + \left(\frac{39}{25} + \frac{78}{N} \right) \frac{5929}{3026} \right) \\ & < 0.007155053, \quad L = \log(N/78). \end{aligned}$$

Thus the displayed values are strict upper bounds at $N = 2 \cdot 10^9$; every analytic term decreases thereafter. The exact rational finite polynomials and restricted diagonal jump scan are reproduced by `scripts/systematic_mixed_full_close.py`; all branch totals are checked by `scripts/verify_four_range_paper_arithmetic.py`.

Theorem 5.17 (Systemic mixed three-pivot close). *The Hall inequality (2), and hence the required bound for Erdős Problem 848, holds for*

$$2 \cdot 10^9 \leq N < 7.67769 \cdot 10^9.$$

Consequently it holds for every $N \geq 2 \cdot 10^9$.

Proof. Suppose that B is a Hall defect and put $A = B \cup T_N(B)$. Then A is compatible, $|A| > |C_N|$, and A^* contains B^* from Corollary 5.16. Let e be the number of even members of A^* and split at $e = \eta_0 N$.

First suppose $e > \eta_0 N$. One even mod-4 class contains more than $0.0011337875N$ members. If that class is not contained in one residue modulo 9, choose three of its members which are nonconstant modulo 9. The prime 2 is inactive and 3 is not common. Since a common-prime product Q satisfies $4Q^2 < N$, at most four further common primes occur; replacing them by 7, 11, 13, 17 only enlarges the finite survivor bound. This is the first row of the table.

If the even class is contained in one residue modulo 9, choose three consecutive members. Their span is at most 1763, so 3 is the only possible common odd prime because $4(3 \cdot 7)^2 = 1764$. The other three mod-4 classes are unrestricted, while the selected class is restricted modulo 9; this is precisely the 161/225 residue union in (30). The seventh row closes this case.

Now suppose $e \leq \eta_0 N$. Charge the even part. If the odd part meets both odd classes modulo 4, choose a triple nonconstant modulo 4; if the odd part is also nonconstant modulo 9, the same triple may be chosen nonconstant for both projections. These are respectively the second and, when all odd members share one residue modulo 9, the third rows. The diagonal set is restricted to odd non-base values, giving $\theta = 23/50$.

It remains that all odd members of A^* occupy one mod-4 class. If $e = 0$ and this class is nonconstant modulo 9, the prime 2 is common but 3 is not; the fourth row applies. If the class is constant modulo 9, three consecutive odd pivots have common primes only 2 and 3. When $e \leq 0.0012N$, charge the even part and use the fifth row. When instead $e > 0.0012N$ and all of A^* is constant modulo 9, select three even pivots; the prime 2 is inactive, and the sixth row allows the common prime 3 and four further common primes.

Finally suppose $0 < e \leq \eta_0 N$ and A^* is not constant modulo 9. Choose two odd pivots and one even pivot so that the three residues modulo 9 are nonconstant. The 2-square event is active on at most two forms and is not common. The asymmetric finite polynomial in the last row, together with the charged even part, closes this final case.

These alternatives are exhaustive. Every row gives

$$\frac{|A|}{N} < \frac{1}{25} - \frac{7}{25N} \leq \frac{|C_N|}{N},$$

contradicting $|A| > |C_N|$. This proves the stated interval. The range $N \geq 7.67769 \cdot 10^9$ is Theorem 6.129, completing the last assertion. $\square$

5.1 Actual-support descent to two hundred million

For $N_0 \leq N < N_1$ we retain the ordered prime support. Put

$$N_0 = 2 \cdot 10^8, \quad N_1 = 2 \cdot 10^9, \quad D = \lfloor N/300 \rfloor, \quad M = 90001.$$

If a pivot has k distinct non-5 odd prime divisors, retain its smallest $k - 1$ of them and require that a larger k -th prime still fit under N_1 . Exact intersections of their nonzero quadratic-residue masks on $m \leq M$ give

k	0	1	2	3	4	5	6	7	8
s_k	90001	90001	45180	22924	11414	5520	2513	1039	353.

The maximizing supports for $k = 2, \dots, 8$ are respectively

$$(38639), (971, 1429), (137, 193, 199), (43, 61, 73, 89), \\ (23, 29, 31, 41, 43), (11, 13, 17, 19, 23, 41), (3, 7, 13, 17, 19, 23, 29).$$

Proposition 5.18 (Uniform actual-support degree). *For $N_0 \leq N < N_1$, every outside vertex has more than*

$$\frac{N}{50} + 0.0001574N$$

squarefree neighbours in C_N , and the identical reverse bound holds from C_N into C_N^{18} . Consequently every mixed Hall defect satisfies

$$|B \setminus C_N^{18}| > 0.0003148N. \quad (31)$$

Proof. Since $D \geq (N - 299)/300$, throughout the interval

$$\frac{N^2 + 25N + 1}{D^2} < 90001.$$

For a valuation factor $c_\nu \in \{1, 2, 4\}$ and k odd prime divisors, put $H = c_\nu 2^{k+1}$. The same Möbius split as in Proposition 5.15, now with the actual-support survivor s_k , gives

$$\deg(b) - \frac{N}{50} \geq \left(\frac{4}{3}\right)^{[2|b]} \frac{\kappa_5(N - 43)}{25} - \frac{N}{300} - \frac{N}{50} - 1 \\ - \frac{N/25 + 1}{(N - 299)/300} - \frac{1}{5(N - 299)/300} \\ - H \left(s_k + \frac{X_D}{2} \left(4 + \frac{s_k}{3026} \right) \right), \quad X_D = \frac{N/25 + 1}{(N - 299)/300}.$$

Every fixed row divided by N is increasing: its leading coefficient is positive, while each remaining positive payment divided by N decreases. It therefore suffices to use $N = N_0$ and the support superset at N_1 . Among all $4 \cdot 9$ rows the minimum is the odd $k = 3$, $H = 16$ row, with survivor 22924 and witness (971, 1429); its margin is

$$31482.2990360 \dots > 0.0001574N_0.$$

The reverse progression has the same support and root calculation. The argument of Corollary 5.16 then gives $|B \setminus C_N^{18}| > 2(0.0001574N) - 1$, proving (31). The exact support enumeration is reproduced by `scripts/systematic_mixed_interval_support.py`; the degree rows are checked by `scripts/verify_four_range_paper_arithmetic.py`. $\square$

The transformed congruence contains one further uniform condition which was not needed for Proposition 5.18. In the notation of the progression split, $d_1 = 25b/g$ with $g \in \{1, 5\}$, so $5 \mid d_1$; moreover c_1 is a unit. Hence

$$d^2 m \equiv c_1 \pmod{d_1}$$

forces m into one fixed nonzero square coset modulo 5. Taking the worse of the two possible cosets and intersecting it with the same actual-support masks gives

k	0	1	2	3	4	5	6	7	8
$s_k^{(5)}$	36001	36001	18234	9314	4680	2290	1055	471	175.

The maximizing retained supports for $k = 2, \dots, 8$ are respectively

$$(44563), (1087, 1117), (101, 241, 263), (53, 59, 71, 83), \\ (19, 29, 37, 41, 43), (11, 13, 17, 19, 29, 31), (3, 7, 11, 13, 17, 23, 37).$$

Corollary 5.19 (Mod-5-coset actual-support degree). *On $N_0 \leq N < N_1$, every forward and reverse outside degree is greater than*

$$\frac{N}{50} + 0.0012427N.$$

Consequently every mixed Hall defect in this interval satisfies

$$|B \setminus C_N^{18}| > 0.0024854N - 1 > 0.002485N.$$

Proof. Repeat Proposition 5.18 with $s_k^{(5)}$. The unique controlling row is now the odd $k = 4$ row, with $H = 32$, survivor 4680, and retained support $(101, 241, 263)$. At $N = N_0$ its margin above $N/50$ is

$$248552.6176834 \dots > 0.0012427N_0.$$

The same normalized monotonicity proves the degree bound on the full interval. Applying the forward–reverse defect argument gives the displayed residual. Both cosets and every feasible support prefix at the upper endpoint are enumerated by `scripts/systematic_mod5_actual_support.py`; the degree rows are checked by `scripts/verify_four_range_paper_arithmetic.py`. $\square$

The same support constraint is retained in the large-square tail. On the following four blocks use $Y = \lfloor N/k_0 \rfloor$. The integer box is $m \leq k_0^2$: indeed $(k_0^2 + 1)(N/k_0 - 1)^2 > N^2 + 25N + 1$ at the left endpoint, and the difference increases. Exact support intersections give the displayed worst root row and the resulting three-pivot prime/root ceiling; the intermediate-prime term uses the three-term Dusart bound and $Y \geq N/(k_0 + 1)$.

$[L, U)$	k_0	H	s	support	three-pivot ceiling
$[2, 3] \cdot 10^8$	75	128	805	(97, 113, 151)	0.008955907
$[3, 5] \cdot 10^8$	85	128	1016	(97, 113, 151)	0.007668494
$[5, 10] \cdot 10^8$	100	128	1391	(73, 199, 257)	0.006336343
$[10, 20] \cdot 10^8$	125	128	2124	(47, 233, 401)	0.004881897.

All support maxima are reproduced by `scripts/systematic_mod5_actual_support.py`. The monotone analytic payments are checked by `scripts/verify_four_range_paper_arithmetic.py`.

Theorem 5.20 (Actual-support pure-18 parity matching). *For every $N \geq 2 \cdot 10^8$, the squarefree-edge graph between C_N^{18} and C_N has a matching saturating C_N^{18} . Consequently every $B \subseteq C_N^{18}$ satisfies the original Hall inequality.*

Proof. It remains only to treat $N_0 \leq N < N_1$. Work in the parity blocks L_i – R_i used in Theorem 5.14. The three-pivot ceiling above is six times a one-form intermediate-prime plus root payment. A parity-block progression has length $N/50 + 1$, rather than $N/25 + 1$, so one sixth of that ceiling is a valid one-form bound. Add the square-density tail $(N/50 + 1)0.000407315$ and retain the exact finite constants of Lemma 5.12. At the four left endpoints, the degree lower bound D and common-non-neighbour upper bound U satisfy

N	D/N	U/N	$D - U$
$2 \cdot 10^8$	0.0152620317	0.0127212769	> 508150
$3 \cdot 10^8$	0.0154767174	0.0093626504	> 1834220
$5 \cdot 10^8$	0.0156988362	0.0065246389	> 4587098
10^9	0.0159413140	0.0041684639	> 11772850 .

On each block D/N increases and U/N decreases. In particular $D > N/100$ and $U < D$. A Hall defect in either parity block therefore contains three consecutive pivots of span at most 200; the reverse degree argument forces its common non-neighbourhood to exceed D , contradicting $U < D$, exactly as in Theorem 5.14. The four endpoint calculations are reproduced by `scripts/verify_four_range_paper_arithmetic.py`. The already proved range $N \geq N_1$ completes the assertion. $\square$

We finally replace the absolute small-square rounding loss by an exact finite prefix. Let $R = 2 \cdot 10^7$. For a union W of non-base residue classes, put

$$Q_W(U) = - \sum_{\substack{2 \leq s \leq R \\ p|s \Rightarrow p \equiv 1 \pmod{4}}} \mu(s) \#\{x \leq U : x \in W, s^2 \mid x^2 + 1\} \\ + \frac{2\theta_W U (\log R + 2)}{R} + 13 \frac{U^2 + 1}{R^2},$$

where θ_W is the density of W . Möbius inversion, the harmonic tail, and the sharp negative-Pell orbit bound show that $Q_W(U)$ is an upper bound for the diagonal non-squarefree count through U .

Partition $[N_0, N_1)$ into 463 adjacent blocks $[L, U]$, each of relative width less than 0.005, with additional breaks at $3 \cdot 10^8, 5 \cdot 10^8, 10^9$. Since the actual diagonal count is monotone, $Q_W(U)/L$ bounds its normalized value throughout the block. The finite computation enumerates 1237599 squarefree support nodes, 4134027 roots, and 51357374 progression events. The eighteen concentrated sets are evaluated simultaneously from

$$U_{25} - E_e + E_{e,c} \quad (e = 0, 2; c \pmod{9}).$$

The controlling block is $[200000000, 200999999]$, where the worst concentrated diagonal is

$$\frac{Q_W(U)}{L} < 0.019673927941605. \quad (32)$$

All integer events and block maxima are reproduced by `scripts/systematic_concentrated_diagonal_blocks.cpp`.

Theorem 5.21 (Actual-support mixed close). *The Hall inequality (2), and hence the exact bound for Erdős Problem 848, holds for every*

$$N \geq 2 \cdot 10^8.$$

Proof. The range $N \geq N_1$ is Theorem 5.17 and the subsequent descent, so let $N_0 \leq N < N_1$. Suppose B is a Hall defect and put $A = B \cup T_N(B)$. Theorem 5.20 excludes $B \subseteq C_N^{18}$. Corollary 5.19 therefore gives

$$|A^*| \geq |B \setminus C_N^{18}| > 0.002485N.$$

Let e be the number of even members of A^* and put $\eta = 0.00016$.

If $e > \eta N$, one even mod-4 class has density greater than $\eta/2$. A closest pair in it has gap less than $2/\eta = 12500$. If the class is nonconstant modulo 9, add a third pivot which makes the triple nonconstant modulo 9; the prime 2 is inactive and 3 is not common. If it is constant modulo 9, the same short pair shows that the only possible common odd primes are 3, 7, since

$$4(3 \cdot 7 \cdot 11)^2 = 213444 > 12500.$$

These are the even-generic and even-concentrated rows below.

Now suppose $e \leq \eta N$ and charge all even members. More than $0.002325N$ odd members remain. If both odd mod-4 classes occur, choose a triple nonconstant modulo 4 and split according as its odd support is nonconstant or constant modulo 9. If only one odd mod-4 class occurs, a closest pair has gap less than 6169.4. In the nonconstant-mod-9 case add a third pivot of another mod-9 residue: 2 is common, 3 is not, and at most one further odd prime is common because $4(7 \cdot 11)^2 = 23716 > 6169.4$. In the constant case the common list is at most 2, 3, 7, since adjoining 11 would again force a gap at least $4(3 \cdot 7 \cdot 11)^2$.

Use cutoff 131, degree-3/2/3 Bonferroni bounds, the integrated tail $6\tau_{131}/25$, the appropriate exact diagonal prefix, and the root ceiling of the containing product block. Exhausting all 463 blocks gives

branch	allowed common primes	$\max A /N$
even generic	7, 11, 13, 17	0.038287487920
even concentrated	3, 7	0.039870783152
two odd classes	7, 11, 13, 17	0.031773851295
two odd, common 3	3, 7, 11, 13, 17	0.035364360630
one odd class	2, 7	0.038964611892
one odd, common 2, 3	2, 3, 7	0.039088651986.

Every maximum occurs in the first block. Its exact target is $1/25 - 7/(25N) > 0.0399999986$, so all six exhaustive alternatives contradict $|A| > |C_N|$. This proves the new interval and hence the theorem. $\square$

Corollary 5.22 (A mixed-base strip). *Let $N \geq 7.67769 \cdot 10^9$ and let B be a compatible outside set. Put*

$$B^* = B \setminus C_N^{18}, \quad \mathcal{R}(N) = 2\mathcal{D}(N) - |C_N^{18}|.$$

If $|B^| \leq \mathcal{R}(N)$, then B satisfies the Hall inequality. Moreover*

$$\mathcal{R}(N) > 0.000000036N \quad (2.2827 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}),$$

and, more sharply,

$$\mathcal{R}(N) > 0.0010038N \quad (7.67769 \cdot 10^9 \leq N < 1.15 \cdot 10^{10}),$$

and

$$\mathcal{R}(N) > 0.0000268N \quad (1.15 \cdot 10^{10} \leq N < 1.2643 \cdot 10^{10}),$$

and

$$\mathcal{R}(N) > 0.000000228N \quad (1.2643 \cdot 10^{10} \leq N < 1.279 \cdot 10^{10}),$$

and

$$\mathcal{R}(N) > 0.00000079N \quad (1.279 \cdot 10^{10} \leq N < 1.29 \cdot 10^{10}),$$

and

$$\mathcal{R}(N) > 0.00000104N \quad (1.29 \cdot 10^{10} \leq N < 1.382 \cdot 10^{10}),$$

and

$$\mathcal{R}(N) > 0.00000236N \quad (1.382 \cdot 10^{10} \leq N < 2.2827 \cdot 10^{10}),$$

and

$$\mathcal{R}(N) > 0.000127N \quad (2.3669 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}),$$

and

$$\mathcal{R}(N) > 0.001162N \quad (3.22715 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}),$$

and

$$\mathcal{R}(N) > 0.001187N \quad (3.2532 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}),$$

and

$$\mathcal{R}(N) > 0.001414N \quad (3.4984 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}),$$

and

$$\mathcal{R}(N) > 0.001424N \quad (3.51 \cdot 10^{10} \leq N < 3.4375 \cdot 10^{11}),$$

and

$$\mathcal{R}(N) > 0.00032N \quad (1.001 \cdot 10^{11} \leq N < N_{25}),$$

and

$$\mathcal{R}(N) > 0.000724N \quad (N \geq 5.5 \cdot 10^{11}).$$

Thus every Hall defect in the new bottom interval must contain more than $0.001627N$ genuinely non-base outside vertices. The subsequent displayed bounds give the strip separately on each later degree piece.

Proof. If $B^* = \emptyset$, use Corollary 5.11. Otherwise fix $x \in B^*$. Repeat the relevant one-pivot degree proof with the variable in the progression $18 \pmod{25}$. Since $x \not\equiv 18 \pmod{25}$, the gcd $(18x + 1, 25x)$ is 1 or 5, so the identical calculation gives at least $\mathcal{D}(N)$ values $y \in C_N^{18}$ for which $xy + 1$ is squarefree. Compatibility forces all of them out of B . Consequently

$$|B \cap C_N^{18}| \leq |C_N^{18}| - \mathcal{D}(N).$$

Under the stated hypothesis,

$$|B| \leq |C_N^{18}| - \mathcal{D}(N) + |B^*| \leq \mathcal{D}(N).$$

The corresponding one-pivot degree bound now proves the Hall inequality.

Finally $|C_N^{18}| \leq N/25 + 1$. Direct substitution in the displayed formula for $\mathcal{M}_{\text{QR2}}$ at $N_0 = 7.67769 \cdot 10^9$ gives

$$\frac{2\mathcal{M}_{\text{QR2}}(N_0)}{N_0} - \frac{1}{25} - \frac{1}{N_0} > 0.0010038.$$

Direct substitution in $\mathcal{M}_{\text{QR}}$ at $1.15 \cdot 10^{10}$ gives

$$\frac{2\mathcal{M}_{\text{QR}}(1.15 \cdot 10^{10})}{1.15 \cdot 10^{10}} - \frac{1}{25} - \frac{1}{1.15 \cdot 10^{10}} > 0.0000268.$$

Direct substitution in $\mathcal{M}_{1/32}$ at $1.2643 \cdot 10^{10}$ gives

$$\frac{2\mathcal{M}_{1/32}(1.2643 \cdot 10^{10})}{1.2643 \cdot 10^{10}} - \frac{1}{25} - \frac{1}{1.2643 \cdot 10^{10}} > 0.000000228.$$

Direct substitution in $\mathcal{M}_{1/16}$ at $1.279 \cdot 10^{10}$ gives the stronger 0.00000079 bound. Direct substitution in $\mathcal{M}_{1/8}$ at $1.29 \cdot 10^{10}$ gives the stronger 0.00000104 bound. Direct substitution in $\mathcal{M}_{1/4}$ at $1.382 \cdot 10^{10}$ gives the stronger 0.00000236 bound. Direct substitution in $\mathcal{M}_{1/2}$ at $2.2827 \cdot 10^{10}$ gives the weaker transition bound 0.000000036, while substitution at $2.3669 \cdot 10^{10}$ gives the sharper 0.000127 bound. Direct substitutions at $3.22715 \cdot 10^{10}$, $3.2532 \cdot 10^{10}$, $3.4984 \cdot 10^{10}$, and $3.51 \cdot 10^{10}$ give the sharper 0.001162, 0.001187, 0.001414, and 0.001424 bounds. The quotient increases on each of the displayed degree pieces. Direct checks from $\mathcal{M}_{1/2}$ to $\mathcal{M}_1$ preserve both displayed bounds on their stated intervals. The later switches to $\mathcal{M}$, across its root-count jump, and from $\mathcal{M}$ to $\mathcal{L}$ keep the stated weaker bounds. Direct evaluation at $1.001 \cdot 10^{11}$ gives more than $0.00032N$, and at $5.5 \cdot 10^{11}$ gives more than $0.0012N$, and the same switch checks prove the $0.000724N$ bound from that point onward. $\square$

6 The valid two-pivot CRT decomposition

The large-prime tail is nonempty in general, but it can be placed into exact CRT classes without changing the Hall target.

Lemma 6.1 (Two-pivot square classes). *Let $b, c \not\equiv 7 \pmod{25}$ be distinct, and let $a \equiv 7 \pmod{25}$. Suppose both $ab + 1$ and $ac + 1$ are non-squarefree. Choose primes p, q with*

$$p^2 \mid ab + 1, \quad q^2 \mid ac + 1.$$

Then $p, q \neq 5$, and exactly one of the following alternatives applies:

1. $p = q$ and $p^2 \mid b - c$;
2. $p \neq q$ and a lies in one uniquely determined residue class modulo $25p^2q^2$.

Proof. If $25 \mid ab + 1$, then $7b + 1 \equiv 0 \pmod{25}$. Since $7^{-1} \equiv 18 \pmod{25}$, this forces $b \equiv 7 \pmod{25}$, contrary to the hypothesis. Hence $p \neq 5$, and similarly $q \neq 5$.

If $p = q$, subtracting the two divisibilities gives $p^2 \mid a(b - c)$. But $p \nmid a$, since $p \mid a$ would imply $ab + 1 \equiv 1 \pmod{p}$. Therefore $p^2 \mid b - c$.

If $p \neq q$, the three congruences

$$a \equiv 7 \pmod{25}, \quad a \equiv -b^{-1} \pmod{p^2}, \quad a \equiv -c^{-1} \pmod{q^2}$$

have pairwise coprime moduli. The Chinese remainder theorem gives one class modulo $25p^2q^2$. $\square$

For a compatible pair $B = \{b, c\}$ define the same-prime and distinct-prime parts of its common non-neighbour set by

$$\begin{aligned} S_N(b, c) &= \{a \in T_N(B) : \text{some prime } p \text{ has } p^2 \mid ab + 1 \text{ and } p^2 \mid ac + 1\}, \\ D_N(b, c) &= T_N(B) \setminus S_N(b, c). \end{aligned}$$

This is an exact disjoint decomposition, not a new cut. The same-prime debit can be bounded independently of what occurs in $D_N(b, c)$.

Proposition 6.2 (Two-point same-prime debit). *Let $N \geq 107$ and let $B = \{b, c\}$ be a compatible outside set. Then*

$$2 + |S_N(b, c)| \leq |C_N|.$$

For $N \geq 500$ the stronger explicit estimate

$$|C_N| - 2 - |S_N(b, c)| > \frac{0.549N}{25} - \log_4(N - 1) - \frac{57}{25} \quad (33)$$

holds. Consequently the two-point Hall inequality follows whenever the distinct-prime part fits inside the right-hand side of (33).

Proof. By Lemma 6.1, every relevant prime satisfies $p \neq 5$ and $p^2 \mid |b - c|$. For a fixed p , the simultaneous conditions $a \equiv 7 \pmod{25}$ and $p^2 \mid ab + 1$ determine one residue class modulo $25p^2$. Hence

$$|S_N(b, c)| \leq \sum_{\substack{p^2 \mid |b-c| \\ p \neq 5}} \left(\frac{N}{25p^2} + 1 \right). \quad (34)$$

First suppose $107 \leq N \leq 499$. Since $(2 \cdot 3 \cdot 7)^2 = 1764 > N - 1$, at most two distinct primes occur in the sum, and the largest two class bounds are those for $p = 2, 3$. For $107 \leq N \leq 167$ the exact class counts are at most 2 and 1, so $2 + |S_N(b, c)| \leq 5 \leq |C_N|$. For $168 \leq N \leq 499$,

$$2 + |S_N(b, c)| \leq 4 + \frac{N}{100} + \frac{N}{225} = 4 + \frac{13N}{900} \leq \frac{N}{25} - \frac{7}{25} \leq |C_N|.$$

The middle inequality holds already at $N = 168$ and its right-minus-left side increases with N .

Now let $N \geq 500$, and let k be the number of primes in (34). Their squared product divides $|b - c| \leq N - 1$, so $4^k \leq N - 1$ and $k \leq \log_4(N - 1)$. Also, every relevant prime is different from 5. Removing 5 and the displayed composite integers from the sum over all integers gives the elementary numerical bound

$$\sum_{p \neq 5} \frac{1}{p^2} < \frac{\pi^2}{6} - 1 - \sum_{m \in \{4, 5, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20\}} \frac{1}{m^2} < 0.451.$$

It follows that

$$2 + |S_N(b, c)| < 2 + \frac{0.451N}{25} + \log_4(N - 1) \leq \frac{N}{25} - \frac{7}{25} \leq |C_N|.$$

At $N = 500$ the middle inequality has positive margin, and its margin is increasing because

$$\frac{0.549}{25} - \frac{1}{(N - 1) \log 4} > 0 \quad (N \geq 500).$$

This closes every case. Subtracting the last strict upper bound from $|C_N| \geq N/25 - 7/25$ gives (33). $\square$

Thus the same-prime contribution has a global debit bound, while every fixed finite distinct-prime skeleton is impossible. What remains is to fit the genuinely unbounded $D_N(b, c)$ contribution into the residual debit (33).

The extreme one-form tail already has a coefficient-uniform square-root bound. This pays the branch in which only one witness is very large, without requiring a second large witness.

Lemma 6.3 (Uniform one-form large-square tail). *For $1 \leq t \leq N$ and $Y \geq 1$, put*

$$\mathcal{H}_t(Y) = \{a \leq N : \text{some prime } p > Y \text{ satisfies } p^2 \mid at + 1\}$$

and

$$K_t = \left\lfloor \frac{tN + 1}{Y^2} \right\rfloor.$$

Then

$$|\mathcal{H}_t(Y)| \leq 2\sqrt{N+1} \sum_{s=1}^{K_t} \frac{1}{\sqrt{s}} + 2K_t\sqrt{t} \leq 4\sqrt{K_t(N+1)} + 2K_t\sqrt{t}. \quad (35)$$

The same bounds hold after restricting a to any progression.

Proof. First note the elementary root bound

$$\#\{x \pmod{m} : x^2 \equiv u \pmod{m}\} \leq 2\sqrt{m} \quad ((u, m) = 1). \quad (36)$$

Indeed, an odd prime power contributes at most two roots, while a power of 2 contributes 1, 2, or at most 4 roots according as its exponent is 1, 2, or at least 3. If the odd part of m has r distinct prime divisors, then

$$\frac{2^r}{\sqrt{\text{rad}(m_{\text{odd}})}} \leq \max \left\{ 1, \frac{2}{\sqrt{3}} \right\};$$

after the possible prime 3, every new odd prime contributes a factor at most $2/\sqrt{5} < 1$. Combining this with the three stated 2-adic cases gives (36). The case $m = 1$ is immediate.

For every counted a , choose a witnessing prime and write

$$p^2 s = at + 1.$$

Then $1 \leq s \leq K_t$, $(s, t) = 1$, and

$$p^2 \equiv s^{-1} \pmod{t}, \quad p \leq \sqrt{\frac{tN + 1}{s}}.$$

For fixed s , (36) bounds the number of possible integers p —and hence the number of possible a —by

$$2\sqrt{t} \left(\frac{1}{t} \sqrt{\frac{tN + 1}{s}} + 1 \right) \leq 2\sqrt{\frac{N + 1}{s}} + 2\sqrt{t}.$$

Summing over s gives the first bound. The second follows from $\sum_{s \leq K} s^{-1/2} \leq 2\sqrt{K}$. Forgetting primality and progression conditions only enlarges the count. $\square$

Corollary 6.4 (The very-large mixed tail is paid). *Let $N \geq 41$ and $Y = \lfloor N/3 \rfloor$. For every $t \leq N$,*

$$|\mathcal{H}_t(Y)| < 27.42\sqrt{N+1}.$$

Consequently, for any three pivots $t_1, t_2, t_3 \leq N$, the number of $a \leq N$ for which at least one of the three forms $at_i + 1$ has a prime-square witness exceeding Y is less than

$$82.26\sqrt{N+1}. \quad (37)$$

This includes, in particular, the case in which exactly one witness exceeds Y .

Proof. The inequality $N^2 + 1 < 10\lfloor N/3 \rfloor^2$ holds for $N \geq 41$, so $K_t \leq 9$. Since

$$\sum_{s=1}^9 s^{-1/2} < 4.705,$$

the first bound in (35) is less than $9.41\sqrt{N+1} + 18\sqrt{N} < 27.42\sqrt{N+1}$. The union bound over the three forms gives (37). $\square$

The root count and tail arithmetic are reproduced by `scripts/uniform_large_square_tail_scan.py`. On its displayed finite box it checks 76116 unit-congruence cases and 173880 tail cases, with no root, tail, or $K_t \leq 9$ failure.

The large-square part of that obligation admits a coefficient-uniform Pell reduction. The point is to retain the determinant $|b - c|$ instead of multiplying one coefficient into the norm, which would introduce a spurious root count depending on the growing pivots.

Lemma 6.5 (Determinant-uniform Pell orbits). *Let A, B, d be positive integers. For $X \geq \sqrt{d}$, let $\mathcal{P}_{A,B,d}(X)$ be the set of positive integer solutions of*

$$Ap^2 - Bq^2 = d, \quad (q, d) = 1, \quad p\sqrt{A} + q\sqrt{B} \leq X.$$

Put

$$R_d(A, B) = \#\{z \pmod{d} : Az^2 \equiv B \pmod{d}\},$$

with $R_1(A, B) = 1$. Then

$$|\mathcal{P}_{A,B,d}(X)| \leq R_d(A, B) \left(1 + \left\lfloor \log_{2+\sqrt{3}} \frac{X}{\sqrt{d}} \right\rfloor \right) \leq d \left(1 + \left\lfloor \log_{2+\sqrt{3}} \frac{X}{\sqrt{d}} \right\rfloor \right). \quad (38)$$

In particular, the orbit count depends on the determinant d , not on the possibly growing product AB .

Proof. Every solution determines

$$z \equiv pq^{-1} \pmod{d}, \quad Az^2 \equiv B \pmod{d}.$$

Take two solutions (p_1, q_1) and (p_2, q_2) with the same z , and put

$$u = \frac{Ap_1p_2 - Bq_1q_2}{d}, \quad v = \frac{p_1q_2 - p_2q_1}{d}.$$

The common-root congruence shows that both numerators are divisible by d . The identity

$$(Ap_1p_2 - Bq_1q_2)^2 - AB(p_1q_2 - p_2q_1)^2 = (Ap_1^2 - Bq_1^2)(Ap_2^2 - Bq_2^2) = d^2$$

therefore gives

$$u^2 - ABv^2 = 1.$$

Writing $\alpha_i = p_i\sqrt{A} + q_i\sqrt{B}$ and $\beta_i = p_i\sqrt{A} - q_i\sqrt{B}$, one also has

$$\frac{\alpha_1}{\alpha_2} = u - v\sqrt{AB}.$$

Here $\alpha_i > 0$, $\beta_i = d/\alpha_i > 0$, and the conjugate of the displayed quotient in $\mathbb{Q}(\sqrt{AB})$ is β_1/β_2 . Hence, after ordering the two solutions by α_i , the larger of a nontrivial quotient and its inverse is a totally positive unit greater than 1. If AB is a square, the displayed unit equation forces $v = 0$ and $u = 1$, so the two solutions coincide. Otherwise every nontrivial integral totally positive unit greater than 1 is at least $2 + \sqrt{3}$: indeed it has the form $x + y\sqrt{AB}$ with $x \geq 2$, $y \geq 1$, and $AB y^2 = x^2 - 1 \geq 3$.

Thus the solutions in one root class form a geometric sequence of ratio at least $2 + \sqrt{3}$. Since α_i has norm d and is larger than its positive conjugate, $\alpha_i \geq \sqrt{d}$. Each root class consequently contains at most the first parenthesis in (38). Summing over the $R_d(A, B) \leq d$ root classes proves the claim. $\square$

Corollary 6.6 (A one-payment two-large-witness tail). *Let $1 \leq b < c \leq N$, put $d = c - b$, and let $Y \geq d$. Define*

$$K = \left\lfloor \frac{N^2 + 1}{Y^2} \right\rfloor, \quad L = 1 + \left\lfloor \log_{2+\sqrt{3}} \left(2\sqrt{\frac{N(N^2 + 1)}{d}} \right) \right\rfloor.$$

The number of $a \leq N$ for which there are primes $p, q > Y$ satisfying

$$p^2 \mid ab + 1, \quad q^2 \mid ac + 1$$

is at most

$$K^2 d L. \tag{39}$$

The same bound holds after interchanging b and c , and after restricting a to any progression.

Proof. For each counted a , choose one pair of witnessing primes and write

$$ab + 1 = p^2 r, \quad ac + 1 = q^2 s.$$

Then $1 \leq r, s \leq K$, while

$$cp^2 r - bq^2 s = c(ab + 1) - b(ac + 1) = d.$$

Since $q > Y \geq d$, one has $(q, d) = 1$. For fixed (r, s) apply Lemma 6.5 with $A = cr$ and $B = bs$. Its height parameter may be taken to be $2\sqrt{N(N^2 + 1)}$, because

$$p\sqrt{cr} = \sqrt{c(ab + 1)} \leq \sqrt{N(N^2 + 1)}$$

and the analogous inequality holds for $q\sqrt{bs}$. Thus each of the K^2 pairs (r, s) contributes at most dL solutions. Forgetting primality and progression conditions only enlarges the count, proving (39). $\square$

The factor d in (39) was paid separately for every quotient pair. For an actual-support argument the quotient sets are sparse, and the root classes should instead be summed before the Pell-orbit payment.

Lemma 6.7 (Determinant square-sieve over quotient supports). *Let $1 \leq b < c \leq N$, $d = c - b$, and $Y \geq d$. Let γ_b, γ_c be positive integers and let*

$$\mathcal{R} \subseteq [1, K_b], \quad \mathcal{S} \subseteq [1, K_c].$$

Consider the integers $a \leq N$ for which there are primes $p, q > Y$ and $r \in \mathcal{R}, s \in \mathcal{S}$ satisfying

$$ab + 1 = \gamma_b p^2 r, \quad ac + 1 = \gamma_c q^2 s.$$

Put

$$A_0 = c\gamma_b, \quad B_0 = b\gamma_c, \quad g_A = (A_0, d), \quad g_B = (B_0, d)$$

and

$$L = 1 + \left\lfloor \log_{2+\sqrt{3}} \left(2\sqrt{\frac{N(N^2+1)}{d}} \right) \right\rfloor.$$

Then their number is at most

$$L \sum_{r \in \mathcal{R}} \sum_{s \in \mathcal{S}} R_d^\times(A_0 r, B_0 s) \leq L \min\{d|\mathcal{R}||\mathcal{S}|, |\mathcal{R}|(g_B K_c + d), |\mathcal{S}|(g_A K_b + d)\}. \quad (40)$$

Here $R_d^\times(A, B)$ counts only roots $z \pmod{d}$ with $(z, d) = 1$. This remains valid after restricting a to any progression. In the transformed $q_0 = 25$ progression one may take $\gamma_t \in \{1, 5\}$ and let $\mathcal{R}, \mathcal{S}$ be the exact actual-support/mod-5 quotient masks.

Proof. For a counted a choose one witness pair. Eliminating a gives

$$A_0 r p^2 - B_0 s q^2 = d.$$

Since $p, q > Y \geq d$, one has $(p, d) = (q, d) = 1$. For fixed (r, s) , Lemma 6.5 therefore gives at most $R_d^\times(A_0 r, B_0 s)L$ solutions, since the associated root $z = pq^{-1}$ is a unit. Its height bound is unchanged because

$$p\sqrt{A_0 r} = \sqrt{c(ab+1)}, \quad q\sqrt{B_0 s} = \sqrt{b(ac+1)}.$$

Summing first proves the first inequality in (40).

For the second, interchange the quotient and root sums. There are at most d possible unit roots z , and a root contributes precisely when

$$B_0 s \equiv A_0 r z^2 \pmod{d}.$$

For fixed (z, r) this linear congruence is either insoluble or is one class modulo d/g_B . Hence among $s \in \mathcal{S} \subseteq [1, K_c]$ it has at most

$$\min \left\{ |\mathcal{S}|, \frac{g_B K_c}{d} + 1 \right\}$$

solutions. Summing over z and r gives both $d|\mathcal{R}||\mathcal{S}|$ and $|\mathcal{R}|(g_B K_c + d)$. Interchanging the two sides gives the third bound because z^2 is a unit, so $(A_0 z^2, d) = g_A$. Taking their minimum proves the assertion. Discarding a progression condition only enlarges the count. $\square$

For $Y = \lfloor N/3 \rfloor$, Corollary 6.4 also pays the branch in which exactly one witness exceeds Y . The remaining large-prime obligation is therefore the medium window between the finite sieve cutoff and $N/3$, including its endpoint discrepancy; it is still part of the same distinct-prime debit.

For the pure 18-class parity blocks, the determinant lemma has the following finite very-large specialization.

Corollary 6.8 (Pure parity determinant debit at the half split). *Let $107 \leq N \leq 2 \cdot 10^8$, put $Y = \lfloor N/2 \rfloor$, and let $t_1 < t_2 < t_3$ be members of C_N^{18} of one parity and of span at most 200. The number of members a of the opposite base parity for which at least two of the three forms $at_i + 1$ have prime-square witnesses greater than Y is at most*

$$16 \left(1 + \left\lfloor \log_{2+\sqrt{3}} \left(2 \sqrt{\frac{N(N^2+1)}{50}} \right) \right\rfloor \right). \quad (41)$$

At $N = 2 \cdot 10^6$ this is at most 256; retaining the actual pivot supports lowers the exhaustive maximum to 224.

Proof. For a pure pivot $t \equiv 18 \pmod{25}$ one has

$$at + 1 = p^2 r, \quad r \equiv 2 \text{ or } 3 \pmod{5}.$$

The product at is even in the opposite-parity block, so $at + 1$ and hence r are odd. Since $p > Y$ and $N \geq 107$, one has $(tN + 1)/p^2 < 5$. Thus the only possible quotient is $r = 3$; every prime divisor of t may only delete that quotient by its quadratic-residue condition.

Three same-parity members of the 18 class and span at most 200 have one of the six offset patterns in the first column below. For each pair apply Lemma 6.7 with its exact quotient set contained in $\{3\}$. Summing the unit root classes over the three pairs, and then maximizing over the finitely many residues $t_1 \equiv 18 \pmod{25}$ modulo the three gaps, gives

$(t_2 - t_1, t_3 - t_1)$	(50, 100)	(50, 150)	(50, 200)	(100, 150)	(100, 200)	(150, 200)
root-class sum	8	10	14	10	16	14.

Every gap is at least 50, so its orbit factor is at most the parenthesis in (41). A union bound over the three pairs proves the assertion. The residue table and the actual-support scan over all 479960 eligible triples at $N = 2 \cdot 10^6$ are reproduced by `scripts/systematic_pure_determinant_tail.py`. $\square$

There is an even sharper one-form observation at this particular split.

Corollary 6.9 (The pure very-large branch is terminally paid). *Under the hypotheses of Corollary 6.8, the number of opposite-parity base vertices for which at least one of the three forms has a prime-square witness greater than Y is at most 192. Consequently the determinant debit is not the controlling term in the pure half-split continuation.*

Proof. The preceding quotient argument leaves only

$$3p^2 = at + 1.$$

It forces $t > (3(Y + 1)^2 - 1)/N > 3N/4 - 2$. The interval

$$Y < p \leq \sqrt{(tN + 1)/3}$$

has length less than $0.079N < t$. Thus each root class of $3x^2 \equiv 1 \pmod{t}$ contains at most one possible p . If $3 \nmid t$ or $4 \nmid t$ there are no roots. Otherwise the prime 5 is absent and every odd prime-power factor contributes at most two roots. Since

$$7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 = 215656441 > 2 \cdot 10^8,$$

t has at most six relevant odd prime divisors. Hence there are at most $2^6 = 64$ witnesses for one pivot, and the union over three pivots is at most 192. $\square$

Corollaries 6.8 and 6.9 unconditionally remove the very-large branch from the pure matching obstruction. The remaining medium window $Q < p \leq N/2$ is bounded jointly by separating “at least two small” from “at least two tail” witnesses. Let

$$\mathcal{P}_{47} = \{p : 3 \leq p \leq 47, p \neq 5\}, \quad z_p = p^{-2}.$$

Put

$$\begin{aligned} \lambda_3 &= 1 - e_1(z) + e_2(z) - e_3(z) > 0.847622705884, \\ \alpha_2 &= 1 - 3\{1 - e_1(2z)\} + 2\{1 - e_1(3z) + e_2(3z)\} < 0.109696396217. \end{aligned}$$

The first polynomial is the degree-3 lower Bonferroni bound for one form. For the second, if S_i is the event that form i avoids every finite square class, the indicator that at least two forms have a finite witness is

$$1 - e_2(S_1, S_2, S_3) + 2e_3(S_1, S_2, S_3).$$

The degree-1 lower bound for each two-form survivor and the degree-2 upper bound for the three-form survivor give precisely α_2 . Their respective endpoint budgets are

$$377, \quad 3e_1(21) + 2\{e_1(31) + e_2(31)\} = 1560. \quad (42)$$

Let $Y = \lfloor N/25 \rfloor$ and

$$\Pi(N) = \frac{Y}{\log Y - 1.1} + 1.34, \quad \tau_{47} = 0.003887.$$

For $G > 0$ define

$$T_G(N) = \left(\frac{N}{50} + 1\right) \tau_{47} + \Pi(N) - 15 + G, \quad (43)$$

$$D_G(N) = \lambda_3 \left(\frac{N}{50} - 1\right) - 377 - T_G(N), \quad (44)$$

$$U_G(N) = \alpha_2 \left(\frac{N}{50} + 1\right) + 1560 + \frac{3}{2}T_G(N). \quad (45)$$

The prime bound $\pi(Y) \leq \Pi(N)$ on the range used below follows from (D) for $Y > 60184$; the added 1.34 covers the remaining integer interval $60000 \leq Y \leq 60184$, which is checked directly.

Lemma 6.10 (Joint pure-tail certificate). *For $1.5 \cdot 10^6 \leq N < 2 \cdot 10^8$, every vertex in a pure opposite-parity block has degree at least $D_G(N)$, and every eligible span-200 triple has common non-neighbourhood at most $U_G(N)$, where*

$$G = \begin{cases} 1541.126, & 1.5 \cdot 10^6 \leq N < 2.3 \cdot 10^6, \\ 3605, & 2.3 \cdot 10^6 \leq N < 2 \cdot 10^8. \end{cases}$$

On both intervals

$$D_G(N) > U_G(N), \quad D_G(N) > \frac{N}{125} + 2. \quad (46)$$

Proof. The one-form finite sieve gives the first term of (44). For $47 < p \leq Y$, the square-density is included in τ_{47} and there is one terminal class per prime. For $p > Y$, write $at + 1 = p^2 m$. Since $Y \geq N/26$ and $t \leq N$, one has $m \leq 676$. Opposite parity makes m odd, the mod-5 calculation makes it a nonsquare modulo 5, and every odd prime divisor of t imposes its exact quadratic-residue condition. The transformed root payment is

$$H_t \left(s_t + \frac{2(N/50 + 1)}{Y} \right), \quad \frac{2(N/50 + 1)}{Y} < 1.04004,$$

where s_t is the exact survivor count. Exhausting all 92000 pure pivots $t \leq 2.3 \cdot 10^6$ gives maximum 1541.12512, attained at

$$t = 2038168, \quad \nu_2(t) = 3, \quad H_t = 128, \quad s_t = 11.$$

For the full upper interval, retain the smallest $k - 1$ of the k non-5 odd prime divisors of t . Requiring that one further prime still fit below $2 \cdot 10^8$ gives the following unconditional relaxation:

k	0	1	2	3
s_k	135	135	83	48
H_k	8	16	32	64
$H_k(s_k + 1.04004)$	1088.321	2176.641	2689.282	3138.563
k	4	5	6	7
s_k	26	13	6	1
H_k	128	256	512	1024
$H_k(s_k + 1.04004)$	3461.126	3594.251	3604.501	2089.001

Thus $G = 3605$ is valid. This proves the degree bound.

Now let a be a common non-neighbour of three pivots. If at least two forms have witnesses in $\mathcal{P}_{47}$, the finite polynomial α_2 and the second error in (42) pay it. Otherwise at least two forms have witnesses exceeding 47. If H_i denotes the latter tail event for form i , then

$$\#\{a : \text{at least two } H_i\} \leq \frac{1}{2} \sum_{i=1}^3 |H_i| \leq \frac{3}{2} T_G(N).$$

This proves (45); in particular the medium terminal events are paid jointly rather than independently.

It remains only to check the two scalar margins. Both are increasing on each displayed interval. Between jumps of Y this is immediate; at a jump, differentiating $y/(\log y - 1.1)$ gives an increment less than 0.091, which is dominated by the linear increment in both margins. More explicitly, an increment of 25 in N increases $D_G - U_G$ by more than

$$\frac{\lambda_3 - \alpha_2}{2} - \frac{5}{4} \tau_{47} - 2.5(0.091) > 0.136,$$

and increases $D_G - (N/125 + 2)$ by more than

$$\frac{\lambda_3}{2} - \frac{1}{5} - \frac{1}{2} \tau_{47} - 0.091 > 0.130.$$

At the two left endpoints the rigorous rounded values are

N	G	D_G	U_G	$D_G - U_G$
1500000	1541.126	> 17347.432	< 16406.103	> 941.330
2300000	3605	> 25936.158	< 25621.101	> 315.058

and $D_G - (N/125 + 2)$ is respectively greater than 5345.43 and 7534.15. The support enumeration, finite polynomials, prime counts, and endpoint margins are reproduced by `scripts/systematic_pure_joint_tail_close.py`. $\square$

Theorem 6.11 (Pure matching down to one and a half million). *For every $N \geq 1.5 \cdot 10^6$, the squarefree-edge graph between C_N^{18} and C_N has a matching saturating C_N^{18} . Hence every $B \subseteq C_N^{18}$ satisfies the original Hall inequality.*

Proof. Theorem 5.20 handles $N \geq 2 \cdot 10^8$. In the remaining range work in one parity block and suppose S is a Hall defect, with $m = |S|$. The forward degree bound gives $m > D_G(N) > N/125 + 2$. If no three consecutive members of S had span at most 200, their spans, being multiples of 50, would all be at least 250, whence

$$250(m-2) \leq \sum_{j=1}^{m-2} (x_{j+2} - x_j) \leq 2(N-1),$$

contrary to $m > N/125 + 2$. Choose such a triple.

As in Theorem 5.14, the reverse degree bound and the assumed Hall defect force the unused right-hand set to have more than $D_G(N)$ members. Every one is a common non-neighbour of the chosen triple, contradicting $U_G(N) < D_G(N)$. Hall's theorem supplies the matching in both parity blocks, and restricting it to B proves the last assertion. $\square$

6.1 Joint mixed tail descent to forty million

The same joint split has a stronger mixed form. Averaging the all-three-finite intersection with the two-of-three event preserves the mixed diagonal balance.

Lemma 6.12 (Half finite-intersection, half two-of-three). *For three pairs of events F_i, H_i one has pointwise*

$$\mathbf{1}_{\bigcap_{i=1}^3 (F_i \cup H_i)} \leq \frac{1}{2} \mathbf{1}_{F_1 \cap F_2 \cap F_3} + \frac{1}{2} \mathbf{1}_{\{\#\{i: F_i\} \geq 2\}} + \frac{1}{2} \sum_{i=1}^3 \mathbf{1}_{H_i}. \quad (47)$$

Proof. If exactly k of the F_i occur and all three unions occur, the right side is at least 1 for $k = 3, 2, 1, 0$, respectively: its minimum in the four cases is $1, 1, 1, 3/2$. This also proves the assertion when some F_i and H_i occur simultaneously. $\square$

We apply the lemma with F_i the existence of a prime-square witness through 47 and H_i the existence of a witness above 47. Put

$$\eta = 0.00015, \quad \tau_{47} = 0.003887.$$

Retaining the nonzero square coset modulo 5 in both the finite and actual-support degree arguments gives the following low block.

Proposition 6.13 (Mixed residual below two hundred million). *For $4 \cdot 10^7 \leq N < 2 \cdot 10^8$, every mixed Hall defect satisfies*

$$|A^*| > 0.00030627N > 2\eta N.$$

Proof. First let $4 \cdot 10^7 \leq N < 8 \cdot 10^7$. Use the one-form degree-3 Bonferroni lower bound through 47, split the remaining primes at $Y = \lfloor N/70 \rfloor$, and retain the actual support and worse nonzero mod-5 coset in the root tail. The global support at $8 \cdot 10^7$ fits in $M = 4902$. Evaluated at the lower endpoint, the least row is

$$\text{odd}, \quad k = 4, \quad H = 32, \quad s_k = 286, \quad (p_1, p_2, p_3) = (61, 73, 79).$$

After the reciprocal-survivor correction the normalized mixed residual is greater than

$$0.0077080.$$

For $8 \cdot 10^7 \leq N < 2 \cdot 10^8$, use $D = \lfloor N/300 \rfloor$ and the common box $M = 90001$ exactly as in Proposition 5.18, but intersect every complementary mask with the worse of the two nonzero square cosets modulo 5. To make the calculation uniform, take the prime-support superset at $2 \cdot 10^8$ and evaluate every fixed row at $N = 8 \cdot 10^7$. The least row is

$$\text{odd}, \quad k = 3, \quad H = 16, \quad s_k = 9176, \quad (p_1, p_2) = (449, 491).$$

Its excess over $N/50$ is greater than 12542.5712308. The same reverse degree subtraction, after including the reciprocal-survivor correction < 291.111 , therefore gives

$$\frac{2(12542.5712308 - 291.111) - 1}{8 \cdot 10^7} > 0.00030627.$$

In both blocks the global support is fixed, the normalized affine main term increases, and all normalized positive payments decrease. Thus the lower endpoints control their intervals. The support enumeration is reproduced by `scripts/systematic_mixed_interval_support.py`; the endpoint subtraction is checked by `scripts/verify_four_range_paper_arithmetic.py`. $\square$

We next record the finite payment. On an odd-prime progression let S_i be the event that the i th form survives every prime through 47. The indicator of at least two finite-bad forms is

$$1 - \sum_{i < j} \mathbf{1}_{S_i S_j} + 2\mathbf{1}_{S_1 S_2 S_3}.$$

Use degree-1 lower bounds for the three pair survivors and a degree-2 upper bound for the triple survivor. At a common prime the pair multiplicities sum to 3 and the triple multiplicity is 1. At a pair-coincident non-common prime they are 5 and 2; without a coincidence they are 6 and 3. If $z_p = p^{-2}$, declaring a non-common prime pair-coincident changes the resulting upper polynomial by

$$z_p(1 - 2S_p) > 0, \quad S_p < 3 \sum_{\substack{3 \leq q \leq 47 \\ q \neq 5}} q^{-2} < \frac{1}{2}.$$

Hence all non-common odd primes may be declared pair-coincident for this upper bound. A common prime only decreases the two-of-three polynomial, so the two finite polynomials are evaluated jointly.

Prime 2 is exact. For a triple meeting both odd classes modulo 4, one mod-4 class makes the close pair automatically bad, the opposite class makes the third form automatically bad, and the other two classes use the odd-prime two-of-three polynomial. For three pivots in one odd mod-4 class, one class makes all three forms automatically bad and the other three use the odd-prime polynomial.

If the even mass exceeds ηN , a close pair in one even mod-4 class has gap less than 13334. After charging the even mass in the other case, the odd mass is greater than $0.0001635517807N$. A close pair in the denser of two odd classes has gap less than 12229; if only one odd class occurs, its close pair has gap less than 6115. For any common odd-prime product Q , the close pair gives $4Q^2 < d$. Exhausting the resulting cutoff-prime supports gives

branch	maximizing common support	$I_3 + I_2$	$\epsilon_3 + \epsilon_2$
even generic	7	< 0.223675978	4874
even concentrated	3, 7	< 0.309830838	4510
two odd classes	$\emptyset$	< 0.541995982	7113
two odd, common 3	3	< 0.543144010	6615
one odd class	2	< 0.679312680	7435
one odd, common 2, 3	2, 3, 7	< 0.739301322	6423.

Here I_3 is the degree-3/2/3 all-three-finite bound used above, and I_2 is the new two-of-three bound. By Lemma 6.12, the normalized finite payment across the two base progressions is therefore

$$\frac{I_3 + I_2}{25} + \frac{\epsilon_3 + \epsilon_2}{N}. \quad (48)$$

The finite polynomial (48) is combined with the ten-branch envelopes below, each of which includes the reciprocal mass beyond 3025. This keeps each root bound paired with the finite row at the same cutoff.

Finally, the eventwise Möbius–CRT diagonal scan with $R = 2 \cdot 10^7$ uses the following entirely rational tail estimate. If P is the exact truncated signed count at the right endpoint U , and θ is the density of the permitted residue set, then throughout $[L, U]$

$$\frac{\Delta(N)}{N} < \frac{1}{L} \left\{ P + \frac{38\theta U}{R} + \frac{13(U^2 + 1)}{R^2} \right\}. \quad (49)$$

Indeed $\log(2 \cdot 10^7) + 2 < 19$: the positive Taylor series gives $e > 2.7$, while $2.7^{17} > 2 \cdot 10^7$. Every comparison in (49) is made by unsigned-integer cross-multiplication. On all nine blocks from $5 \cdot 10^6$ to $2 \cdot 10^8$ this gives the uniform strict ceilings

support	all	even cell	odd or 8 a	E_2 cell
$10^6 \Delta/N <$	25294	19681	15812	16164

support	odd union	odd+cell	one odd	one odd cell
$10^6 \Delta/N <$	12652	7036	6330	711.

The parameterized scan is reproduced by

`scripts/systematic_concentrated_diagonal_blocks.cpp`.

Theorem 6.14 (Joint mixed close to forty million). *The Hall inequality (2), and hence the exact bound in Erdős Problem 848, holds for every*

$$N \geq 4 \cdot 10^7.$$

Proof. Theorem 5.21 handles $N \geq 2 \cdot 10^8$. In the remaining interval take a mixed defect and put $A = B \cup T_N(B)$. Proposition 6.13 supplies more than three sparse-class charges. Apply the ten exhaustive valuation/parity alternatives used in Proposition 6.18; their selection proof uses only the residual inequality and the gap lemma, and is therefore valid unchanged on this larger interval.

For the six blocks

$$[40, 50), [50, 70), [70, 80), [80, 100), [100, 150), [150, 200) \quad \text{million,}$$

the transformed-root ceilings for E_1, E_2, E_3 , and the odd classes are

$[L, U)$	E_1	E_2	E_3	O
[40, 50)M	.009677836	.009973185	.010991872	.010396089
[50, 70)M	.009350108	.009648325	.010405463	.009926965
[70, 80)M	.008921868	.009135839	.009679089	.009335765
[80, 100)M	.008771555	.008959044	.009511773	.009134226
[100, 150)M	.008534837	.008684996	.009127671	.008825297
[150, 200)M	.008175215	.008275070	.008569448	.008368371.

These ceilings include the reciprocal-survivor correction. Combine them with the exact diagonal grid, the ten finite payments

$$.008685, .012616, .019420, .020878, .026643, .029459$$

in their respective generic/common branches, the sparse-class charge $1/20001 + 1/L$, and the cutoff-23 or cutoff-19 square tail. Exact rational comparison shows that E_2 -generic is the largest row in every block:

$[L, U)$	$\max A /N$	$1/25 - 7/(25L) - \max A /N$
[40, 50)M	.039972160176	$> .0000278328$
[50, 70)M	.039810601176	$> .0001893932$
[70, 80)M	.039554376176	$> .0004456198$
[80, 100)M	.039466546676	$> .0005334498$
[100, 150)M	.039330914676	$> .0006690825$
[150, 200)M	.039127498676	$> .0008724994.$

For example, with $\widehat{\tau}_{23}$ as in the ten-million proof, the controlling lower-endpoint total is

$$C_{40} = \frac{25288306}{10^9} + \frac{8685}{10^6} + \frac{3}{25} \widehat{\tau}_{23} + \frac{1}{2} \frac{9973185}{10^9} < \frac{1}{25} - \frac{7}{25(4 \cdot 10^7)} - \frac{27}{10^6}.$$

Thus every branch contradicts the Hall defect. The rows and exact rational inequalities are checked by `scripts/verify_four_range_paper_arithmetic.py`. $\square$

6.2 Valuation, cell, and fibre descent to five million

The remaining factor-of-eight descent refines the same mixed defect by the 2-adic valuation of its even pivots and by the actual residue cells occupied by the odd pivots. We first isolate the two finite combinatorial facts used below.

Lemma 6.15 (Valuation support and the four-pivot inequality). *Let a be even, p odd, and $p^2m = at + 1$. If $\nu_2(a) = 1, 2$, or at least 3, then respectively*

$$m \equiv 1 \pmod{2}, \quad m \equiv 1 \pmod{4}, \quad m \equiv 1 \pmod{8}.$$

Moreover, for four pairs of events F_i, H_i ,

$$\mathbf{1}_{\cap_{i=1}^4 (F_i \cup H_i)} \leq \frac{1}{2} \sum_{|S|=3} \mathbf{1}_{\cap_{i \in S} F_i} + \frac{1}{2} \sum_{i=1}^4 \mathbf{1}_{H_i}. \quad (50)$$

More generally, for n pairs and $1 \leq k \leq n$,

$$\mathbf{1}_{\cap_{i=1}^n (F_i \cup H_i)} \leq \mathbf{1}_{\#\{i: F_i\} \geq k} + \frac{1}{n - k + 1} \sum_{i=1}^n \mathbf{1}_{H_i}. \quad (51)$$

Proof. Every odd square is 1 modulo 8. Since $2^{\nu_2(a)} \mid at$ and $p^2 \equiv 1 \pmod{8}$, division by p^2 gives the three displayed quotient restrictions. For (50), if exactly j of the F_i occur, the right side is at least $2, 3/2, 1, 1, 2$ for $j = 4, 3, 2, 1, 0$, respectively. For (51), either at least k finite events occur, or at least $n - k + 1$ of the unions must be supplied by their H_i terms. $\square$

Lemma 6.16 (Two-coordinate fibres and capacitated cells). *Let X be a finite set and let $r_q : X \rightarrow \mathbb{Z}/q^2\mathbb{Z}$.*

1. *If every pair of elements of X agrees in at least one of the two coordinates r_7, r_{11} , then X is contained in one mod-49 fibre or in one mod-121 fibre.*
2. *Let S be a set of s occupied mod-9 cells, joined to the mod-49 residues which they contain, and give every right vertex capacity two. If there is no matching of size eight, then some $Y \subseteq S$ satisfies*

$$|Y| - 2|N(Y)| \geq s - 7. \quad (52)$$

Proof. For the first assertion, choose two points with different first coordinates. They have the same second coordinate. Any third point which had a different second coordinate would have to share both distinct first coordinates, which is impossible. The second assertion is the max-flow/min-cut formula for a bipartite matching with right capacities two:

$$\nu = \min_{Y \subseteq S} (s - |Y| + 2|N(Y)|).$$

If $\nu \leq 7$, rearranging gives (52). $\square$

Lemma 6.17 (Gap selection and cell charge). *Let $X \subseteq [1, N]$ and let $G \geq 1$. If no two members of X have difference at most G , then*

$$|X| \leq \frac{N-1}{G+1} + 1 < N \left(\frac{1}{G+1} + \frac{1}{N} \right).$$

Consequently, if X is split into r residue cells, each nondense cell of cardinality at most $(N-1)/(G+1) + 1$ may be discarded at total normalized cost at most

$$r \left(\frac{1}{G+1} + \frac{1}{N} \right).$$

Proof. If $x_1 < \dots < x_m$ are the members of X , then $x_{i+1} - x_i \geq G+1$, so $(m-1)(G+1) \leq N-1$. Summing this estimate over the cells proves the second assertion. $\square$

We now close the two upper intervals. In both proofs split the even pivots into E_1, E_2, E_3 according as $\nu_2 = 1, 2$, or at least 3. The quotient restrictions in Lemma 6.15 reduce their transformed root supports.

Proposition 6.18 (Valuation close on the twenty-million block). *The Hall inequality (2) holds for*

$$2 \cdot 10^7 \leq N < 4 \cdot 10^7.$$

Proof. The cutoff-19 degree-3 one-form certificate gives

$$|A^*| > 0.006910733259914470N.$$

Put

$$\delta_{20} = \frac{1}{20001} + \frac{1}{2 \cdot 10^7}.$$

The four six-form root ceilings for E_1, E_2, E_3 , and an odd pivot are

$$.008622089, \quad .010006474, \quad .012915119, \quad .010177869.$$

If $|E_1| > \delta_{20}N$, Lemma 6.17 supplies a pair of gap at most 20000. Add a third E_1 pivot. If the class is nonconstant modulo 9, choose the third pivot outside the pair's cell and use the generic row; otherwise use the common-3 row. This is the three-pivot inequality (47); no four-pivot claim is used in this interval. If $|E_1| \leq \delta_{20}N$, charge it and repeat with E_2 . If both have been charged, repeat with E_3 ; the off-base diagonal is then supported on the odd-or-divisible-by-8 set. If all three even classes have been charged, their total cost is at most $3\delta_{20}N$.

The remaining odd mass is greater than

$$(0.006910733259914470 - 3\delta_{20})N.$$

If both odd mod-4 classes occur, its denser class contains a pair of gap less than twice the reciprocal of this density; add a pivot from the other class. If only one odd class occurs, its close-pair bound is the reciprocal of the same density and the third pivot is taken in that class. Splitting each alternative according as the three pivots are constant modulo 9 gives the final four rows. Thus the following ten rows are exhaustive:

row	E_1G	E_{13}	E_2G	E_{23}	E_3G
total <	.039257774	.037068015	.039949967	.034297022	.032021074
row	E_{33}	O_2G	O_{23}	O_1G	O_{13}
total <	.035449069	.039924070	.034577881	.039438565	.036137568

The controlling row is now the generic E_2 row. With the exact finite and square-tail payments, its total is

$$C_{20} = \frac{25289550}{10^9} + \frac{1572362642548502371982744227834515457}{171081196711186531200844968164610000000} \quad (53)$$

$$+ \frac{11661}{25000000} + \frac{1}{2} \frac{10006474}{10^9}. \quad (54)$$

Exact common-denominator arithmetic gives

$$C_{20} < \frac{1}{25} - \frac{7}{25(2 \cdot 10^7)} - \frac{50}{10^6}.$$

All other rows have larger slack, and every normalized endpoint payment decreases while the Hall target increases. This proves the full interval. The support exhaustion is reproduced by the accompanying finite enumerations, and the ten rational totals are checked by `scripts/verify_four_range_paper_arithmetic.py`. $\square$

Proposition 6.19 (Cell close on the ten-million block). *The Hall inequality (2) holds for*

$$10^7 \leq N < 2 \cdot 10^7.$$

Proof. Call a mod-9 cell dense if its cardinality is more than $(N - 1)/1000001 + 1$, and put

$$\delta_{10} = \frac{1}{1000001} + \frac{1}{10^7}.$$

The cutoff-17 degree certificate gives $|A^*| > .005826196348744825N$. The relevant root ceilings are

$$.010545872, \quad .011845571, \quad .014894092, \quad .013231580.$$

In E_1 , two dense cells supply two close pairs and hence the four-pivot row (50); one dense cell supplies three pivots and the other eight cells cost at most $8\delta_{10}N$. If there is no dense cell, retain the whole class in the unrestricted diagonal and continue to E_2 . The same alternatives for E_2 give the unrestricted two-cell row or, after charging the nine E_1 cells and eight unused E_2 cells, the E_2 -cell row with charge $17\delta_{10}N$. If neither class supplies a row, charge both and use a dense E_3 cell; this costs $18\delta_{10}N$ and leaves the odd-or-divisible-by-8 diagonal. If no even class supplies a row, all 27 even cells cost at most $27\delta_{10}N$.

For the odd remainder use the cutoff-7 threshold allocation described below for the five-million block. Discarding odd cells of size at most five costs at most 90 pivots. The alternatives are: disjoint triples of active cells in the two odd mod-4 classes; at most eleven active cells; one concentrated residue; or one odd mod-4 class with $s = 1, \dots, 9$ active cells. Lemma 6.16 and the exact recurrence (59) cover the $s = 8, 9$ matching failures. This avoids the invalid inference that crossed common supports at 7 and 11 must put all pivots into one fixed prime-square fibre.

Using the five-million diagonal ceilings, which also dominate the direct scan on this interval, every odd row is at most 0.039247000. The five even rows are, respectively,

row	E_1 two	E_1 one	E_2 two	E_2 one	E_3
total <	.039077928	.039259999	.039944394	.036406507	.037579342.

Thus the controlling row is E_2 with two dense cells. Its four three-form finite intersections admit the conservative cutoff-23 density

$$I_{10} = \frac{5355507302216}{165901428816123}, \quad \epsilon_{10} = 607.$$

Put also

$$\widehat{\tau}_{23} = \frac{64081802747648035629863}{7596668444022826249000000}.$$

This upper bound uses cutoff 23. The decimal-free controlling total is

$$C_{10} = \frac{25289862}{10^9} + \frac{4I_{10}}{25} + \frac{4\epsilon_{10}}{10^7} + \frac{4}{25}\hat{\tau}_{23} + \frac{2}{3}\frac{11845571}{10^9}. \quad (55)$$

Direct common-denominator arithmetic yields

$$C_{10} < \frac{1}{25} - \frac{7}{25 \cdot 10^7} - \frac{55}{10^6}. \quad (56)$$

The exact rational calculation above is reproduced by

`scripts/verify_four_range_paper_arithmetic.py.`

The diagonal ceilings come from

`scripts/systematic_concentrated_diagonal_blocks.cpp.`

□

For the lower block the support constraints are retained jointly. On $5 \cdot 10^6 \leq N < 10^7$, the cutoff-17 degree calculation gives

$$|A^*| > 0.003997632216030162N. \quad (57)$$

The relevant cutoff root ceilings are

$$T_{E_1,23} < .013878159, \quad T_{E_2,7} < .015472245, \quad T_{E_3,23} < .020270750, \quad T_{O,7} < .017398287. \quad (58)$$

For E_1 , call a paired mod-9 cell good if it is contained in neither one mod-49 fibre nor one mod-121 fibre. Lemma 6.16(1) supplies a pair which breaks both primes. Otherwise label the cell 7-only, 11-only, or both. Every cutoff-23 common support compatible with the label is enumerated; in particular, the primes 13, 17, 19, 23 are not silently declared non-common. If there are two good cells, use four pivots. If there is one, pair it with the best bad cell. If there is none, every 7-only or both cell is inserted into the eventwise diagonal through its actual mod-49 fibre. The maxima for $E_{1,0}$ plus $k = 0, \dots, 9$ such fibres have the uniform strict rational ceiling

$$.019113,$$

instead of the incompatible raw-fibre/unrestricted hybrid. The one-paired cell case uses three pivots. The resulting four E_1 envelopes are

$$.039267556, \quad .039360535, \quad .035990495, \quad .039569683.$$

For E_2 , the two-cell and one-cell rows are below .037494782 and .038211102. If E_1, E_2 are both charged, nineteen E_3 pivots force three in one mod-9 cell; its envelope is below .039335952.

For the odd part, discard every mod-9 cell which contains at most five pivots and call the other cells active. Together with the preceding even charges this costs at most 128 pivots. If both odd mod-4 classes admit disjoint triples of active mod-9 cells, choose three pivots from each. Otherwise an elementary three-set check gives at most eleven active odd cells: if one class has at most two cells the assertion is immediate, while if both have at least three, the failure of disjoint triples forces both cell sets to have size at most five. A pair from each class in different mod-9 cells then gives a four-pivot row below .037135620; if all active cells have the same mod-9 residue, the concentrated three-pivot row is below .033200482.

For one odd mod-4 class, let s be its number of active mod-9 cells. When $s = 8$ or 9 , give the mod-49 residues capacity two. A matching of size eight selects eight pivots with no mod-49 collision block larger than two. If it fails, (52) restricts the diagonal. The eventwise maxima in the two failure cases are

$$.003572, \quad .003587.$$

When $s = 1$, either four mod-49 fibres each supply a pair, or all but at most 49 pivots lie in three such fibres; the latter diagonal is below .000050. The cases $2 \leq s \leq 7$ use six pivots distributed as evenly as possible among the active cells.

Here the finite calculation is genuinely finite, with no hidden limiting step. Put $M = 4 \cdot 9 \cdot 49 = 1764$ and let $V_p(S)$ be the worst conditional probability, after the primes above p have been exposed, that the terminal set of finite-bad forms has the required cardinality. If the collision partition at p is $\mathcal{Q}$, then backwards induction uses

$$V_{p^-}(S) = \max_{\mathcal{Q}} \left(\frac{p^2 - |\mathcal{Q}|}{p^2} V_p(S) + \frac{1}{p^2} \sum_{Q \in \mathcal{Q}} V_p(S \cup Q) \right). \quad (59)$$

Allowing the maximizing partition to depend on S only enlarges the true density. If the resulting density is R/M , then in either base progression every prefix has excess at most

$$R \left(1 - \frac{R}{M} \right), \quad (60)$$

because a prefix of length u contains at most $\min(u, R)$ favourable residues. Thus both the main density and the integer endpoint are certified.

The controlling periodic rows are

branch	R/M	tail factor	diagonal	total
$O_1, s = 2$	40/147	2/3	.001415	.039866076
$O_1, s = 7$	55/196	1/2	.004928	.039932809
$O_1, s = 9, \text{ matching}$	1/4	8/15	.006330	.039701523
$O_1, s = 9, \text{ failure}$	55/196	1/2	.003587	.038591809
$O_2, \text{ balanced}$	1/42	1	.012652	.039374127.

The controlling row admits a decimal-free exact comparison. Define

$$\hat{\tau}_7 = \frac{3887}{10^6} + \sum_{p \in \{11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}} \frac{1}{p^2}.$$

The cutoff-7 dynamic programme gives exactly $R/M = 55/196$ and prefix excess $69795/196$. With the upward rational ceilings

$$D_{5,7} < \frac{4928}{10^6}, \quad T_{O,7} \leq \frac{17398287}{10^9},$$

the complete $O_1, s = 7$ row is at most

$$\begin{aligned} C_5 &= \frac{4928}{10^6} + \frac{2}{25} \frac{55}{196} + \frac{2}{5 \cdot 10^6} \left(\frac{55}{196} + \frac{69795}{196} \right) + \frac{1}{2} \frac{17398287}{10^9} \\ &\quad + \frac{3}{25} \hat{\tau}_7 + \frac{128}{5 \cdot 10^6}. \end{aligned} \quad (61)$$

Exact common-denominator arithmetic yields

$$C_5 < \frac{1}{25} - \frac{7}{25(5 \cdot 10^6)} - \frac{67}{10^6}.$$

Consequently the controlling five-million row retains rational slack greater than $6.7 \cdot 10^{-5}$ after outward rounding.

Every omitted subrow is smaller. The exact rational dynamic programme, prefix payments, root jumps, fibre classifications, and all diagonal maxima are reproduced by

`scripts/systematic_mixed_cell_5m_close.py,`
`scripts/systematic_concentrated_diagonal_blocks.cpp.`

Proposition 6.20 (Cell–fibre close on the five-million block). *The Hall inequality (2) holds for*

$$5 \cdot 10^6 \leq N < 10^7.$$

Proof. Suppose that a strict Hall defect exists. The residual estimate (57) supplies more than $0.003997632216030162N$ pivots. Call an E_1 mod-9 cell paired when it contains at least two pivots. If at least two cells are paired, classify them as good, 7-only, 11-only, or both. Two good cells, one good cell, and no good cell give respectively the first three E_1 envelopes displayed above; Lemma 6.16(1) proves that these cases exhaust the possible 7–11 collision patterns. If exactly one cell is paired and contains at least three pivots, the four one-cell rows apply. Otherwise $|E_1| \leq 10$, which is charged.

After that charge, the identical paired-cell split in E_2 gives its two-cell or one-cell row. If neither applies, $|E_2| \leq 10$ and it too is charged. Nineteen remaining E_3 pivots force three into one of the nine mod-9 cells; if there are fewer, $|E_3| \leq 18$. Thus failure to select an even row costs at most $10 + 10 + 18 = 38$ pivots.

In the odd part, cells containing at most five pivots cost at most $2 \cdot 9 \cdot 5 = 90$ more. Hence the total raw charge is at most 128. If both odd mod-4 classes have active cells and admit disjoint triples of cell residues, use the balanced six-pivot row. If not, the argument above gives at most eleven active cells. Two different residues give the four-pivot row, while one residue gives the concentrated three-pivot row. These are all possibilities with both odd classes active.

It remains that only one odd mod-4 class is active. Let s be its number of active cells. For $2 \leq s \leq 7$, six pivots distributed as evenly as possible give the listed periodic rows. For $s = 8, 9$, a capacity-two matching either selects eight pivots with collision blocks of size at most two or Lemma 6.16(2) gives the displayed restricted diagonal. For $s = 1$, either four mod-49 fibres supply pairs, or at most three fibres are nonsingletons and the remaining at most 49 pivots are charged. Therefore the branch list used in `scripts/systematic_mixed_cell_5m_close.py` is exhaustive.

For every branch, (51), the exact periodic recurrence (59), the prefix bound (60), and the appropriate diagonal and root ceilings bound the Hall completion by its listed total. The largest is the $O_1, s = 7$ row, and (61) puts it strictly below $1/25 - 7/(25N)$. Hence $|B| + |T_N(B)| \leq |C_N|$, contradicting the strict Hall defect. $\square$

Theorem 6.21 (Direct interval splice at five million). *The Hall inequality (2), and hence the exact upper bound in Erdős Problem 848, holds for every*

$$N \geq 5 \cdot 10^6.$$

Proof. Proposition 6.20 handles $5 \cdot 10^6 \leq N < 10^7$, Proposition 6.19 handles $10^7 \leq N < 2 \cdot 10^7$, Proposition 6.18 handles $2 \cdot 10^7 \leq N < 4 \cdot 10^7$, and Theorem 6.14 handles $N \geq 4 \cdot 10^7$. The four intervals are disjoint and exhaustive. $\square$

6.3 Full-support continuation below five million

The low degree calculation can be strengthened because every prime divisor of a pivot, rather than only its smallest prefix, may be retained in the quadratic-residue mask. For $M = 4902$ the exact survivor maxima at product bound $3 \cdot 10^6$ are

	$k = 0$	1	2	3	4	5	6
odd	1961	1194	612	312	153	73	28
$\nu_2 = 1$	981	594	312	156	80	38	15
$\nu_2 = 2$	491	312	171	92	54	28	13
$\nu_2 \geq 3$	246	169	102	58	37	22	12.

These maxima include the worse nonzero square coset modulo 5 and the quotient congruences modulo 2, 4, 8. For example, the controlling support is (73, 149, 211) in the odd $k = 3$ row.

Proposition 6.22 (Full-support low residual and bounded clusters). *On $1.5 \cdot 10^6 \leq N < 3 \cdot 10^6$, every mixed Hall defect has more than*

$$\alpha N, \quad \alpha = 0.000249932448083113 \dots \quad (62)$$

non-base residual pivots. Among them there are three of span at most 8045 and eight of span at most 28540.

Proof. Put $Y = \lfloor N/70 \rfloor$, $X = N/25 + 1$, and sieve through 17. For each displayed support row retain the active degree-3 Bonferroni lower bound, subtract

$$X\tau_{17} + \pi(Y) - \pi(17) + H \left(s + \frac{2X}{Y} \right) + \frac{N}{50},$$

including its exact finite discrepancy. The normalized minimum is the odd $k = 3$, $H = 16$, $s = 312$ row at $N = 1.5 \cdot 10^6$; before division by N its reverse-degree residual is

$$374.898672124669531 \dots$$

For fixed Y every row is affine in $1/N$ and increases. Between consecutive prime jumps the normalized positive payments decrease; exact rational checks at every prime jump show that the same lower endpoint is the global minimum.

If $x_1 < \dots < x_m$ are the residual pivots, then for $r = 3$ or 8,

$$\sum_{i=1}^{m-r+1} (x_{i+r-1} - x_i) \leq (r-1)(N-1).$$

Using $m > \alpha N$ and exhausting the fewer than 376 possible integer values of $\lfloor \alpha N \rfloor$ gives the two stated spans. The support search, rational degree rows, prime jumps, and integer span exhaustion are reproduced by `scripts/systematic_full_support_scan.cpp`, `scripts/systematic_one_prime_support_scan.cpp`, and `scripts/systematic_full_support_degree_low.py`. $\square$

We use the following exact one-form tail.

Lemma 6.23 (Exact complete one-form tail). *For $1.5 \cdot 10^6 \leq N < 3 \cdot 10^6$, $1 \leq t \leq N$, and*

$$\mathcal{M}_t(N) = \left\{ a \leq N : \begin{array}{l} a \equiv 7 \text{ or } 18 \pmod{25}, \text{ and some prime } p \text{ satisfies } \\ D < p \leq N, \quad p^2 \mid at + 1 \end{array} \right\},$$

one has

$$|\mathcal{M}_t(N)| \leq 40. \quad (63)$$

Proof. Every event on the stated interval embeds in the upper box $N_* = 3 \cdot 10^6$. Writing $at + 1 = p^2r$, every event occurs in the finite list

$$D < p \leq N_*, \quad 1 \leq r \leq \left\lfloor \frac{N_*^2 + 1}{p^2} \right\rfloor, \quad p^2r - 1 = at.$$

There are 213713 primes and exactly 27898239 pairs (p, r) in this list. For each pair, factor $p^2r - 1$ and retain every divisor $a \leq N_*$ with $a \equiv 7, 18 \pmod{25}$ whose complementary divisor t is also at most N_* . This is an equivalence, not a relaxation. Moreover $D^4 > N_*^2 + 1$, so two different retained primes cannot represent the same pair (a, t) . Exact counters over $1 \leq t \leq N_*$ have maximum

$$40 \quad \text{at } t = 10920.$$

The prime sieve, complete quotient range, all 3063864 retained divisor events, deterministic 64-bit factorization with factor-product verification, and divisor enumeration are implemented by

`scripts/systematic_one_form_medium_exhaustion.cpp`.

Since every smaller- N event is in the enumerated upper box, (63) follows. □

The resulting uniform one-form bounds are as follows.

Lemma 6.24 (Low one-form incidence envelopes). *For $1.5 \cdot 10^6 \leq N < 3 \cdot 10^6$, put*

$$E(N) = \sum_{\substack{3 \leq p \leq 300 \\ p \neq 5}} 2 \left\lceil \frac{N}{25p^2} \right\rceil + 182 + 40,$$

$$O(N) = \sum_{\substack{3 \leq p \leq 300 \\ p \neq 5}} 2 \left\lceil \frac{N}{100p^2} \right\rceil + 268 + 40,$$

where the sums are over primes. If t is even and non-base, then

$$\#\{a \leq N : a \equiv 7, 18 \pmod{25}, at + 1 \text{ is nonsquarefree}\} \leq E(N). \quad (64)$$

If t is odd and non-base and $r \pmod{4}$ is fixed, then the number of such $a \equiv r \pmod{4}$ for which $at + 1$ has an odd-prime-square divisor is at most $O(N)$. At the lower endpoint,

$$E(1.5 \cdot 10^6) = 19694, \quad O(1.5 \cdot 10^6) = 5224.$$

Proof. For a fixed odd prime $p \neq 5$, the congruence $p^2 \mid at + 1$ has at most one a -class modulo p^2 . Intersecting it with the two base classes gives two progressions modulo $25p^2$, and imposing one class modulo 4 gives two progressions modulo $100p^2$. These are the two ceiling sums.

On the complete upper box $a, t \leq 3 \cdot 10^6$, exhaustive modular inversion over $300 < p \leq 28540$ gives total prime-incidence maximum 268, and maximum 182 when t is even and non-base. The latter is attained at $t = 120$. Lemma 6.23 supplies the final 40 for $p > 28540$. If t is even, prime 2 is inactive. Prime 5 is always inactive for a non-base pivot: from $25 \mid at + 1$ and $a \equiv 7$ or $18 \pmod{25}$ one obtains respectively $t \equiv 7$ or $18 \pmod{25}$. This proves both bounds. The upper-box enumeration and its exhaustive check are reproduced by `scripts/systematic_low_high_prime_degree.cpp`. □

Lemma 6.25 (First-low parity diagonal ceilings). *On $1.5 \cdot 10^6 \leq N < 3 \cdot 10^6$, the nonsquarefree diagonal among the non-base classes has the following uniform upper bounds:*

<i>allowed support</i>	<i>count</i>
<i>unrestricted</i>	$0.025297N$
<i>the two odd classes modulo 4</i>	$0.012657N$
<i>one odd class modulo 4</i>	$0.006337N$.

Proof. The eventwise Möbius–CRT diagonal enumeration on 139 blocks applies (49) by exact integer cross-multiplication. Decimal renderings of the largest rational envelopes are

$$0.0252968815411\dots, \quad 0.0126562333156\dots, \quad 0.0063363175457\dots$$

They occur respectively on the blocks beginning at 1665621, 1812996, and 1724796. The square-free semigroup enumeration has 1237599 nodes, 4134027 root terms, and 77041 retained events; the harmonic and negative-Pell tails are then added monotonically at the upper block end-point. The complete producer and the three asserted upward decimal ceilings are in `scripts/systematic_concentrated_diagonal_blocks.cpp`. $\square$

Theorem 6.26 (Direct close of the first low block). *The Hall inequality (2), and hence the exact bound in Erdős Problem 848, holds for every*

$$1.5 \cdot 10^6 \leq N < 3 \cdot 10^6.$$

Proof. Theorem 6.11 closes the pure opposite-base case. Suppose therefore that a mixed Hall defect exists, put $A = B \cup T_N(B)$, and let A^* be its non-base residual part. By Proposition 6.22, the residual pivot set contains more than

$$0.000249932448083113N > 374$$

members throughout the interval. Every base member of A is a common non-neighbour of every selected residual pivot, while every member of A^* belongs to the nonsquarefree diagonal. We use these two facts in three exhaustive parity branches.

If there are at least eight even residual pivots, select eight of them. For each common base non-neighbour, all eight associated forms are nonsquarefree. Averaging their eight indicators and then taking the union bound over prime-square witnesses gives the common-base count at most $E(N)$. Lemma 6.25 therefore gives

$$|A| \leq 0.025297N + E(N).$$

Otherwise charge the at most seven even residual pivots. Suppose first that both odd classes modulo 4 contain at least eight residual pivots, and select four from each class. Split the base variable a into its four classes modulo 4. When a is even, prime 2 witnesses none of the eight forms. When a is odd, it witnesses exactly one of the two four-pivot classes, so the other four forms still require odd-prime-square witnesses. Averaging the required four or eight indicators on each a -class and applying Lemma 6.24 gives common-base count at most $4O(N)$. Hence

$$|A| \leq 0.012657N + 7 + 4O(N).$$

Finally, one odd pivot class has at most seven members. Charge it together with the at most seven even pivots and select eight pivots from the remaining odd class. On exactly one odd a -class

modulo 4, prime 2 witnesses all eight forms; that class contains at most $2\lceil N/100 \rceil$ base values. On the other three a -classes prime 2 is inactive for all selected pivots, so the averaged odd-prime count is at most $3O(N)$. Thus

$$|A| \leq 0.006337N + 14 + 2 \left\lceil \frac{N}{100} \right\rceil + 3O(N).$$

Each right side is strictly smaller than $N/25 - 7/25$ on the full interval. Indeed it is affine and increasing between ceiling jumps, so it suffices to check the lower endpoint and the points immediately following a multiple of $25p^2$, $100p^2$, or 100. The exact scan of all 21999 such points gives

branch	worst N	count margin
eight even	1500751	2359.261953
two odd classes	1500401	20106.184543
one odd class	1500401	4799.718863.

This finite arithmetic is reproduced by `scripts/systematic_low_parity_close.py`. Every branch contradicts the assumed Hall defect, proving the theorem. $\square$

Lemma 6.27 (Second-low support and one-form envelopes). *For $3 \cdot 10^6 \leq N < 5 \cdot 10^6$, every mixed Hall defect has more than*

$$0.00157645N \tag{65}$$

non-base residual pivots. Put

$$E_5(N) = \sum_{\substack{3 \leq p \leq 300 \\ p \neq 5}} 2 \left\lceil \frac{N}{25p^2} \right\rceil + 278 + 49,$$

$$O_5(N) = \sum_{\substack{3 \leq p \leq 300 \\ p \neq 5}} 2 \left\lceil \frac{N}{100p^2} \right\rceil + 383 + 49,$$

where the sums are over primes. If t is even and non-base, its nonsquarefree degree into the two base classes is at most $E_5(N)$. If t is odd and non-base and one class of a modulo 4 is fixed, its odd-prime-square degree into that class and the two base classes is at most $O_5(N)$. At the lower endpoint,

$$E_5(3 \cdot 10^6) = 39211, \quad O_5(3 \cdot 10^6) = 10192.$$

Proof. For the residual assertion, repeat the reverse-degree calculation of Proposition 6.22 with upper support $5 \cdot 10^6$, cutoff 17, split 70, and support modulus 4902. For each possible number $0 \leq k \leq 6$ of retained prime divisors and each 2-adic type, take independently the smallest degree-3 Bonferroni density, the largest endpoint error, and the largest full-support quadratic-residue survivor count. This enlarges every actual debit. On a fixed cell $Y = \lfloor N/70 \rfloor$, the resulting normalized residual has the form $A + B/N$, so its minimum is at a cell endpoint. Exact rational exhaustion of all cells and all 28 relaxed rows has minimum

$$0.001576458682760064\dots > 0.00157645$$

at $N = 3000130$. The controlling relaxed row is odd, has $k = 3$, root factor 16, and survivor maximum 528. This proves (65); in particular the residual contains more than 4729 pivots throughout the interval.

The low-prime terms follow from the same two CRT progressions as in Lemma 6.24. Exhaustive modular inversion on the complete box $a, t \leq 5 \cdot 10^6$ gives, over $300 < p \leq 28540$, unrestricted prime-incidence maximum 383 and even non-base maximum 278. They occur at $t = 168$ and $t = 120$, respectively. For $p > 28540$, enumerate the complete quotient identity

$$p^2 r - 1 = at, \quad a, t \leq 5 \cdot 10^6.$$

There are 77829785 prime-quotient terms and 7742361 retained divisor events. Since $28540^4 > (5 \cdot 10^6)^2 + 1$, distinct witness primes cannot represent the same (a, t) ; the exact maximum degree is 49, at $t = 9240$. Prime 2 is inactive for even t , and prime 5 is inactive for every non-base pivot exactly as in the preceding lemma. The support calculation, middle-prime inversion, and complete tail enumeration are reproduced by `scripts/systematic_second_low_parity_close.py`, `scripts/systematic_low_high_prime_degree.cpp`, and `scripts/systematic_one_form_medium_exhaustion.cpp`. $\square$

Lemma 6.28 (Second-low parity diagonal ceilings). *On $3 \cdot 10^6 \leq N < 5 \cdot 10^6$, the nonsquarefree diagonal among the non-base classes is bounded by*

<i>allowed support</i>	<i>count</i>
<i>unrestricted</i>	$0.025295N$
<i>the two odd classes modulo 4</i>	$0.012655N$
<i>one odd class modulo 4</i>	$0.006333N$.

Proof. The complete eventwise Möbius-CRT producer on this interval again uses (49) and exact integer comparisons. Decimal renderings of its three largest rational envelopes are

$$0.0252947682147\dots, \quad 0.0126547423852\dots, \quad 0.0063325725290\dots$$

The asserted decimals are strict upward ceilings. The enumeration and the three assertions are in `scripts/systematic_concentrated_diagonal_blocks.cpp`. $\square$

Theorem 6.29 (Direct close of the second low block). *The Hall inequality (2), and hence the exact bound in Erdős Problem 848, holds for every*

$$3 \cdot 10^6 \leq N < 5 \cdot 10^6.$$

Proof. The pure opposite-base case is already closed by Theorem 6.11. In a mixed defect, Lemma 6.27 supplies more than 4729 non-base residual pivots, in particular more than the 22 needed below.

Apply the same exhaustive parity trichotomy as in Theorem 6.26. Eight even pivots give

$$|A| \leq 0.025295N + E_5(N).$$

If there are at most seven even pivots and both odd classes modulo 4 have at least eight members, choose four from each class; then

$$|A| \leq 0.012655N + 7 + 4O_5(N).$$

Otherwise charge the at most seven even pivots and the at most seven pivots in the smaller odd class. More than 22 residual pivots ensure that eight remain in the other odd class. Selecting them gives

$$|A| \leq 0.006333N + 14 + 2 \left\lceil \frac{N}{100} \right\rceil + 3O_5(N).$$

Thus no parity allocation is omitted.

Each right side is affine and increasing between its ceiling jumps. Checking the lower endpoint and every point immediately after a multiple of $25p^2$, $100p^2$, or 100 gives 29322 exact candidates and

branch	worst N	count margin
eight even	3000151	4901.940455
two odd classes	3000000	41259.720000
one odd class	3000801	10407.687267.

All three margins are against the unchanged Hall target $N/25 - 7/25$ and are strictly positive. The exact scan is `scripts/systematic_second_low_parity_close.py`, so every possible mixed defect is contradicted and the block is closed. $\square$

The diagonal large-square tail has a further uniform improvement which is specific to norm -1 . It is stronger than the generic totally positive unit gap used above.

Lemma 6.30 (Sharp negative-Pell gap). *Let $d \geq 2$ be nonsquare. The number of positive solutions of*

$$x^2 - dy^2 = -1, \quad x \leq N,$$

is at most

$$P_-(N) := \left\lfloor 1 + \log_{3+2\sqrt{2}} N \right\rfloor. \quad (66)$$

In particular,

$$P_-(N) = 13 \quad (2 \cdot 10^9 \leq N < 7.67769 \cdot 10^9). \quad (67)$$

Thus the last factor in every aggregate diagonal Pell tail on this interval may be replaced by 13.

Proof. Write $\alpha = x + y\sqrt{d}$ for a solution. Its conjugate is $-\alpha^{-1}$. If $\alpha_2 > \alpha_1$ are two solutions, then $\alpha_2/\alpha_1 = u + v\sqrt{d}$ is an integral totally positive unit of norm 1. Hence $u \geq 2$. The case $u = 2$ would force $dv^2 = 3$, hence $d = 3$; but $x^2 \equiv -1 \pmod{3}$ has no solution. Therefore $u \geq 3$, and

$$\frac{\alpha_2}{\alpha_1} = u + \sqrt{u^2 - 1} \geq 3 + 2\sqrt{2}.$$

Also

$$1 + \sqrt{2} \leq \alpha = x + \sqrt{x^2 + 1} \leq (1 + \sqrt{2})x,$$

so the ratio of the largest to the smallest allowed α is at most N . The geometric gap proves (66). Finally direct squaring gives

$$(3 + 2\sqrt{2})^{12} < 2 \cdot 10^9 < 7.67769 \cdot 10^9 < (3 + 2\sqrt{2})^{13},$$

which proves (67). $\square$

For an actual residual Hall defect, the mixed-base strip forces a much more rigid pair of pivots.

Proposition 6.31 (Bounded-gap residual pivots). *Let $N \geq 3.7 \cdot 10^{11}$, and suppose that a compatible outside set B violates the Hall inequality. Then there are distinct*

$$b, c \in B \setminus C_N^{18}, \quad 1 \leq |b - c| \leq 5000.$$

If $N \geq 5.5 \cdot 10^{11}$, the pair may be chosen with $d := |b - c| \leq 1381$; if also $N < N_{25}$, one may take $d \leq 834$. For this sharper pair,

$$|S_N(b, c)| \leq \frac{13N}{900} + 2. \quad (68)$$

Moreover, for every $a \in D_N(b, c)$,

$$\gcd(ab + 1, ac + 1)$$

is a squarefree divisor of d . The original Hall inequality follows if

$$|T_N(B) \cap D_N(b, c)| \leq \frac{23N}{900} - |B| - \frac{57}{25}. \quad (69)$$

Thus above $2 \cdot 10^{12}$ the distinct-prime contribution has determinant at most 1381 and linear debit $23N/900$.

Proof. Put $B^* = B \setminus C_N^{18}$ and $m = |B^*|$. By Corollary 5.22,

$$m > \mathcal{R}(N).$$

List the elements of B^* increasingly. First suppose $3.7 \cdot 10^{11} \leq N < 5.5 \cdot 10^{11}$. Then

$$m - 1 > 0.0002N - 1 > \frac{N - 1}{5001}.$$

The least consecutive gap is therefore less than 5001, hence at most 5000.

If $5.5 \cdot 10^{11} \leq N < N_{25}$, then instead

$$m - 1 > 0.0012N - 1 > \frac{N - 1}{835},$$

so the least gap is at most 834.

Finally suppose $N \geq N_{25}$. The second part of Corollary 5.22 gives

$$m > 0.000724N = \frac{181N}{250000}.$$

The least consecutive gap is at most $(N - 1)/(m - 1)$. The identity

$$\frac{181}{250000} - \frac{1}{1382} = \frac{142}{345500000}$$

and $N \geq 2 \cdot 10^{12}$ give

$$m - 1 > \frac{181N}{250000} - 1 > \frac{N - 1}{1382}.$$

Hence the least gap is strictly smaller than 1382, and, being integral, is at most 1381.

If a prime contributes to $S_N(b, c)$, then its square divides d and the prime is not 5. Three distinct such primes would force

$$(2 \cdot 3 \cdot 7)^2 = 1764 \mid d,$$

which is impossible. There are therefore at most two, and the two largest residue-class budgets occur for $p = 2, 3$. This proves (68).

For $u = ab + 1$ and $v = ac + 1$ one has

$$cu - bv = c - b,$$

so $(u, v) \mid d$. If (u, v) were divisible by a prime square, then a would belong to $S_N(b, c)$, contrary to $a \in D_N(b, c)$. Thus the gcd is squarefree.

Finally $T_N(B) \subseteq T_N(\{b, c\})$ and the latter is the disjoint union of its same- and distinct-prime parts. If (69) holds, then

$$\begin{aligned} |B| + |T_N(B)| &\leq |B| + \left(\frac{13N}{900} + 2\right) + \left(\frac{23N}{900} - |B| - \frac{57}{25}\right) \\ &= \frac{N}{25} - \frac{7}{25} \leq |C_N|. \end{aligned}$$

This is exactly the original Hall inequality. $\square$

Theorem 6.32 (Parity-pair descent to the matching threshold). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 1.409 \cdot 10^{12}$.*

Proof. Theorem 4.11 handles $N \geq 8.858 \cdot 10^{12}$, so assume

$$1.409 \cdot 10^{12} \leq N < 8.858 \cdot 10^{12}$$

and suppose that B is a Hall defect. Put $A = B \cup T_N(B)$ and $A^* = B \setminus C_N^{18}$. Corollary 5.22 and Proposition 6.31 give

$$|A^*| > \left(\frac{2}{9018009} + 5 \cdot 10^{-7}\right) N \quad \text{and a pair in } A^* \text{ at distance at most } 1400000.$$

We use the same thirteen-prime set $\mathcal{P}$ as in Lemma 4.10. Put

$$J_1 = 1 - \prod_{p \in \mathcal{P}} \left(1 - \frac{1}{p^2}\right) < 0.152377, \quad I_2 < 0.129757, \quad I_3 < 0.136690.$$

Here I_2 is the two-form intersection density when at most two residue pairs coincide, and I_3 is the corresponding density for at most three. If the difference of two pivots is arbitrary but at most N , no more than six pairs can coincide, because

$$(3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23)^2 = 497704286448369 > N.$$

The corresponding finite product gives

$$I_6 < 0.146927.$$

Write $\epsilon_1 = 2^{13} - 1 = 8191$ and $\epsilon_2 = 2(2^{13} - 1) + (3^{13} - 1) = 1610704$ for the one-form and two-form inclusion–exclusion discrepancies.

The large-prime tail can be reused with parity-sensitive root counts. For $k \geq 2$ and root bounds H_1, H_2 , set

$$\begin{aligned} \mathfrak{T}_{H_1, H_2, k}(N) &= \frac{4(0.004751)}{25} + \frac{4(1.25506)}{k \log(N/(k+1))} \\ &\quad + \frac{2(H_1 + H_2)}{N} \left(\frac{2(N/25 + 2)}{N/(k+1)} + \frac{N^2 + 25N + 1}{(N/(k+1))^2} \right). \end{aligned}$$

This is the bound obtained from (16) with $Y = \lfloor N/k \rfloor$: the first two terms pay the intermediate primes in both forms and both base classes, and the last pays their transformed progression tails.

For an odd pivot the effective root modulus is odd and, after removal of the redundant factor 5, is at most $5N < P_{12}$; hence $H = 2048$. For an even pivot we retain $H = 8192$. Direct substitution at the left endpoint gives

$$\mathfrak{T}_{8192,8192,168}(N) < 0.002732802, \quad \mathfrak{T}_{2048,2048,269}(N) < 0.002018195.$$

Both displayed envelopes decrease. At the same endpoint,

$$\Delta_{25}^{(0)} < 0.001098525, \quad \Delta_{50}^{(0)} < 0.001098460, \quad \Delta_{100}^{(0)} < 0.001098428.$$

Let e be the number of even elements of A^* . First suppose

$$e - 1 > \frac{N - 1}{9018009}.$$

Two even pivots then lie at distance at most 9018008. The prime 2 cannot witness either linear form, and at most three odd-prime residue pairs coincide. The unrestricted aggregate diagonal estimate and the two-pivot base sieve give, after division by N ,

$$\frac{|A|}{N} < \frac{23}{25}C_{\text{quad}} + \Delta_{25}^{(0)} + \frac{2I_3}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{8192,8192,168} < 0.039928.$$

It remains to suppose

$$e \leq \frac{N - 1}{9018009} + 1 < \frac{N}{9018009} + 1.$$

Let O be the odd part of A^* . If O meets both odd residue classes modulo 4, choose one pivot from each. On each of the two base classes modulo 25, two of the four progressions modulo 100 have an automatic factor 4 for exactly one pivot, and two have none. Thus the small-prime base density is at most $4(J_1 + I_6)/100$. Charging the even part by e and using the odd diagonal estimate gives

$$\frac{|A|}{N} < \frac{1}{9018009} + \frac{1}{N} + \frac{23}{50}C_{\text{quad}} + \Delta_{50}^{(0)} + \frac{4(J_1 + I_6)}{100} + \frac{4\epsilon_1 + 4\epsilon_2}{N} + \mathfrak{T}_{2048,2048,269} < 0.027674.$$

Finally suppose that O lies in one odd residue class modulo 4. Since

$$|O| > |A^*| - e > N \left(\frac{1}{9018009} + 5 \cdot 10^{-7} \right) - 2,$$

two elements of O lie at distance at most 9018008; the extra $5 \cdot 10^{-7}N$ absorbs the two endpoint losses. At most three odd-prime residue pairs therefore coincide. In each base class one progression modulo 100 has an automatic factor 4 for both pivots, while the other three are governed by the two-form density I_2 . The restricted diagonal estimate now gives

$$\frac{|A|}{N} < \frac{1}{9018009} + \frac{1}{N} + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^{(0)} + \frac{2}{100} + \frac{6I_3}{100} + \frac{2 + 6\epsilon_2}{N} + \mathfrak{T}_{2048,2048,269} < 0.037615.$$

All three bounds are strictly below $1/25 - 7/(25N)$ on the stated range, contradicting $|A| > |C_N|$. This closes the interval and proves the theorem. $\square$

Theorem 6.33 (Primorial matching descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 5.5 \cdot 10^{11}$.*

Proof. It remains only to treat

$$5.5 \cdot 10^{11} \leq N < 1.409 \cdot 10^{12}.$$

Suppose B is a Hall defect and put $A = B \cup T_N(B)$ and $A^* = B \setminus C_N^{18}$. Corollary 5.22 gives

$$|A^*| > 0.0012N.$$

On this interval an even pivot has root count at most 4096, since $5N < 8P_{11} = 29682952539240$; an odd pivot has root count at most 2048, since $5N < P_{12} = 152125131763605$.

Retain $J_1, I_2, I_6, \epsilon_1, \epsilon_2$ and the tail function $\mathfrak{T}$ from the preceding proof. At the left endpoint,

$$\mathfrak{T}_{4096,4096,154} < 0.002958674, \quad \mathfrak{T}_{2048,2048,194} < 0.002515972,$$

and

$$\Delta_{25}^{(0)} < 0.001465437, \quad \Delta_{50}^{(0)} < 0.001465317, \quad \Delta_{100}^{(0)} < 0.001465258.$$

All these envelopes decrease.

Let e count the even elements of A^* . If $e - 1 > (N - 1)/53361$, two even pivots lie within 53360 and at most two odd-prime residue pairs coincide. Therefore

$$\frac{|A|}{N} < \frac{23}{25}C_{\text{quad}} + \Delta_{25}^{(0)} + \frac{2I_2}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{4096,4096,154} < 0.039970.$$

Otherwise $e < N/53361 + 1$. If the odd part of A^* meets both odd classes modulo 4, choose one pivot from each. As before at most six small-prime residue pairs can coincide, and

$$\frac{|A|}{N} < \frac{1}{53361} + \frac{1}{N} + \frac{23}{50}C_{\text{quad}} + \Delta_{50}^{(0)} + \frac{4(J_1 + I_6)}{100} + \frac{4\epsilon_1 + 4\epsilon_2}{N} + \mathfrak{T}_{2048,2048,194} < 0.028564.$$

If instead the odd part lies in one class modulo 4, its size is greater than $(0.0012 - 1/53361)N - 2$. Thus two odd pivots lie within 53360, and

$$\frac{|A|}{N} < \frac{1}{53361} + \frac{1}{N} + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^{(0)} + \frac{2}{100} + \frac{6I_2}{100} + \frac{2 + 6\epsilon_2}{N} + \mathfrak{T}_{2048,2048,194} < 0.038093.$$

Each bound is below $1/25 - 7/(25N)$, a contradiction. The preceding theorem handles all larger N . $\square$

Theorem 6.34 (Two-adic primordial descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 3.7 \cdot 10^{11}$.*

Proof. By Theorem 6.33, it remains to consider

$$3.7 \cdot 10^{11} \leq N < 5.5 \cdot 10^{11}.$$

Suppose that B is a Hall defect, put $A = B \cup T_N(B)$ and $A^* = B \setminus C_N^{18}$, and let e be the number of even elements of A^* . Corollary 5.22 gives

$$|A^*| > 0.0002N. \tag{70}$$

We first determine the required root counts. Let

$$m = 2^\nu m_0 \leq 5N, \quad m_0 \text{ odd},$$

be the effective modulus in Lemma 4.1. Since

$$P_{11} = 3710369067405, \quad P_{12} = 152125131763605,$$

an odd pivot has root count at most 2048. For an even pivot, respectively,

$\nu_2(b)$	m_0 is at most	$\rho(m)$ is at most
1	$5N/2 < P_{11}$	1024
2	$5N/4 < P_{11}$	2048
≥ 3	$5N/8 < P_{11}$	4096.

Indeed these are exactly the factors $c_1 = 1, c_2 = 2, c_\nu = 4$ in the proof of Lemma 4.1, multiplied by the largest possible $2^{\omega(m_0)}$.

Retain $J_1, I_2, I_6, \epsilon_1, \epsilon_2$ and $\mathfrak{T}$ from the proof of Theorem 6.32. At $N = 3.7 \cdot 10^{11}$,

$$\begin{aligned} \Delta_{25}^{(0)} &< 0.001654479, & \Delta_{50}^{(0)} &< 0.001654326, & \Delta_{100}^{(0)} &< 0.001654249, \\ \Delta_8^{(0)} &< 0.002795228, & \mathfrak{T}_{1024,1024,218} &< 0.002375113, & \mathfrak{T}_{2048,2048,172} &< 0.002781711, \\ & & \mathfrak{T}_{4096,4096,136} &< 0.003291527. \end{aligned}$$

All these envelopes decrease on the interval.

First suppose $e > 3N/100000$. Partition the even pivots as

$$E_1 = \{b : \nu_2(b) = 1\}, \quad E_2 = \{b : \nu_2(b) = 2\}, \quad E_3 = \{b : \nu_2(b) \geq 3\}.$$

If

$$|E_i| - 1 > \frac{N-1}{100001} \quad (i = 1 \text{ or } 2),$$

two pivots of that type have distance at most 100000. The prime 2 cannot witness either associated linear form. Moreover at most two odd-prime residue pairs can coincide: three would make the even difference divisible by

$$2(3 \cdot 7 \cdot 11)^2 = 106722.$$

The unrestricted diagonal estimate and the two-pivot base sieve therefore give, for $i = 1$ and $i = 2$, respectively,

$$\begin{aligned} \frac{|A|}{N} &< \frac{23}{25}C_{\text{quad}} + \Delta_{25}^{(0)} + \frac{2I_2}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{1024,1024,218} < 0.039578, \\ \frac{|A|}{N} &< \frac{23}{25}C_{\text{quad}} + \Delta_{25}^{(0)} + \frac{2I_2}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{2048,2048,172} < 0.039985. \end{aligned}$$

It remains in the even case to suppose that neither E_1 nor E_2 is dense. Then

$$|E_1| + |E_2| \leq \frac{2(N-1)}{100001} + 2$$

and, using $e > 3N/100000$ and $N \geq 3.7 \cdot 10^{11}$,

$$|E_3| - 1 > \frac{3N}{100000} - \frac{2(N-1)}{100001} - 3 > \frac{N-1}{100001}.$$

Thus two members of E_3 again lie within 100000. Apart from the raw charge for $E_1 \cup E_2$, every member of A^* is odd or divisible by 8, so Corollary 4.4 applies. Consequently

$$\begin{aligned} \frac{|A|}{N} &< \frac{23}{40}C_{\text{quad}} + \Delta_8^{(0)} + \frac{2}{100001} + \frac{2}{N} + \frac{2I_2}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{4096,4096,136} \\ &< 0.032221. \end{aligned}$$

We may therefore assume $e \leq 3N/100000$. Let O be the odd part of A^* . By (70),

$$|O| > 0.00017N.$$

If O meets both odd classes modulo 4, select one pivot from each. At most six small-prime residue pairs coincide, just as in Theorem 6.32; charging all even pivots directly gives

$$\begin{aligned} \frac{|A|}{N} &< \frac{3}{100000} + \frac{23}{50}C_{\text{quad}} + \Delta_{50}^{(0)} + \frac{4(J_1 + I_6)}{100} + \frac{4\epsilon_1 + 4\epsilon_2}{N} + \mathfrak{T}_{2048,2048,172} \\ &< 0.029036. \end{aligned}$$

Finally suppose that O lies in one odd class modulo 4. The inequality $|O| > 17N/100000$ implies, throughout the interval, that two of its elements lie at distance at most 5882. Their difference is even, so at most two odd-prime residue pairs coincide. In each base class one progression modulo 100 has an automatic factor 4 for both pivots, while the other three are governed by I_2 . Hence

$$\begin{aligned} \frac{|A|}{N} &< \frac{3}{100000} + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^{(0)} + \frac{2}{100} + \frac{6I_2}{100} + \frac{2 + 6\epsilon_2}{N} + \mathfrak{T}_{2048,2048,172} \\ &< 0.038568. \end{aligned}$$

These five cases exhaust the parity and two-adic possibilities. Every bound is strictly below $1/25 - 7/(25N)$ on the stated interval, contradicting $|A| > |C_N|$. Thus the interval closes inside the original Hall inequality. $\square$

Theorem 6.35 (Three-class matching descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 3.4375 \cdot 10^{11}$.*

Proof. Theorem 6.34 leaves only

$$3.4375 \cdot 10^{11} \leq N < 3.7 \cdot 10^{11}.$$

Suppose that B is a Hall defect, put $A = B \cup T_N(B)$ and $A^* = B \setminus C_N^{18}$, and let e count the even elements of A^* . Corollary 5.22 gives

$$|A^*| > 2.5 \cdot 10^{-7}N. \tag{71}$$

On this interval every odd pivot has root count at most 1024, because its effective modulus is odd and at most $5N < P_{11}$. The three even valuation classes retain the respective bounds 1024, 2048, 4096 from the preceding theorem.

In addition to the constants already defined, let I_1 denote the two-form small-prime intersection density when at most one residue pair coincides. Its worst case is the coincident prime 3, and direct rational multiplication in the product defining I_J gives

$$I_1 < 0.112570.$$

At the left endpoint,

$$\begin{aligned} \Delta_{25}^{(0)} &< 0.001692156, & \Delta_{50}^{(0)} &< 0.001691995, & \Delta_{100}^{(0)} &< 0.001691915, \\ \Delta_W &< 0.002859240, & \mathfrak{T}_{1024,1024,211} &< 0.002417843, & \mathfrak{T}_{4096,4096,132} &< 0.003358600. \end{aligned}$$

Every displayed envelope decreases on the interval.

First suppose $e > 0.004N$. One of the three even valuation classes contains more than $N/750$ elements, so two of its members lie at distance at most 750. Two coincident odd-prime residue pairs would make this even difference divisible by

$$2(3 \cdot 7)^2 = 882;$$

hence at most one pair coincides. Using the worst even root count 4096 gives

$$\frac{|A|}{N} < \frac{23}{25}C_{\text{quad}} + \Delta_{25}^{(0)} + \frac{2I_1}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{4096,4096,132} < 0.039225.$$

We may assume $e \leq 0.004N$. Let O be the odd part of A^* . If O meets both odd classes modulo 4, choose one pivot from each. Charging the even part directly and using the cross-class calculation from Theorem 6.32 gives

$$\begin{aligned} \frac{|A|}{N} &< 0.004 + \frac{23}{50}C_{\text{quad}} + \Delta_{50}^{(0)} + \frac{4(J_1 + I_6)}{100} + \frac{4\epsilon_1 + 4\epsilon_2}{N} + \mathfrak{T}_{1024,1024,211} \\ &< 0.032681. \end{aligned}$$

It remains to suppose that O occupies at most one odd class modulo 4. If $e > 10^{-7}N$, two even pivots lie at distance at most 10^7 . Four coincident odd-prime residue pairs would force their even difference to be at least

$$2(3 \cdot 7 \cdot 11 \cdot 13)^2 = 18036018,$$

so at most three pairs coincide. Choose an odd class $\varepsilon \in \{1, 3\}$ not occupied by O . Then $A^* \subseteq W_\varepsilon$, and Lemma 4.5 gives

$$\frac{|A|}{N} < \frac{69}{100}C_{\text{quad}} + \Delta_W + \frac{2I_3}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{4096,4096,132} < 0.036032.$$

Finally let $e \leq 10^{-7}N$. By (71), the single odd class contains more than $1.5 \cdot 10^{-7}N$ elements. Two lie at distance at most 6699999, again allowing at most three coincident odd-prime residue pairs. The one-class odd calculation therefore gives

$$\begin{aligned} \frac{|A|}{N} &< 10^{-7} + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^{(0)} + \frac{2}{100} + \frac{6I_3}{100} + \frac{2 + 6\epsilon_2}{N} + \mathfrak{T}_{1024,1024,211} \\ &< 0.038630. \end{aligned}$$

The four alternatives are exhaustive, and every upper bound is strictly less than $1/25 - 7/(25N)$. This contradicts the original Hall defect and closes the whole interval without introducing another cut. $\square$

Theorem 6.36 (Five-free Euler-product descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 1.8263 \cdot 10^{11}$.*

Proof. Theorem 6.35 leaves the interval

$$1.8263 \cdot 10^{11} \leq N < 3.4375 \cdot 10^{11}.$$

Suppose that B is a Hall defect, put $A = B \cup T_N(B)$ and $A^* = B \setminus C_N^{18}$, and split its even part into

$$E_1 = \{b : \nu_2(b) = 1\}, \quad E_2 = \{b : \nu_2(b) = 2\}, \quad E_3 = \{b : \nu_2(b) \geq 3\}.$$

Write $e = |E_1| + |E_2| + |E_3|$. Corollary 5.22 gives

$$|A^*| > 2.5 \cdot 10^{-7}N. \quad (72)$$

As in the preceding theorem, odd pivots have root count at most 1024, and the three even classes have respective root bounds 1024, 2048, 4096. Retain $I_1 < 0.112570$ and all the other finite-sieve constants defined above.

At $N = 1.8263 \cdot 10^{11}$,

$$\begin{aligned} \Delta_{25}^{(0)} &< 0.002053324, & \Delta_{50}^{(0)} &< 0.002053084, & \Delta_{100}^{(0)} &< 0.002052964, \\ \Delta_W &< 0.003472940, & \Delta_8^{(0)} &< 0.003472880, & \mathfrak{T}_{1024,1024,172} &< 0.002836481, \\ \mathfrak{T}_{2048,2048,136} &< 0.003359437, & \mathfrak{T}_{4096,4096,108} &< 0.004015367. \end{aligned}$$

All these envelopes decrease.

If $|E_1| > 0.0012N$, two members of E_1 lie at distance at most 834. Since two coincident odd-prime residue pairs would force the even difference to be at least $2(3 \cdot 7)^2 = 882$, at most one pair coincides. Hence

$$\frac{|A|}{N} < \frac{23}{25}C_{\text{quad}} + \Delta_{25}^{(0)} + \frac{2I_1}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{1024,1024,172} < 0.039072.$$

Suppose next that $|E_1| \leq 0.0012N$ but $|E_2| > 0.0012N$. Choose the same kind of pair in E_2 . After charging E_1 directly, every remaining non-base diagonal element lies in W_2 . Thus

$$\frac{|A|}{N} < 0.0012 + \frac{69}{100}C_{\text{quad}} + \Delta_W + \frac{2I_1}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{2048,2048,136} < 0.035925.$$

Now suppose $|E_1|, |E_2| \leq 0.0012N$ but $|E_3| > 0.0012N$. A pair in E_3 again has distance at most 834 and at most one coincident residue pair. Apart from the raw charge for $E_1 \cup E_2$, the diagonal lies in V_8 , so

$$\frac{|A|}{N} < 0.0024 + \frac{23}{40}C_{\text{quad}} + \Delta_8^{(0)} + \frac{2I_1}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{4096,4096,108} < 0.034636.$$

We may therefore assume $e \leq 0.0036N$. Let O be the odd part of A^* . If O meets both odd classes modulo 4, the cross-class pair calculation, with the even part charged directly, gives

$$\frac{|A|}{N} < 0.0036 + \frac{23}{50}C_{\text{quad}} + \Delta_{50}^{(0)} + \frac{4(J_1 + I_6)}{100} + \frac{4\epsilon_1 + 4\epsilon_2}{N} + \mathfrak{T}_{1024,1024,172} < 0.033077.$$

It remains that O occupies at most one odd class. If $e > 10^{-7}N$, two even pivots lie within 10^7 , so at most three odd-prime residue pairs coincide. Choosing an unoccupied odd class ε gives $A^* \subseteq W_\varepsilon$, and therefore

$$\frac{|A|}{N} < \frac{69}{100}C_{\text{quad}} + \Delta_W + \frac{2I_3}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{4096,4096,108} < 0.037311.$$

Finally, if $e \leq 10^{-7}N$, equation (72) puts more than $1.5 \cdot 10^{-7}N$ pivots in the single odd class. Two lie within 6699999, allowing at most three coincident residue pairs. Hence

$$\frac{|A|}{N} < 10^{-7} + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^{(0)} + \frac{2}{100} + \frac{6I_3}{100} + \frac{2 + 6\epsilon_2}{N} + \mathfrak{T}_{1024,1024,172} < 0.039434.$$

The six cases exhaust all possible mass allocations. Each bound is strictly below $1/25 - 7/(25N)$, contradicting the same original Hall inequality. $\square$

Theorem 6.37 (Root-product correlated descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 1.001 \cdot 10^{11}$.*

Proof. Theorem 6.36 leaves

$$1.001 \cdot 10^{11} \leq N < 1.8263 \cdot 10^{11}.$$

Suppose that B is a Hall defect and use the decomposition $A^* = O \sqcup E_1 \sqcup E_2 \sqcup E_3$ from the preceding theorem. Put $e = |E_1| + |E_2| + |E_3|$. The correlated degree and Corollary 5.22 give

$$|A^*| > 0.00032N. \quad (73)$$

The parity-sensitive root bounds remain 1024 for odd pivots and 1024, 2048, 4096 for E_1, E_2, E_3 . At the left endpoint,

$$\begin{aligned} \Delta_{25}^{(0)} &< 0.002467727, & \Delta_{50}^{(0)} &< 0.002467376, & \Delta_{100}^{(0)} &< 0.002467200, \\ \Delta_W &< 0.004178034, & \Delta_8^{(0)} &< 0.004177947, & \mathfrak{T}_{1024,1024,142} &< 0.003333255, \\ \mathfrak{T}_{2048,2048,112} &< 0.003981577, & \mathfrak{T}_{4096,4096,88} &< 0.004795138. \end{aligned}$$

All envelopes decrease.

The first four mass cases are identical to those in the preceding theorem: if $|E_1| > 0.0012N$, if instead $|E_2| > 0.0012N$, if instead $|E_3| > 0.0012N$, or if all three are at most $0.0012N$ and O meets both odd classes, the respective bounds are

$$\begin{aligned} \frac{|A|}{N} &< \frac{23}{25}C_{\text{quad}} + \Delta_{25}^{(0)} + \frac{2I_1}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{1024,1024,142} < 0.039998, \\ \frac{|A|}{N} &< 0.0012 + \frac{69}{100}C_{\text{quad}} + \Delta_W + \frac{2I_1}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{2048,2048,112} < 0.037267, \\ \frac{|A|}{N} &< 0.0024 + \frac{23}{40}C_{\text{quad}} + \Delta_8^{(0)} + \frac{2I_1}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{4096,4096,88} < 0.036136, \\ \frac{|A|}{N} &< 0.0036 + \frac{23}{50}C_{\text{quad}} + \Delta_{50}^{(0)} + \frac{4(J_1 + I_6)}{100} + \frac{4\epsilon_1 + 4\epsilon_2}{N} + \mathfrak{T}_{1024,1024,142} < 0.034017. \end{aligned}$$

It remains that O occupies at most one odd class. If $e > 10^{-5}N$, two even pivots lie within 100000, so at most two odd-prime residue pairs coincide. Choosing an unoccupied odd class gives

$$\frac{|A|}{N} < \frac{69}{100}C_{\text{quad}} + \tilde{\Delta}_W + \frac{2I_2}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{4096,4096,88} < 0.038256.$$

If $e \leq 10^{-5}N$, equation (73) leaves more than $0.00031N$ pivots in the single odd class. Two lie within 3226, again allowing at most two coincident residue pairs, and

$$\frac{|A|}{N} < 10^{-5} + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^{(0)} + \frac{2}{100} + \frac{6I_2}{100} + \frac{2 + 6\epsilon_2}{N} + \mathfrak{T}_{1024,1024,142} < 0.039983.$$

All six exhaustive bounds are below $1/25 - 7/(25N)$, closing the interval in the original Hall balance. $\square$

The last descent uses a sharper elementary payment for the same prime tail. Exact rational summation over the primes through 10^5 , followed by the integral bound for all remaining integers, gives

$$\sum_{47 < p \leq 10^5} \frac{1}{p^2} < 0.003876557, \quad \sum_{n > 10^5} \frac{1}{n^2} < 10^{-5}.$$

Consequently

$$\sum_{p>47} \frac{1}{p^2} < 0.003887 =: \tau_{\sharp}.$$

Define $\mathfrak{T}_{H_1, H_2, k}^{\sharp}$ by the formula for $\mathfrak{T}$ with 0.004751 replaced by $\tau_{\sharp}$.

For the intermediate-prime count, Dusart proved the explicit estimate [4]

$$\pi(y) \leq \frac{y}{\log y - 1.1} \quad (y > 60184) \quad (\text{D})$$

If $Y = \lfloor N/k \rfloor$, then $N/(k+1) \leq Y \leq N/k$ in every use below. Consequently define

$$\begin{aligned} \mathfrak{T}_{H_1, H_2, k}^{D, \sharp}(N) &= \frac{4\tau_{\sharp}}{25} + \frac{4}{k(\log(N/(k+1)) - 1.1)} \\ &\quad + \frac{2(H_1 + H_2)}{N} \left(\frac{2(N/25 + 2)}{N/(k+1)} + \frac{N^2 + 25N + 1}{(N/(k+1))^2} \right). \end{aligned}$$

This is the same two-pivot union bound as $\mathfrak{T}^{\sharp}$, with only the prime-count payment replaced by (D). For the three values of k used below, already at the left endpoint $N/(k+1) > 6 \cdot 10^8$, so the hypothesis in (D) holds with ample room.

Theorem 6.38 (Short-block correlated descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 6.7 \cdot 10^{10}$.*

Proof. It remains after Theorem 6.37 to treat

$$6.7 \cdot 10^{10} \leq N < 1.001 \cdot 10^{11}.$$

Use the same decomposition $A^* = O \sqcup E_1 \sqcup E_2 \sqcup E_3$. Corollary 5.22 gives

$$|A^*| > 0.00127N. \quad (74)$$

At the left endpoint,

$$\begin{aligned} \tilde{\Delta}_{25} &< 0.002514345, & \tilde{\Delta}_{50} &< 0.002513892, & \tilde{\Delta}_{100} &< 0.002513665, \\ \tilde{\Delta}_W &< 0.004175714, & \tilde{\Delta}_8 &< 0.004175602, & \mathfrak{T}_{1024, 1024, 117}^{D, \sharp} &< 0.003267694, \\ \mathfrak{T}_{2048, 2048, 93}^{D, \sharp} &< 0.003933514, & \mathfrak{T}_{4096, 4096, 73}^{D, \sharp} &< 0.004768991. \end{aligned}$$

All envelopes decrease.

The first four allocation cases from Theorem 6.37, now using Lemma 4.6 and $\mathfrak{T}^{D, \sharp}$, give

case	E_1	E_2	E_3	two odd classes
$ A /N$	< 0.039995	< 0.037233	< 0.036123	< 0.034030 .

respectively. For clarity, these are the cases $|E_1| > 0.0012N$, then $|E_2| > 0.0012N$, then $|E_3| > 0.0012N$, and finally both odd classes occupied after all three even classes are sparse; their formulas are unchanged.

It remains that the odd part occupies at most one odd class. If $e > 10^{-5}N$, an even pair lies within 100000, at most two residue pairs coincide, and

$$\frac{|A|}{N} < \frac{69}{100}C_{\text{quad}} + \Delta_W + \frac{2I_2}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{4096, 4096, 73}^{D, \sharp} < 0.038243.$$

Otherwise (74) puts more than $0.00126N$ pivots in the single odd class. Two lie within 794. Their difference is divisible by 4; if two residue pairs coincided, then $(3 \cdot 7)^2$ would also divide it, forcing a difference of at least 1764. Thus at most one residue pair coincides, and

$$\frac{|A|}{N} < 10^{-5} + \frac{23}{100}C_{\text{quad}} + \tilde{\Delta}_{100} + \frac{2}{100} + \frac{6I_1}{100} + \frac{2+6\epsilon_2}{N} + \mathfrak{T}_{1024,1024,117}^{D,\sharp} < 0.038980.$$

All six bounds directly contradict the original Hall balance. $\square$

For the last short interval put

$$h_{\dagger} := \sum_{\substack{1 \leq a \leq 6293 \\ (a, 43890)=1}} \frac{1}{a} < 2.302594, \quad K_{\dagger}(R) = 2\eta_* R h_{\dagger} + 128\sqrt{R}.$$

This is again an exact finite rational inequality. For $6.45 \cdot 10^{10} \leq N \leq 6.7 \cdot 10^{10}$ define

$$\begin{aligned} \Delta_q^{\dagger} &= \tilde{\Delta}_q - \frac{3K_*(r_N+1)}{N} + \frac{3K_{\dagger}(r_N+1)}{N}, \\ \Delta_W^{\dagger} &= \tilde{\Delta}_W - \frac{6K_*(r_N+1)}{N} + \frac{6K_{\dagger}(r_N+1)}{N}, \\ \Delta_8^{\dagger} &= \tilde{\Delta}_8 - \frac{6K_*(r_N+1)}{N} + \frac{6K_{\dagger}(r_N+1)}{N}. \end{aligned}$$

Since $\sqrt{r_N+1} < 6293$ on this interval, the proof of Lemma 4.6 gives the same three diagonal conclusions with the daggered errors.

There is also a much smaller rounding budget for the one-coincidence finite sieve. For a vector $u = (u_p)_{p \in \mathcal{P}}$, write $e_j(u)$ for its j th elementary symmetric function. Put $x_p = p^{-2}$ and, for a possible coincident prime p_0 , put

$$z_p(p_0) = \begin{cases} p^{-2}, & p = p_0, \\ 2p^{-2}, & p \neq p_0. \end{cases}$$

Define

$$L_5 = \sum_{j=0}^5 (-1)^j e_j(x), \quad U_4(p_0) = \sum_{j=0}^4 (-1)^j e_j(z(p_0)).$$

Lemma 6.39 (Truncated one-coincidence sieve). *If at most one small-prime residue pair coincides, the two-form intersection density in one base progression is at most*

$$\widehat{I}_1 := 1 - 2L_5 + \max_{p_0 \in \mathcal{P}} U_4(p_0) < 0.112569312,$$

and its endpoint discrepancy is at most

$$\widehat{\epsilon}_1 = 18824.$$

Thus in two base classes the corresponding contribution is at most

$$\frac{2\widehat{I}_1}{25}N + 2\widehat{\epsilon}_1.$$

Proof. For either individual form, the degree-5 Bonferroni polynomial is a lower bound for avoiding all thirteen forbidden residues. For the union of the two residue sets at each prime, the degree-4 polynomial is an upper bound for avoiding them. Subtracting the two individual lower bounds from this joint upper bound proves the displayed density. Exact rational evaluation over the thirteen possible p_0 shows that the maximum occurs at $p_0 = 3$ and is below 0.112569312; the no-coincidence case is smaller.

The two individual degree-5 polynomials contain $\sum_{j=1}^5 \binom{13}{j} = 2379$ nonconstant residue counts each. In the zero-coincidence joint case, which has the largest endpoint count, the numbers of residue classes in degrees 0, 1, 2, 3, 4 are 1, 26, 312, 2288, 11440, so there are 14066 nonconstant counts. Charging one endpoint error to each gives the uniform budget

$$2(2379) + 14066 = 18824.$$

The progression has length at most $N/25 + 1$, and repeating it in the two base classes proves the last assertion. $\square$

Lemma 6.40 (Truncated two-coincidence sieve). *If at most two small-prime residue pairs coincide, the same truncation gives*

$$\widehat{I}_2 < 0.129756863, \quad \widehat{\epsilon}_2 = 18824.$$

In six classes modulo 100, the resulting contribution is at most

$$\frac{6\widehat{I}_2}{100}N + 6\widehat{\epsilon}_2.$$

Proof. Use the same individual degree-5 lower bounds as in Lemma 6.39. In the joint degree-4 upper bound let two local residue counts be 1 and the other eleven be 2. Exact rational evaluation over the possible coincident pairs is maximal for $\{3, 7\}$ and gives the displayed density. The joint numbers of residue classes in degrees 0, 1, 2, 3, 4 are

$$1, 24, 265, 1782, 8140.$$

Thus the exact-two-coincidence endpoint count is

$$2(2379) + (24 + 265 + 1782 + 8140) = 14969 < 14972.$$

For the stated “at most two” assertion, however, zero coincidences have the larger uniform endpoint budget 18824 from the preceding lemma. Repeating in the six base progressions modulo 100 proves the claim. $\square$

Theorem 6.41 (Bonferroni short-block descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 6.45 \cdot 10^{10}$.*

Proof. Theorem 6.38 leaves only

$$6.45 \cdot 10^{10} \leq N < 6.7 \cdot 10^{10}.$$

At the left endpoint the daggered diagonal errors and optimized tails satisfy

$$\begin{array}{lll} \Delta_{25}^\dagger < 0.002526953, & \Delta_{50}^\dagger < 0.002526489, & \Delta_{100}^\dagger < 0.002526257, \\ \Delta_W^\dagger < 0.004191286, & \Delta_8^\dagger < 0.004191171, & \mathfrak{T}_{1024,1024,116}^{D,\#} < 0.003304054, \\ \mathfrak{T}_{2048,2048,92}^{D,\#} < 0.003979107, & \mathfrak{T}_{4096,4096,72}^{D,\#} < 0.004826133. & \end{array}$$

All these envelopes decrease. In the first three allocation cases replace (I_1, ϵ_2) by $(\hat{I}_1, \hat{\epsilon}_1)$ using Lemma 6.39; in the fourth case retain the two-odd-class formula. Their respective bounds are

case	E_1	E_2	E_3	two odd classes
$ A /N$	< 0.039997	< 0.037246	< 0.036148	< 0.034083 .

It remains that the odd part occupies at most one odd class. If $e > 10^{-5}N$, the even-pair formula gives

$$\frac{|A|}{N} < \frac{69}{100}C_{\text{quad}} + \Delta_W^\dagger + \frac{2I_2}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{4096,4096,72}^{D,\#} < 0.038318.$$

Otherwise the mixed-strip bound leaves more than $0.00114N$ pivots in one odd class, so two lie within 878. As in Theorem 6.38, at most one residue pair coincides. Hence

$$\frac{|A|}{N} < 10^{-5} + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^\dagger + \frac{2}{100} + \frac{6\hat{I}_1}{100} + \frac{2+6\hat{\epsilon}_1}{N} + \mathfrak{T}_{1024,1024,116}^{D,\#} < 0.038886.$$

All six exhaustive bounds are strictly below $1/25 - 7/(25N)$, so they close directly in the original Hall balance. $\square$

The dense E_1 branch admits a further root-small-prime correlation. For an E_1 pivot b , let ρ_b denote the effective root count in its transformed $q_0 = 25$ progression.

Lemma 6.42 (At most one high-root valuation-one pivot). *Let $2.2827 \cdot 10^{10} \leq N \leq 6.45 \cdot 10^{10}$ and let distinct $b, c \in E_1$ satisfy $|b - c| \leq 834$. Then at most one of ρ_b, ρ_c exceeds 512.*

If, say, $\rho_b > 512$, at least nine primes of $\mathcal{P}$ divide b . Consequently at most four small-prime witness events remain active for that form. In a high-low pair the finite intersection density and endpoint error may be bounded by

$$I_{HL} < 0.001457, \quad \epsilon_{HL} = 50000.$$

Proof. Because $\nu_2(b) = 1$, the root formula has $c_1 = 1$. A root count above 512 therefore requires at least ten distinct odd prime factors in the effective modulus. At most one of them is the modulus prime 5, so at least nine distinct primes other than 5 divide b . If one were at least 53, then

$$b \geq 2 \cdot 53 \cdot 3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29 = 68578748238 > 6.45 \cdot 10^{10}.$$

Thus at least nine members of $\mathcal{P}$ divide b .

If both b and c had high root count, their two nine-element divisor sets inside the thirteen-element set $\mathcal{P}$ would intersect in at least five primes. Hence

$$3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 = 51051 \mid b - c,$$

contrary to $0 < |b - c| \leq 834$.

For the last assertion, enumerate the nine-element subsets $D \subseteq \mathcal{P}$ satisfying $2 \prod_{p \in D} p \leq 6.45 \cdot 10^{10}$. There are exactly eight. Delete those inactive events from the high form and retain all thirteen for the low form. Exact rational enumeration over these eight sets and the possible single coincident residue pair gives $I_{HL} < 0.001457$; the maximum has active set $\{29, 37, 41, 47\}$ and coincident prime 29. Full inclusion-exclusion has at most

$$(2^4 - 1) + (2^{13} - 1) + (3^4 2^9 - 1) = 49677 < 50000$$

nonconstant CRT residue counts, proving the endpoint bound. $\square$

It follows that the normalized small-prime-plus-large-tail contribution in the dense E_1 branch is at most the larger of

$$\frac{2\widehat{I}_1}{25} + \frac{2\widehat{\epsilon}_1}{N} + \mathfrak{T}_{512,512,k}^{D,\#}(N), \quad (75)$$

$$\frac{2I_{HL}}{25} + \frac{100000}{N} + \mathfrak{T}_{512,1024,k'}^{D,\#}(N). \quad (76)$$

Indeed the first line treats two low-root pivots, while Lemma 6.42 reduces every remaining pair to the second line.

Theorem 6.43 (Root–small-prime correlated descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 5.5 \cdot 10^{10}$.*

Proof. Theorem 6.41 leaves

$$5.5 \cdot 10^{10} \leq N < 6.45 \cdot 10^{10}.$$

At the left endpoint,

$$\begin{aligned} \Delta_{25}^\dagger &< 0.002661441, & \Delta_{50}^\dagger &< 0.002660928, & \Delta_{100}^\dagger &< 0.002660671, \\ \Delta_W^\dagger &< 0.004418674, & \Delta_8^\dagger &< 0.004418546, & \mathfrak{T}_{512,512,139}^{D,\#} &< 0.002891955, \\ \mathfrak{T}_{512,1024,121}^{D,\#} &< 0.003209719, & \mathfrak{T}_{1024,1024,110}^{D,\#} &< 0.003462020, \\ \mathfrak{T}_{2048,2048,87}^{D,\#} &< 0.004176882, & \mathfrak{T}_{4096,4096,69}^{D,\#} &< 0.005074210. \end{aligned}$$

All envelopes decrease. If $|E_1| > 0.0012N$, select the usual pair within 834 and use (75)–(76); both alternatives, including the diagonal term, give

$$\frac{|A|}{N} < 0.039999.$$

The next three allocation cases and the one-odd-class even-pair case give, respectively,

$$\frac{|A|}{N} < 0.037671, \quad 0.036624, \quad 0.034393, \quad 0.038802.$$

Finally suppose $e \leq 10^{-5}N$ and the odd part lies in one class modulo 4. The mixed-strip bound leaves more than $0.00063N$ odd pivots, so two lie within $1588 < 1764$. At most one small-prime residue pair therefore coincides, and Lemma 6.39 gives

$$\frac{|A|}{N} < 10^{-5} + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^\dagger + \frac{2}{100} + \frac{6\widehat{I}_1}{100} + \frac{2+6\widehat{\epsilon}_1}{N} + \mathfrak{T}_{1024,1024,110}^{D,\#} < 0.039179.$$

These six cases are exhaustive and each is strictly below $1/25 - 7/(25N)$, giving a direct contradiction in the original Hall cut. $\square$

The same correlation can be used when the last pivots are odd, but now a single prime above 47 is arithmetically possible. The resulting finite family can be enumerated exactly.

Lemma 6.44 (Odd high-root finite patterns). *Let $N \leq 4.7 \cdot 10^{10}$ and let b be odd with effective root count above 512. Then at least eight members of $\mathcal{P}$ divide b , so at most five small-prime events remain active. For a pair with at most two coincident residue pairs, exact finite enumeration gives*

$$I_{OL} < 0.004200 \quad \text{in the high–low case,} \quad I_{OO} < 0.004054 \quad \text{in the high–high case,}$$

with endpoint discrepancy at most 71000 in either case.

Proof. A high root count again requires nine non-5 prime divisors of b . Two of them cannot both exceed 47, since

$$53 \cdot 59 \cdot 3 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 = 69761140449 > 4.7 \cdot 10^{10}.$$

Thus at least eight lie in $\mathcal{P}$. Enumerate subsets $D \subseteq \mathcal{P}$ of size at least nine with $\prod_{p \in D} p \leq N$, and subsets of size eight with $53 \prod_{p \in D} p \leq N$. There are thirty possible active complements. Exact rational evaluation of the two-form local products over these complements and zero, one, or two coincident pairs gives the displayed bounds. The high–low maximum has active set $\{17, 41, 43, 47\}$ with coincidences $\{17, 41\}$; the high–high maximum has the same two active sets and coincidences. With at most five active events in a high form, full inclusion–exclusion uses at most

$$(2^5 - 1) + (2^{13} - 1) + (3^5 2^8 - 1) = 70429 < 71000$$

nonconstant CRT classes even against all thirteen events in the other form. $\square$

Theorem 6.45 (Two-sided root–small-prime descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 4.7 \cdot 10^{10}$.*

Proof. Theorem 6.43 leaves

$$4.7 \cdot 10^{10} \leq N < 5.5 \cdot 10^{10}.$$

At the left endpoint,

$$\begin{aligned} \Delta_{25}^\dagger &< 0.002801247, & \Delta_{50}^\dagger &< 0.002800681, & \Delta_{100}^\dagger &< 0.002800397, \\ \Delta_W^\dagger &< 0.004655324, & \Delta_8^\dagger &< 0.004655183, & \mathfrak{T}_{512,512,132}^{D,\#} &< 0.003023853, \\ \mathfrak{T}_{512,1024,115}^{D,\#} &< 0.003360097, & \mathfrak{T}_{1024,1024,105}^{D,\#} &< 0.003627133, \\ \mathfrak{T}_{2048,2048,83}^{D,\#} &< 0.004383725, & \mathfrak{T}_{4096,4096,65}^{D,\#} &< 0.005333451. \end{aligned}$$

All envelopes decrease. In the dense E_1 case, Lemma 6.42 and (75)–(76), now with the sharpened $I_{HL} < 0.001457$, give

$$\frac{|A|}{N} < 0.039991.$$

The next four allocations give

$$\frac{|A|}{N} < 0.038115, \quad 0.037120, \quad 0.034718, \quad 0.039308.$$

In the final single-odd-class case, the mixed strip leaves more than $0.00009N$ odd pivots, so two lie within 11112. Three coincident residue pairs would force

$$\text{lcm}(4, (3 \cdot 7 \cdot 11)^2) = 213444$$

to divide their difference; hence at most two coincide. If both roots are at most 512, use Lemma 6.40 and the $H = 512, 512$ tail. If one or both roots are high, use Lemma 6.44 with the corresponding $H = 512, 1024$ or $H = 1024, 1024$ tail. The low–low alternative is the largest and gives

$$\frac{|A|}{N} < 10^{-5} + \frac{23}{100} C_{\text{quad}} + \Delta_{100}^\dagger + \frac{2}{100} + \frac{6\widehat{I}_2}{100} + \frac{2 + 6\widehat{\epsilon}_2}{N} + \mathfrak{T}_{512,512,132}^{D,\#} < 0.039912.$$

Thus all six exhaustive allocations lie strictly below $1/25 - 7/(25N)$ and close directly in the original Hall balance. $\square$

The last rounding loss in the short block can itself be evaluated exactly. Put

$$h_{\dagger} := \sum_{\substack{1 \leq a \leq 5591 \\ (a, 43890)=1}} \frac{1}{a} < 2.279170, \quad K_{\dagger}(R) = 2\eta_* R h_{\dagger} + 6.744\sqrt{R}.$$

For $4.6 \cdot 10^{10} \leq N \leq 4.7 \cdot 10^{10}$ define

$$\begin{aligned} \Delta_q^{\dagger} &= \Delta_q^{(0)} - \frac{3K_0(r_N + 1)}{N} + \frac{3K_{\dagger}(r_N + 1)}{N} \quad (q = 25, 50, 100), \\ \Delta_W^{\dagger} &= \Delta_W - \frac{6K_0(r_N + 1)}{N} + \frac{6K_{\dagger}(r_N + 1)}{N}, \\ \Delta_8^{\dagger} &= \Delta_8^{(0)} - \frac{6K_0(r_N + 1)}{N} + \frac{6K_{\dagger}(r_N + 1)}{N}. \end{aligned}$$

Lemma 6.46 (Exact short-block discrepancy). *On this interval the three aggregate diagonal conclusions hold with the double-daggered errors. These envelopes decrease. At the left endpoint,*

$$\begin{aligned} \Delta_{25}^{\dagger} &< 0.002758521, \quad \Delta_{50}^{\dagger} < 0.002757946, \quad \Delta_{100}^{\dagger} < 0.002757658, \\ \Delta_W^{\dagger} &< 0.004563837, \quad \Delta_8^{\dagger} < 0.004563695. \end{aligned}$$

Proof. Write $C(y) = \#\{a \leq y : (a, 43890) = 1\}$. The exact finite recurrence

$$C(y) = C(y-1) + \mathbf{1}_{(y, 43890)=1}$$

over $0 \leq y \leq 5591$ gives

$$\max_{0 \leq y \leq 5591} \left(C(y) - \frac{288}{1463}y \right) = 828 - \frac{288}{1463}4189 = \frac{4932}{1463} < 3.372.$$

The harmonic inequality defining $h_{\dagger}$ is likewise an exact rational summation. Since $\sqrt{r_N + 1} < 5591$ throughout the interval, the hyperbola step in Lemma 4.6 now gives

$$\sum_{\substack{s \leq R \\ (s, 43890)=1}} d(s) \leq 2\eta_* R h_{\dagger} + 2(3.372)\sqrt{R} = K_{\dagger}(R).$$

Substitution in the three diagonal arguments proves the formulas. Elementary differentiation gives monotonicity, and direct left-endpoint evaluation gives the displayed strict bounds. $\square$

Theorem 6.47 (Exact-discrepancy descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 4.6 \cdot 10^{10}$.*

Proof. Theorem 6.45 leaves

$$4.6 \cdot 10^{10} \leq N < 4.7 \cdot 10^{10}.$$

At the left endpoint, optimizing the same integer split parameter gives

$$\begin{aligned} \mathfrak{Z}_{512, 512, 131}^{D, \#} &< 0.003042501, \quad \mathfrak{Z}_{512, 1024, 115}^{D, \#} < 0.003381367, \quad \mathfrak{Z}_{1024, 1024, 104}^{D, \#} < 0.003650423, \\ \mathfrak{Z}_{2048, 2048, 82}^{D, \#} &< 0.004412878, \quad \mathfrak{Z}_{4096, 4096, 65}^{D, \#} < 0.005370064. \end{aligned}$$

All terms decrease. Use Lemma 6.46 in the same six exhaustive allocations as Theorem 6.45. The first five endpoint bounds are, respectively,

$$\frac{|A|}{N} < 0.039967, \quad 0.0382, \quad 0.0372, \quad 0.0349, \quad 0.0394.$$

For completeness, the matching strip now satisfies $\mathcal{R}(N) > 0.0000268N$. In the final branch $e \leq 10^{-5}N$, it leaves more than $0.0000168N$ odd pivots in one class modulo 4, so two lie within 59524. Three coincident residue pairs would again force 213444 to divide their nonzero difference. Hence at most two coincide, and the low-low alternative remains the largest. Lemma 6.40 and the $H = 512, 512$ tail give

$$\frac{|A|}{N} < 10^{-5} + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^{\dagger} + \frac{2}{100} + \frac{6\hat{I}_2}{100} + \frac{2 + 6\hat{\epsilon}_2}{N} + \mathfrak{T}_{512,512,131}^{D,\#} < 0.039888.$$

Every bound is strictly below $1/25 - 7/(25N)$, giving a direct contradiction inside the original Hall cut. $\square$

One more short block is available on the interval just below this endpoint. Put

$$h_{\natural} := \sum_{\substack{1 \leq a \leq 5551 \\ (a, 43890)=1}} \frac{1}{a} < 2.277735, \quad K_{\natural}(R) = 2\eta_* R h_{\natural} + 6.744\sqrt{R},$$

and for $4.5715 \cdot 10^{10} \leq N \leq 4.6 \cdot 10^{10}$ define

$$\begin{aligned} \Delta_q^{\natural} &= \Delta_q^{(0)} - \frac{3K_0(r_N + 1)}{N} + \frac{3K_{\natural}(r_N + 1)}{N} \quad (q = 25, 50, 100), \\ \Delta_W^{\natural} &= \Delta_W - \frac{6K_0(r_N + 1)}{N} + \frac{6K_{\natural}(r_N + 1)}{N}, \\ \Delta_8^{\natural} &= \Delta_8^{(0)} - \frac{6K_0(r_N + 1)}{N} + \frac{6K_{\natural}(r_N + 1)}{N}. \end{aligned}$$

The proof of Lemma 6.46 applies unchanged: $\sqrt{r_N + 1} < 5551$, while the exact maximum coprime-count discrepancy is still attained at 4189. The naturalled envelopes decrease, and at the left endpoint

$$\begin{aligned} \Delta_{25}^{\natural} &< 0.002762881, \quad \Delta_{50}^{\natural} < 0.002762304, \quad \Delta_{100}^{\natural} < 0.002762016, \\ \Delta_W^{\natural} &< 0.004570809, \quad \Delta_8^{\natural} < 0.004570666. \end{aligned}$$

Theorem 6.48 (Vanishing-strip descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 4.5715 \cdot 10^{10}$.*

Proof. Theorem 6.47 leaves

$$4.5715 \cdot 10^{10} \leq N < 4.6 \cdot 10^{10}.$$

At the left endpoint the optimized tails are

$$\begin{aligned} \mathfrak{T}_{512,512,131}^{D,\#} &< 0.003047890, \quad \mathfrak{T}_{512,1024,114}^{D,\#} < 0.003387517, \quad \mathfrak{T}_{1024,1024,104}^{D,\#} < 0.003657224, \\ \mathfrak{T}_{2048,2048,82}^{D,\#} &< 0.004421371, \quad \mathfrak{T}_{4096,4096,65}^{D,\#} < 0.005380780. \end{aligned}$$

The first four allocations from Theorem 6.45, with the naturalled diagonal errors, are respectively below

$$0.039977, \quad 0.0382, \quad 0.0372, \quad 0.0349.$$

It remains to split according to the total even mass e . If $e > 0.000000056N$, two even pivots lie within 17857143. Four coincident small-prime pairs would force

$$2(3 \cdot 7 \cdot 11 \cdot 13)^2 = 18036018$$

to divide their nonzero difference. Thus at most three pairs coincide. The full finite product $I_3 < 0.136690$, its endpoint budget $\epsilon_2 = 1610704$, and the $H = 4096, 4096$ tail give

$$\frac{|A|}{N} < \frac{69}{100}C_{\text{quad}} + \Delta_W^{\natural} + \frac{2I_3}{25} + \frac{2\epsilon_2}{N} + \mathfrak{T}_{4096,4096,65}^{D,\sharp} < 0.039827.$$

Otherwise the mixed strip $\mathcal{R}(N) > 0.0000047808N$ leaves more than $0.0000047248N$ odd pivots in one class modulo 4. Two lie within 211650, which is below

$$4(3 \cdot 7 \cdot 11)^2 = 213444.$$

Hence at most two pairs coincide. The low-low alternative is again largest, and gives

$$\frac{|A|}{N} < 0.000000056 + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^{\natural} + \frac{2}{100} + \frac{6\widehat{I}_2}{100} + \frac{2 + 6\widehat{\epsilon}_2}{N} + \mathfrak{T}_{512,512,131}^{D,\sharp} < 0.039888.$$

All six exhaustive bounds are strictly below $1/25 - 7/(25N)$ and therefore close directly in the original Hall balance. $\square$

We now take the first of those two terminal alternatives: enlarge the finite prime window while subtracting the added primes from the same explicit tail. Let

$$\mathcal{P}_{113} = \{p : 3 \leq p \leq 113, p \neq 5\}, \quad |\mathcal{P}_{113}| = 28.$$

Define the degree-5 individual and degree-4 joint Bonferroni polynomials exactly as before, but over $\mathcal{P}_{113}$.

Lemma 6.49 (Extended finite Bonferroni window). *For at most r coincident residue pairs, $r = 1, 3, 4$, the corresponding two-form intersection densities satisfy*

$$\widehat{I}_1^{113} < 0.112772853, \quad \widehat{I}_3^{113} < 0.136772805, \quad \widehat{I}_4^{113} < 0.141729589.$$

A single endpoint discrepancy valid uniformly for all three assertions is

$$\widehat{\epsilon}_{113} = 600250.$$

Moreover

$$\sum_{p > 113} \frac{1}{p^2} < 0.001386157 =: \tau_{113}.$$

Proof. Exact rational evaluation of the Bonferroni polynomials gives the displayed densities; for each r the maximum occurs when the coincident primes are the r smallest members of $\mathcal{P}_{113}$. Each individual degree-5 polynomial has

$$\sum_{j=1}^5 \binom{28}{j} = 122437$$

nonconstant counts. The zero-coincidence joint polynomial, which maximizes the endpoint count, has degree counts

$$56, \quad 1512, \quad 26208, \quad 327600.$$

Thus the uniform total is

$$2(122437) + 56 + 1512 + 26208 + 327600 = 600250.$$

Finally subtract the exact rational sum $\sum_{47 < p \leq 113} p^{-2}$ from the proved bound $\sum_{p > 47} p^{-2} < 0.003887$ to obtain τ_{113} . $\square$

Define $\mathfrak{T}_{H_1, H_2, k}^{D, 113}$ by replacing $\tau_{\sharp}$ with τ_{113} in $\mathfrak{T}_{H_1, H_2, k}^{D, \sharp}$. For the last short block put

$$h_{\flat} := \sum_{\substack{1 \leq a \leq 5540 \\ (a, 43890)=1}} \frac{1}{a} < 2.277374, \quad K_{\flat}(R) = 2\eta_* R h_{\flat} + 6.744\sqrt{R},$$

and define $\Delta_q^{\flat}, \Delta_W^{\flat}, \Delta_8^{\flat}$ by replacing K_0 with $K_{\flat}$ in the same manner as above. On $4.5654 \cdot 10^{10} \leq N \leq 4.5715 \cdot 10^{10}$ the exact short-block proof applies, and at the left endpoint

$$\begin{aligned} \Delta_{25}^{\flat} &< 0.002763777, & \Delta_{50}^{\flat} &< 0.002763200, & \Delta_{100}^{\flat} &< 0.002762911, \\ \Delta_W^{\flat} &< 0.004572225, & \Delta_8^{\flat} &< 0.004572081. \end{aligned}$$

Theorem 6.50 (Extended-window descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 4.5654 \cdot 10^{10}$.*

Proof. Theorem 6.48 leaves

$$4.5654 \cdot 10^{10} \leq N < 4.5715 \cdot 10^{10}.$$

At the left endpoint,

$$\begin{aligned} \mathfrak{T}_{512, 512, 131}^{D, 113} &< 0.002648917, & \mathfrak{T}_{512, 1024, 114}^{D, 113} &< 0.002988705, & \mathfrak{T}_{1024, 1024, 104}^{D, 113} &< 0.003258556, \\ \mathfrak{T}_{2048, 2048, 82}^{D, 113} &< 0.004023067, & \mathfrak{T}_{4096, 4096, 65}^{D, 113} &< 0.004982955. \end{aligned}$$

Use Lemma 6.49 in the low-low alternative. The high-root alternatives remain smaller. The first four exhaustive allocations are below

$$0.039620, \quad 0.0382, \quad 0.0372, \quad 0.0349.$$

Split at $e = 0.0000000002N$. Above that cutoff, two even pivots lie within 5000000000 . Five coincidences would force

$$2(3 \cdot 7 \cdot 11 \cdot 13 \cdot 17)^2 = 5212409202$$

to divide their difference, so at most four coincide. Hence

$$\frac{|A|}{N} < \frac{69}{100} C_{\text{quad}} + \Delta_W^{\flat} + \frac{2\hat{I}_4^{113}}{25} + \frac{2\hat{\epsilon}_{113}}{N} + \mathfrak{T}_{4096, 4096, 65}^{D, 113} < 0.039790.$$

Below the cutoff, the directly evaluated strip $\mathcal{R}(N) > 0.00000003669N$ leaves more than $0.00000003649N$ odd pivots in one class modulo 4. Two lie within 27405000 , below

$$4(3 \cdot 7 \cdot 11 \cdot 13)^2 = 36072036.$$

Thus at most three pairs coincide, and

$$\frac{|A|}{N} < 0.0000000002 + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^b + \frac{2}{100} + \frac{6\widehat{I}_3^{113}}{100} + \frac{2 + 6\widehat{\epsilon}_{113}}{N} + \mathfrak{T}_{512,512,131}^{D,113} < 0.039987.$$

Every one of the six bounds is strictly below $1/25 - 7/(25N)$, closing the original Hall balance. $\square$

For the longer descent use

$$\mathcal{P}_{131} = \{p : 3 \leq p \leq 131, p \neq 5\}, \quad |\mathcal{P}_{131}| = 30.$$

Lemma 6.51 (Thirty-prime Bonferroni window). *The degree-5/degree-4 truncation over $\mathcal{P}_{131}$ gives*

$$\widehat{I}_1^{131} < 0.112782897, \quad \widehat{I}_2^{131} < 0.129880546, \quad \widehat{\epsilon}_{131} = 821632,$$

and the remaining prime-square tail satisfies

$$\sum_{p>131} \frac{1}{p^2} < 0.001265886 =: \tau_{131}.$$

Proof. Exact rational evaluation is maximal at coincidence sets $\{3\}$ and $\{3, 7\}$, respectively. Each individual polynomial now has $\sum_{j=1}^5 \binom{30}{j} = 174436$ nonconstant counts. The zero-coincidence joint degree counts are

$$60, \quad 1740, \quad 32480, \quad 438480,$$

so the uniform endpoint total is

$$2(174436) + 60 + 1740 + 32480 + 438480 = 821632.$$

Exact subtraction of $\sum_{47 < p \leq 131} p^{-2}$ from 0.003887 proves the tail bound. $\square$

Define $\mathfrak{T}^{D,131}$ by using τ_{131} in the same tail formula. The flat short-block proof remains valid on $3.71 \cdot 10^{10} \leq N \leq 4.5654 \cdot 10^{10}$. At the new left endpoint,

$$\begin{aligned} \Delta_{25}^b &< 0.002953746, & \Delta_{50}^b &< 0.002953088, & \Delta_{100}^b &< 0.002952758, \\ \Delta_W^b &< 0.004891857, & \Delta_8^b &< 0.004891694, \end{aligned}$$

and

$$\begin{aligned} \mathfrak{T}_{512,512,123}^{D,131} &< 0.002817694, & \mathfrak{T}_{512,1024,107}^{D,131} &< 0.003183824, & \mathfrak{T}_{1024,1024,97}^{D,131} &< 0.003474609, \\ \mathfrak{T}_{2048,2048,77}^{D,131} &< 0.004298780, & \mathfrak{T}_{4096,4096,60}^{D,131} &< 0.005333610. \end{aligned}$$

Theorem 6.52 (Half-root descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 3.71 \cdot 10^{10}$.*

Proof. Theorem 6.50 leaves

$$3.71 \cdot 10^{10} \leq N < 4.5654 \cdot 10^{10}.$$

Use Proposition 5.4, the flat diagonal errors, and Lemma 6.51. The first four allocation bounds are

$$\frac{|A|}{N} < 0.039998, \quad 0.038327, \quad 0.037417, \quad 0.034754.$$

If $e > 10^{-5}N$, two even pivots lie within 100000. Three coincidences would force

$$2(3 \cdot 7 \cdot 11)^2 = 106722$$

to divide their difference, so at most two coincide. Therefore

$$\frac{|A|}{N} < \frac{69}{100}C_{\text{quad}} + \Delta_W^b + \frac{2\widehat{I}_2^{131}}{25} + \frac{2\widehat{\epsilon}_{131}}{N} + \mathfrak{T}_{4096,4096,60}^{D,131} < 0.039530.$$

Otherwise $\mathcal{R}(N) > 0.001593N$ leaves more than $0.001583N$ odd pivots in one class modulo 4, so two lie within 632. Two coincidences would force $4(3 \cdot 7)^2 = 1764$ to divide their difference. Thus at most one coincides, and

$$\frac{|A|}{N} < 10^{-5} + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^b + \frac{2}{100} + \frac{6\widehat{I}_1^{131}}{100} + \frac{2 + 6\widehat{\epsilon}_{131}}{N} + \mathfrak{T}_{512,512,123}^{D,131} < 0.038971.$$

All six bounds decrease and lie strictly below $1/25 - 7/(25N)$, closing directly in the original Hall balance. $\square$

We next sharpen the first two of those terms.

Lemma 6.53 (Three-term prime count and integrated square tail). *For $y \geq 355991$, Dusart's explicit estimate gives*

$$\pi(y) \leq \frac{y}{\log y} \left(1 + \frac{1}{\log y} + \frac{2.51}{\log^2 y} \right).$$

Moreover

$$\sum_{p > 131} \frac{1}{p^2} < 0.001257623 =: \tau_{131}^+.$$

Consequently, if $L = \log(N/(k+1))$, the intermediate-prime payment in the two-pivot tail is at most

$$\frac{4}{k} \left(\frac{1}{L} + \frac{1}{L^2} + \frac{2.51}{L^3} \right).$$

Proof. The displayed prime-count estimate is Theorem 6.9 of [4]. Exact rational summation gives

$$\sum_{131 < p \leq 10^5} \frac{1}{p^2} < 0.0012554421.$$

Partial summation and the classical bound $\pi(t) < 1.25506t/\log t$ give

$$\sum_{p > 10^5} \frac{1}{p^2} \leq 2(1.25506) \int_{10^5}^{\infty} \frac{dt}{t^2 \log t} < \frac{2(1.25506)}{10^5 \log(10^5)} < 0.0000021803.$$

This proves the square-tail constant. Finally $N/(k+1) \leq \lfloor N/k \rfloor \leq N/k$; monotonicity of the displayed prime-count majorant gives the normalized payment. $\square$

Define $\mathfrak{T}_{H_1, H_2, k}^{+, 131}$ by

$$\begin{aligned} \mathfrak{T}_{H_1, H_2, k}^{+, 131}(N) &= \frac{4\tau_{131}^+}{25} + \frac{4}{k} \left(\frac{1}{L} + \frac{1}{L^2} + \frac{2.51}{L^3} \right) \\ &\quad + \frac{2(H_1 + H_2)}{N} \left(\frac{2(N/25 + 2)}{N/(k+1)} + \frac{N^2 + 25N + 1}{(N/(k+1))^2} \right), \end{aligned}$$

where $L = \log(N/(k+1))$.

The remaining diagonal divisor sum can be evaluated rather than enveloped. Put

$$\mathcal{S}(R) = \sum_{\substack{s \leq R \\ (s, 43890)=1}} d(s), \quad C(y) = \#\{a \leq y : (a, 43890) = 1\}.$$

Lemma 6.54 (Exact hyperbola certificate). *For $3.51 \cdot 10^{10} \leq N \leq 3.71 \cdot 10^{10}$,*

$$\frac{\mathcal{S}(r_N + 1)}{N} < 0.000623885.$$

Hence the three aggregate diagonal estimates hold with

$$\begin{aligned} \Delta_q^{\text{ex}}(N) &= \frac{23}{N} + 3(0.000623885) + \frac{46(\log(r_N + 1) + 2)}{qr_N} + \frac{N^2 + 1}{Nr_N^2}(1 + \log_{2+\sqrt{3}} N), \\ \Delta_W^{\text{ex}}(N) &= \frac{69}{N} + 6(0.000623885) + \frac{69(\log(r_N + 1) + 2)}{50r_N} + \frac{N^2 + 1}{Nr_N^2}(1 + \log_{2+\sqrt{3}} N), \\ \Delta_8^{\text{ex}}(N) &= \frac{115}{N} + 6(0.000623885) + \frac{23(\log(r_N + 1) + 2)}{20r_N} + \frac{N^2 + 1}{Nr_N^2}(1 + \log_{2+\sqrt{3}} N). \end{aligned}$$

At the left endpoint these are below, respectively,

$$0.002903682, \quad 0.002903000, \quad 0.002902659, \quad 0.004774997, \quad 0.004774828.$$

Proof. The exact hyperbola identity is

$$\mathcal{S}(R) = 2 \sum_{\substack{a \leq \sqrt{R} \\ (a, 43890)=1}} C(\lfloor R/a \rfloor) - C(\lfloor \sqrt{R} \rfloor)^2,$$

where

$$C(y) = 8640 \left\lfloor \frac{y}{43890} \right\rfloor + C(y \bmod 43890).$$

On the stated interval, $\lfloor r_N + 1 \rfloor$ runs from 25728590 to 26696881. Divide this range into consecutive blocks of length 1000. On each block use the exact value of $\mathcal{S}$ at the upper endpoint and the least integer N mapping into the block. The maximum occurs on the first block and equals

$$\frac{21898353}{35100000000} < 0.000623885.$$

This finite calculation uses only the displayed identity and the 43890 residue prefix. Substitution proves the diagonal formulas; all remaining terms decrease, and direct endpoint evaluation gives the five numbers. $\square$

Theorem 6.55 (Exact-diagonal descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 3.51 \cdot 10^{10}$.*

Proof. Theorem 6.52 leaves

$$3.51 \cdot 10^{10} \leq N < 3.71 \cdot 10^{10}.$$

At the left endpoint the optimized new tails are

$$\begin{aligned} \mathfrak{T}_{512,512,120}^{+,131} &< 0.002865809, & \mathfrak{T}_{512,1024,105}^{+,131} &< 0.003238848, & \mathfrak{T}_{1024,1024,95}^{+,131} &< 0.003535139, \\ \mathfrak{T}_{2048,2048,75}^{+,131} &< 0.004374804, & \mathfrak{T}_{4096,4096,59}^{+,131} &< 0.005429329. \end{aligned}$$

Use Lemmas 6.51, 6.53, and 6.54. The first four allocations give

$$\frac{|A|}{N} < 0.039998, \quad 0.038289, \quad 0.037398, \quad 0.034775.$$

If $e > 10^{-5}N$, the even pair again has at most two coincidences and gives

$$\frac{|A|}{N} < \frac{69}{100}C_{\text{quad}} + \Delta_W^{\text{ex}} + \frac{2\hat{I}_2^{131}}{25} + \frac{2\hat{\epsilon}_{131}}{N} + \mathfrak{T}_{4096,4096,59}^{+,131} < 0.039511.$$

Otherwise $\mathcal{R}(N) > 0.001424N$ leaves more than $0.001414N$ odd pivots in one class modulo 4, so two lie within $707 < 1764$ and at most one pair coincides. Thus

$$\frac{|A|}{N} < 10^{-5} + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^{\text{ex}} + \frac{2}{100} + \frac{6\hat{I}_1^{131}}{100} + \frac{2 + 6\hat{\epsilon}_{131}}{N} + \mathfrak{T}_{512,512,120}^{+,131} < 0.038976.$$

All six bounds are strictly below $1/25 - 7/(25N)$ and close the original Hall balance directly. $\square$

Let

$$\mathcal{P}_{137} = \{p : 3 \leq p \leq 137, p \neq 5\}, \quad |\mathcal{P}_{137}| = 31.$$

Define the individual degree-5 lower polynomial and the joint degree-4 upper polynomial as in Lemma 6.39.

Lemma 6.56 (Conditional endpoint certificate at 137). *If no residue pair coincides, the two-form intersection density is below*

$$\hat{I}_0^{137} < 0.012349183$$

and its endpoint discrepancy is at most 954056. If exactly one pair coincides, the corresponding bounds are

$$\hat{I}_1^{137} < 0.112787353, \quad \hat{\epsilon}_1^{137} = 919775.$$

Consequently, whenever at most one pair coincides and $N \geq 3.4984 \cdot 10^{10}$, the contribution from the two base progressions is at most

$$\frac{2\hat{I}_1^{137}}{25}N + 2\hat{\epsilon}_1^{137}.$$

Moreover

$$\sum_{p>137} \frac{1}{p^2} < 0.001204343 =: \tau_{137}^+.$$

Proof. Exact rational evaluation of $1 - 2L_5 + U_4$ gives the first two density bounds; in the one-coincidence case the maximum is attained at the prime 3. Each individual polynomial has

$$\sum_{j=1}^5 \binom{31}{j} = 206367$$

nonconstant counts. The joint degree counts are

$$62, 1860, 35960, 503440$$

with no coincidence and

$$61, 1800, 34220, 470960$$

with exactly one. Hence the two endpoint totals are respectively 954056 and 919775. At the least allowed N , the two normalized alternatives are below

$$\frac{2\hat{I}_0^{137}}{25} + \frac{2(954056)}{N} < 0.001042478, \quad \frac{2\hat{I}_1^{137}}{25} + \frac{2(919775)}{N} < 0.009075571.$$

Both endpoint terms decrease, so the second alternative dominates on the whole range. This proves the combined assertion while retaining the larger endpoint count in the zero-coincidence calculation.

Finally, exact rational summation gives

$$\sum_{137 < p \leq 10^5} \frac{1}{p^2} < 0.0012021627.$$

The integrated tail in Lemma 6.53 is below 0.0000021803; their strict sum is below 0.001204343. $\square$

Lemma 6.57 (Exact bottom short-interval blocks). *For*

$$3.4984 \cdot 10^{10} \leq N \leq 3.51 \cdot 10^{10}$$

one has

$$\frac{\mathcal{S}(\lfloor r_N + 1 \rfloor)}{N} < 0.000624485942, \quad \frac{4\pi(\lfloor N/120 \rfloor)}{N} < 0.001808618752.$$

Thus the exact diagonal estimates hold with

$$\begin{aligned} \Delta_q^{\text{ex},0}(N) &= \frac{23}{N} + 3(0.000624485942) + \frac{46(\log(r_N + 1) + 2)}{qr_N} + \frac{N^2 + 1}{Nr_N^2}(1 + \log_{2+\sqrt{3}} N), \\ \Delta_W^{\text{ex},0}(N) &= \frac{69}{N} + 6(0.000624485942) + \frac{69(\log(r_N + 1) + 2)}{50r_N} + \frac{N^2 + 1}{Nr_N^2}(1 + \log_{2+\sqrt{3}} N), \\ \Delta_8^{\text{ex},0}(N) &= \frac{115}{N} + 6(0.000624485942) + \frac{23(\log(r_N + 1) + 2)}{20r_N} + \frac{N^2 + 1}{Nr_N^2}(1 + \log_{2+\sqrt{3}} N). \end{aligned}$$

At the left endpoint the five relevant values are below

$$0.002906492, \quad 0.002905809, \quad 0.002905468, \quad 0.004779610, \quad 0.004779440.$$

Proof. Here $\lfloor r_N + 1 \rfloor$ runs from 25671873 to 25728590. Partition that integer interval into blocks of length 10 and use the exact hyperbola identity of Lemma 6.54 at each upper endpoint. The largest normalized value is

$$\frac{\mathcal{S}(25671892)}{34984018923} = \frac{21847028}{34984018923} < 0.000624485942.$$

For the prime block, $Y = \lfloor N/120 \rfloor$ runs from 291533333 to 292500000. An Eratosthenes sieve gives

$$\pi(291533332) = 15818171, \quad \pi(292500000) - \pi(291533332) = 49764.$$

Partition the Y -interval into blocks of length 10, using the prime count at the upper endpoint and the least N mapping into the block. The maximum is

$$\frac{4\pi(291534732)}{34984166760} = \frac{4(15818255)}{34984166760} < 0.001808618752.$$

Both scans are finite integer certificates. Substitution in the aggregate diagonal formulas gives the remaining assertions. $\square$

On this short interval put

$$\begin{aligned} \mathfrak{T}_{512,512,120}^{\text{pc},137}(N) &= \frac{4\tau_{137}^+}{25} + 0.001808618752 \\ &\quad + \frac{2048}{N} \left(\frac{2(N/25 + 2)}{N/121} + \frac{N^2 + 25N + 1}{(N/121)^2} \right). \end{aligned}$$

Lemma 6.57 and the original two-pivot tail proof show that this is a valid upper bound; at the left endpoint it is below 0.002858980.

Theorem 6.58 (Conditional-endpoint descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 3.4984 \cdot 10^{10}$.*

Proof. Theorem 6.55 leaves

$$3.4984 \cdot 10^{10} \leq N < 3.51 \cdot 10^{10}.$$

At the left endpoint Proposition 5.4 gives a matching margin greater than $2.4 \cdot 10^7$, and Corollary 5.22 gives $\mathcal{R}(N) > 0.001414N$. The first four exhaustive allocations, using Lemma 6.57, are below

$$0.039999998, \quad 0.038299, \quad 0.037409, \quad 0.034782.$$

For the dense E_1 branch, two pivots lie within 834, hence at most one small-prime residue pair coincides. The low-low alternative is

$$\begin{aligned} \frac{|A|}{N} &< \frac{23}{25}C_{\text{quad}} + \Delta_{25}^{\text{ex},0} + \frac{2\hat{I}_1^{137}}{25} + \frac{2\hat{\epsilon}_1^{137}}{N} + \mathfrak{T}_{512,512,120}^{\text{pc},137} \\ &< 0.039999998. \end{aligned}$$

The high-low alternative from Lemma 6.42, with its original cutoff-47 tail with $k = 105$, is below 0.031852 and is therefore smaller.

If $e > 10^{-5}N$, the even pair has at most two coincidences. Keeping the cutoff-131 bounds gives

$$\frac{|A|}{N} < \frac{69}{100}C_{\text{quad}} + \Delta_W^{\text{ex},0} + \frac{2\hat{I}_2^{131}}{25} + \frac{2\hat{\epsilon}_{131}}{N} + \mathfrak{T}_{4096,4096,59}^{+,131} < 0.039522.$$

Otherwise more than $0.001404N$ odd pivots remain in one class modulo 4. Two lie within $713 < 1764$, so at most one pair coincides, and

$$\frac{|A|}{N} < 10^{-5} + \frac{23}{100}C_{\text{quad}} + \Delta_{100}^{\text{ex},0} + \frac{2}{100} + \frac{6\hat{I}_1^{131}}{100} + \frac{2 + 6\hat{\epsilon}_{131}}{N} + \mathfrak{T}_{512,512,120}^{+,131} < 0.038983.$$

Every term not covered by a uniform finite block decreases on this interval. All six bounds are therefore strictly below $1/25 - 7/(25N)$ and directly close the original Hall balance. $\square$

Let

$$\mathcal{P}_{509} = \{p : 3 \leq p \leq 509, p \neq 5\}, \quad |\mathcal{P}_{509}| = 95.$$

Lemma 6.59 (Conditional degree-3/2 certificate at 509). *Use the degree-3 individual lower polynomial and degree-2 joint upper polynomial on $\mathcal{P}_{509}$. With no coincidence the intersection density and endpoint discrepancy are bounded by*

$$\widehat{J}_0^{509} < 0.013594002, \quad \zeta_0^{509} = 304000.$$

With exactly one coincidence they are bounded by

$$\widehat{J}_1^{509} < 0.113400471, \quad \zeta_1^{509} = 303811.$$

For $N \geq 3.2532 \cdot 10^{10}$ and at most one coincidence, the contribution from both base progressions is therefore at most

$$\frac{2\widehat{J}_1^{509}}{25}N + 2\zeta_1^{509}.$$

The remaining square tail satisfies

$$\sum_{p>509} \frac{1}{p^2} < 0.000266795 =: \tau_{509}^+.$$

Proof. Exact rational evaluation of $1 - 2L_3 + U_2$ gives the displayed densities; the exactly-one maximum again occurs at 3. Each individual polynomial has

$$\sum_{j=1}^3 \binom{95}{j} = 142975$$

nonconstant counts. The zero-coincidence joint degree counts are 190, 17860, while the exactly-one counts are 189, 17672. Hence

$$2(142975) + 190 + 17860 = 304000, \quad 2(142975) + 189 + 17672 = 303811.$$

At the least N , the two normalized alternatives are below 0.001106210 and 0.009090716, respectively; the latter dominates thereafter. Finally exact rational summation gives

$$\sum_{509 < p \leq 10^5} \frac{1}{p^2} < 0.0002646140.$$

Adding the strict integrated tail 0.0000021803 proves the last assertion. $\square$

Lemma 6.60 (Exact 509-window interval blocks). *For*

$$3.2532 \cdot 10^{10} \leq N \leq 3.4984 \cdot 10^{10}$$

one has

$$\frac{\mathcal{S}(\lfloor r_N + 1 \rfloor)}{N} < 0.000638391308, \quad \frac{4\pi(\lfloor N/117 \rfloor)}{N} < 0.001859823072.$$

Write $\Delta_q^{\text{ex},509}, \Delta_W^{\text{ex},509}, \Delta_8^{\text{ex},509}$ for the exact diagonal formulas with 0.000638392 in place of the normalized divisor sum. Their five left-endpoint errors are below

$$0.002970568, \quad 0.002969853, \quad 0.002969495, \quad 0.004885388, \quad 0.004885211.$$

Proof. The integer $\lfloor r_N + 1 \rfloor$ runs from 24457858 to 25671873. Using length-100 blocks and the exact hyperbola identity, the maximum is

$$\frac{S(24457957)}{32532000000} = \frac{20768146}{32532000000} < 0.000638391308.$$

For $Y = \lfloor N/117 \rfloor$, the range is $278051282 \leq Y \leq 299008547$. The exact sieve gives

$$\pi(278051281) = 15125934, \quad \pi(299008547) - \pi(278051281) = 1075477.$$

The same length-100 scan is maximal at the block ending at 278052281:

$$\frac{4\pi(278052281)}{32532105294} = \frac{4(15125990)}{32532105294} < 0.001859823072.$$

Substitution proves the five diagonal bounds. $\square$

On this interval define

$$\begin{aligned} \mathfrak{T}_{512,512,117}^{\text{pc},509}(N) &= \frac{4\tau_{509}^+}{25} + 0.001859823072 \\ &\quad + \frac{2048}{N} \left(\frac{2(N/25 + 2)}{N/118} + \frac{N^2 + 25N + 1}{(N/118)^2} \right). \end{aligned}$$

This is below 0.002779668 at the left endpoint and decreases apart from the already uniform prime block.

Theorem 6.61 (Order-optimized conditional descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 3.2532 \cdot 10^{10}$.*

Proof. Theorem 6.58 leaves

$$3.2532 \cdot 10^{10} \leq N < 3.4984 \cdot 10^{10}.$$

At the left endpoint the matching margin exceeds $1.9 \cdot 10^7$ and $\mathcal{R}(N) > 0.001187N$. Lemmas 6.59 and 6.60 give the first four allocation bounds

$$0.039999904, \quad 0.038521, \quad 0.037660, \quad 0.034950.$$

In the dense E_1 branch, the low-low alternative is

$$\frac{|A|}{N} < \frac{23}{25} C_{\text{quad}} + \Delta_{25}^{\text{ex},509} + \frac{2\hat{J}_1^{509}}{25} + \frac{2\zeta_1^{509}}{N} + \mathfrak{T}_{512,512,117}^{\text{pc},509} < 0.039999904,$$

where $\Delta^{\text{ex},509}$ denotes the formulas of Lemma 6.60. The original high-low alternative, optimized at $k = 102$, is below 0.031996.

If $e > 10^{-5}N$, the cutoff-131 two-coincidence estimate gives

$$\frac{|A|}{N} < 0.039772.$$

Otherwise the single odd class contains more than $0.001177N$ pivots; two lie within $850 < 1764$, so at most one residue pair coincides, and the cutoff-131 estimate gives

$$\frac{|A|}{N} < 0.039129.$$

All six bounds are uniform on the interval and lie strictly below $1/25 - 7/(25N)$. Hence every allocation directly contradicts the original Hall balance. $\square$

Lemma 6.62 (Terminal cutoff comparison). *Put $N_{\dagger} = 3253000000$. Consider the following family in the dense- E_1 branch: an arbitrary prime cutoff, any valid odd individual lower Bonferroni degree and even joint upper degree (including full inclusion-exclusion), separate zero/one-coincidence endpoint counts, an arbitrary integer split $k \geq 2$ with Y replaced by its lower bound $N/(k+1)$ in the root term, and the exact diagonal sum. At $N = N_{\dagger}$ its unique minimum uses cutoff 509, Bonferroni degrees $3/2$, and $k = 117$, and the resulting envelope equals*

$$0.040000010216\dots > \frac{1}{25} - \frac{7}{25N_{\dagger}}.$$

No larger cutoff or split parameter gives a smaller envelope.

Proof. At $N_{\dagger}$ the exact values are

$$\lfloor r_N + 1 \rfloor = 24456856, \quad \mathcal{S}(24456856) = 20767164,$$

and

$$\lfloor N/117 \rfloor = 278034188, \quad \pi(278034188) = 15125057.$$

For cutoffs through 509, exact enumeration of all 77223 cutoff/order triples is minimized by cutoff 509 and degrees $3/2$. Exact enumeration of $2 \leq k \leq 800$ is minimized by $k = 117$; for $k \geq 801$, the last root-tail term alone exceeds

$$\frac{2048(802)^2}{N_{\dagger}} > 0.0405.$$

It remains to make the cutoff enumeration terminal. For the minimizing degree- $3/2$ pair, adjoining a new noncoincident prime p when n primes are already present changes the endpoint-plus-density-plus-square-tail payment by

$$\frac{2(n^2 + 5n + 2)}{N_{\dagger}} - \frac{4}{25p^2} \left(1 - \left(S_1 + S_2 - \frac{1}{9} \right) \right),$$

where $S_1 = \sum q^{-2}$ and $S_2 = \sum_{q < q'} q^{-2} q'^{-2}$ over the current window. At the first step, $n = 95$, $p = 521$ and $S_1 + S_2 - 1/9 > 0.05754$, so this increment is greater than $2.87 \cdot 10^{-8}$. Thereafter its positive endpoint term increases while the magnitude of its negative term decreases. The eight other potentially competitive degree pairs are checked exactly through cutoff 4999. For degree $1/0$ the finite-density increment exactly cancels the square-tail saving, leaving a positive endpoint increment. For the other seven, beyond 4999 the endpoint increment exceeds the maximal possible $4/(25p^2)$ density-tail saving. Higher degrees are already worse at cutoff 509, and the same increment comparison only strengthens thereafter.

Finally

$$\frac{1}{25} - \frac{7}{25N_{\dagger}} = 0.039999999991\dots,$$

which proves the stated comparison. □

Lemma 6.63 (Exact split-point prime-root block). *Put $Y = \lfloor N/118 \rfloor$. On*

$$3.2275 \cdot 10^{10} \leq N \leq 3.2532 \cdot 10^{10},$$

the combined intermediate-prime and transformed-root payment satisfies

$$\frac{4\pi(Y)}{N} + \frac{2048}{N} \left(\frac{2(N/25 + 2)}{Y} + \frac{N^2 + 25N + 1}{Y^2} \right) < 0.002729824014.$$

Also

$$\frac{\mathcal{S}(\lfloor r_N + 1 \rfloor)}{N} < 0.000639920595.$$

With 0.000639921 in the exact diagonal formulas, the five left-endpoint errors are below

$$0.002977624, \quad 0.002976905, \quad 0.002976546, \quad 0.004897029, \quad 0.004896851.$$

Proof. For the prime-root term divide

$$273516949 \leq Y \leq 275694915$$

into blocks $u \leq Y \leq v$ of length 10. On each block use $\pi(v)$ and the least mapped N in the prime term. In the root term use $Y \geq u$ and the largest mapped N : after expansion its only increasing part is $2048N/u^2$, so this is a valid upper bound. The maximum block is

$$u = 273517049, \quad v = 273517058,$$

with

$$\pi(v) = 14892354, \quad 32275011782 \leq N \leq 32275012961,$$

and gives 0.0027298240131...

The diagonal range is $24328878 \leq \lfloor r_N + 1 \rfloor \leq 24457858$. The length-10 hyperbola scan is maximal at

$$\frac{\mathcal{S}(24328897)}{32275018458} = \frac{20653449}{32275018458} < 0.000639920595.$$

Substitution gives the five displayed errors. □

Define

$$\mathfrak{V}_{118}^{509} := \frac{4\tau_{509}^+}{25} + 0.002729824014 < 0.002772511214.$$

Theorem 6.64 (Exact split-point descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 3.2275 \cdot 10^{10}$.*

Proof. Theorem 6.61 leaves

$$3.2275 \cdot 10^{10} \leq N < 3.2532 \cdot 10^{10}.$$

At the left endpoint the matching margin exceeds $1.8 \cdot 10^7$ and $\mathcal{R}(N) > 0.001162N$. Lemmas 6.59 and 6.63 give the first four allocation bounds

$$0.039999953, \quad 0.038546, \quad 0.037688, \quad 0.034969.$$

The dense E_1 low-low alternative is

$$\frac{|A|}{N} < \frac{23}{25} C_{\text{quad}} + \Delta_{25}^{\text{ex}, Y} + \frac{2\hat{J}_1^{509}}{25} + \frac{2\zeta_1^{509}}{N} + \mathfrak{V}_{118}^{509} < 0.039999953,$$

where $\Delta^{\text{ex},Y}$ uses the diagonal constant from Lemma 6.63; the high-low alternative remains smaller. If $e > 10^{-5}N$, the even branch is below 0.039800. Otherwise more than $0.001152N$ pivots remain in the single odd class, so a pair lies within $869 < 1764$ and the last branch is below 0.039145. Every bound is uniform and strictly below $1/25 - 7/(25N)$, closing the original Hall balance. $\square$

Lemma 6.65 (Boundary-corrected prime-square tail). *One has*

$$\sum_{p>509} \frac{1}{p^2} < 0.000265420 =: \tau_{509}^{++}.$$

Proof. Exact rational summation through 10^7 gives

$$\sum_{509 < p \leq 10^7} \frac{1}{p^2} < 0.000265410358.$$

Partial summation retains the usually discarded lower boundary:

$$\sum_{p>X} \frac{1}{p^2} = -\frac{\pi(X)}{X^2} + 2 \int_X^\infty \frac{\pi(t)}{t^3} dt.$$

With $X = 10^7$, $\pi(X) = 664579$, and $\pi(t) < 1.25506t/\log t$, this is below

$$-\frac{664579}{10^{14}} + \frac{2(1.25506)}{10^7 \log(10^7)} < 0.000000008928.$$

The two strict bounds sum to less than 0.000265420. $\square$

Lemma 6.66 (Final boundary-tail block). *On*

$$3.22715 \cdot 10^{10} \leq N \leq 3.2275 \cdot 10^{10},$$

the exact- Y combined prime-root term with $k = 118$ is below 0.002729928779, and

$$\frac{\mathcal{S}(\lfloor r_N + 1 \rfloor)}{N} < 0.000639941094.$$

At the left endpoint the corresponding dense- E_1 envelope, using τ_{509}^{++} , is below 0.039999935.

Proof. The same length-10 block calculation as Lemma 6.63 is maximal at

$$273487318 \leq Y \leq 273487327, \quad \pi(273487327) = 14890805,$$

with

$$32271503524 \leq N \leq 32271504703.$$

It gives 0.0027299287782... The diagonal scan is maximal at its first block:

$$\frac{\mathcal{S}(24327128)}{32271500000} = \frac{20651859}{32271500000} < 0.000639941094.$$

Using 0.000639942 in the diagonal formula, the degree-3/2 conditional survivor of Lemma 6.59, and Lemma 6.65 gives 0.03999993450... $\square$

Theorem 6.67 (Boundary-tail descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 3.22715 \cdot 10^{10}$.*

Proof. Theorem 6.64 leaves only the interval in Lemma 6.66. The dense E_1 branch is below 0.039999935. The other five bounds remain, respectively, below

$$0.038546, \quad 0.037688, \quad 0.034969, \quad 0.039800, \quad 0.039145.$$

All are uniform and strictly below $1/25 - 7/(25N)$, so the original Hall balance closes directly. $\square$

Lemma 6.68 (Root count versus inactive 509-events). *Let two pivots have at most r coincident residue pairs in $\mathcal{P}_{509}$, where $r = 1, 2, 3, 4$. For two low-root pivots the degree-3/2 density and endpoint pairs are bounded by*

r	1	2	3	4
I_r^{LL}	0.113400471	0.130297684	0.137108297	0.142005658
ϵ_r^{LL}	303811	303623	303436	303250.

If one pivot has the upper root count and the other is low-root, the bounds improve to

r	1	2	3
I_r^{HL}	0.111673122	0.114566348	0.115175656
ϵ_r^{HL}	269183	269003	268824.

For $r = 4$, reactivating every event of a high-root pivot shows that the last low-low pair is also a valid upper bound for every mixed root alternative. For two valuation-one upper-root pivots at distance at most 834, the $r = 1$ bounds improve further to

$$I_1^{HH} < 0.008402835, \quad \epsilon_1^{HH} \leq 234616.$$

Proof. For a valuation-one pivot, root count 512 requires at least nine odd prime factors in the effective modulus. At most one is 5, so the pivot has at least eight non-5 odd prime divisors. Two of those cannot both exceed 509, since already

$$2(521)(523)(3)(7)(11)(13)(17)(19) > 5.28 \cdot 10^{11}.$$

Thus at least seven events in $\mathcal{P}_{509}$ are inactive. The same eight-inactive-event relaxation is valid for a root-4096 high pivot in the higher 2-adic classes.

At the upper endpoint $3.22715 \cdot 10^{10}$, enumerate the eight-subsets $D \subseteq \mathcal{P}_{509}$ with $2 \prod_{p \in D} p \leq N$ and the seven-subsets with room for one further prime at least 521. There are 9223 resulting active-event patterns. For each pattern evaluate the individual L_3 polynomials and the joint U_2 polynomial, with the r smallest active common primes coincident. The high-low maxima give the second table; for $r = 1$ the maximizing inactive set is

$$\{7, 11, 13, 19, 23, 29, 31, 41\}.$$

The first table is the exact full-active evaluation. The fourth column is obtained by making the four smallest active local counts equal to one; exact rational evaluation gives the displayed density and endpoint count. Any forced inactive event can only decrease this full-active upper bound.

If both valuation-one pivots are high, every common odd divisor divides their even difference, so the product of the common divisors is at most 417. Exact evaluation of all gap-compatible pattern pairs gives the last two bounds. Without this small-gap condition, reactivating every event of the second pivot can only enlarge both the intersection upper bound and its endpoint count, so the high-low table is a valid upper bound for the other high-high uses below. $\square$

Lemma 6.69 (Uniform root-inactive interval block). *On*

$$2.631 \cdot 10^{10} \leq N \leq 3.22715 \cdot 10^{10},$$

the exact prime-root term with $k = 121$ and root sum 768 is below 0.002678109464, and

$$\frac{\mathcal{S}(\lfloor r_N + 1 \rfloor)}{N} < 0.000680804714.$$

With 0.000680805 in the diagonal formulas, the five left-endpoint errors are below

$$0.003165882, \quad 0.003165064, \quad 0.003164655, \quad 0.005207890, \quad 0.005207687.$$

Proof. For $Y = \lfloor N/121 \rfloor$, use length-1000 blocks exactly as in Lemma 6.63. The maximum is the first block:

$$217438016 \leq Y \leq 217439015, \quad \pi(217439015) = 11989378,$$

with $2631000000 \leq N \leq 26310120935$, and gives 0.0026781094634... The length-1000 diagonal scan is also maximal on its first block:

$$\frac{\mathcal{S}(21231414)}{2631000000} = \frac{17911972}{2631000000} < 0.000680804714.$$

Substitution gives the five errors. □

Theorem 6.70 (Root-inactive-event descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 2.631 \cdot 10^{10}$.*

Proof. Theorem 6.67 leaves the interval of Lemma 6.69. In the dense E_1 branch, split according to whether each selected pivot has root count at most 256 or exactly 512. The low-low, high-low, and high-high alternatives, using Lemma 6.68, are respectively below

$$0.039826, \quad 0.039999725, \quad 0.032014.$$

The high-low alternative uses the exact block in the preceding lemma and is the largest. The next three allocation bounds are

$$0.039232, \quad 0.038464, \quad 0.035494.$$

It remains to treat the one-odd-class allocation. Put $e = |E_1| + |E_2| + |E_3|$. If $e > 10^{-5}N$, one valuation class contains more than $e/3$, hence supplies a pair within 300000. Four coincidences would force $2(3 \cdot 7 \cdot 11 \cdot 13)^2 > 1.8 \cdot 10^7$ to divide the difference, so $r \leq 3$. Splitting each pivot at root count 2048 and applying the $r = 3$ rows of Lemma 6.68 gives three root alternatives, all below 0.039743.

If $e \leq 10^{-5}N$, the mixed strip leaves more than $0.000476N$ pivots in the single odd class. Two lie within $2101 < 4(3 \cdot 7 \cdot 11)^2$, so $r \leq 2$. Splitting at root count 256 and using the $r = 2$ rows gives all three root alternatives below 0.039758. Thus all six exhaustive allocations lie strictly below $1/25 - 7/(25N)$ and directly close the original Hall balance. □

Lemma 6.71 (Unit-block root-inactive endpoint). *On $2.6305 \cdot 10^{10} \leq N \leq 2.631 \cdot 10^{10}$, unit Y -blocks give the high-low prime-root term below 0.002678282125, while unit diagonal blocks give*

$$\frac{\mathcal{S}(\lfloor r_N + 1 \rfloor)}{N} < 0.000680809352.$$

The dense E_1 branch is therefore below 0.039999978.

Proof. The prime-root maximum occurs at

$$Y = 217396723, \quad \pi(Y) = 11987194, \quad 26305003483 \leq N \leq 26305003603.$$

The diagonal maximum is

$$\frac{\mathcal{S}(21227725)}{26305000000} = \frac{17908690}{26305000000} < 0.000680809352.$$

Substitution in the high-low formula of Theorem 6.70 gives 0.03999997766... $\square$

Theorem 6.72 (Terminal root-inactive descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 2.6305 \cdot 10^{10}$.*

Proof. Theorem 6.70 leaves only the interval in Lemma 6.71. Its dense E_1 bound is below 0.039999978; the other five bounds in the preceding theorem have much larger slack and decrease. This closes every allocation in the original Hall balance. $\square$

Lemma 6.73 (Dynamic cutoff-pattern endpoint). *On*

$$2.6262 \cdot 10^{10} \leq N \leq 2.6305 \cdot 10^{10},$$

the high-low valuation-one branch admits the cutoff-491 bounds

$$I_1^{HL} < 0.111634279, \quad \epsilon_1^{HL} \leq 244029, \quad \sum_{p>491} \frac{1}{p^2} < 0.000277248.$$

For $Y = \lfloor N/121 \rfloor$, the exact intermediate-prime and root payment is below 0.002679811009, while

$$\frac{\mathcal{S}(\lfloor r_N + 1 \rfloor)}{N} < 0.000681147190.$$

The resulting dense E_1 envelope is below 0.039999965.

Proof. A root-512 valuation-one pivot has at least eight non-5 odd prime divisors. At most one can exceed 491, since

$$2(499)(503)(3)(7)(11)(13)(17)(19) = 486918618186 > 2.6305 \cdot 10^{10}.$$

At the upper endpoint enumerate the eight-subsets of $\mathcal{P}_{491}$ with $2 \prod p \leq N$ and the seven-subsets with room for one further prime at least 499. There are 6167 distinct feasible inactive-event patterns. Exact evaluation of the individual degree-3 and joint degree-2 polynomials, including their endpoint counts, is maximized by

$$D = \{7, 11, 13, 17, 23, 29, 31, 37\}, \quad 2 \prod_{p \in D} p = 26037677666,$$

and gives the first two displayed bounds. Exact summation from 491 through 10^7 , followed by the boundary-corrected tail estimate of Lemma 6.65, gives the third.

Unit Y -blocks are maximal at $Y = 217041322$; after subtracting the 94 primes already paid by the finite cutoff, the exact prime count is 11968438. The corresponding normalized prime-root payment is below 0.002679811009. Unit diagonal blocks are maximal at

$$\frac{\mathcal{S}(21204587)}{26262002235} = \frac{17888289}{26262002235} < 0.000681147190.$$

Using this ratio in the exact diagonal formula gives $\Delta_{25} < 0.003167512089$. Substitution of all the preceding strict bounds gives 0.03999996408... $\square$

Theorem 6.74 (Dynamic-pattern descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 2.6262 \cdot 10^{10}$.*

Proof. Theorem 6.72 leaves only the interval of Lemma 6.73. The dense E_1 branch is below 0.039999965. The other five allocation bounds in Theorem 6.70 have larger slack. The only structural change at the new endpoint is that the mixed strip is $> 0.000486N$; after removing an even part of size at most $10^{-5}N$, the final odd class still contains more than $0.000476N$ pivots and supplies a pair within 2101. Thus the same at-most-two-coincidence argument applies, and all six allocations close directly in the original Hall balance. $\square$

Lemma 6.75 (Quadratic-product endpoint block). *On*

$$2.6260 \cdot 10^{10} \leq N \leq 2.6262 \cdot 10^{10},$$

the cutoff-491 prime-root payment is below 0.002679890050, and

$$\frac{\mathcal{S}(\lfloor r_N + 1 \rfloor)}{N} < 0.000681163934.$$

With $C_{\text{quad}} < 0.027346508$, the dense E_1 envelope is below 0.039999953.

Proof. The inactive-pattern family and its density/endpoint pair are those of Lemma 6.73. Unit Y -blocks are maximal at

$$Y = 217024793, \quad \pi(Y) - \pi(491) = 11967617,$$

and give 0.0026798900498... Unit diagonal blocks are maximal at

$$\frac{\mathcal{S}(21203533)}{26260044182} = \frac{17887395}{26260044182} < 0.000681163934.$$

The exact diagonal formula is then below 0.003167587519. Substitution, now using the sharpened quadratic Euler product, gives 0.03999995253... $\square$

Theorem 6.76 (Quadratic-product descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 2.6260 \cdot 10^{10}$.*

Proof. The preceding theorem leaves only the interval of Lemma 6.75. Its dense branch is below 0.039999953; the other five allocations retain larger slack. The mixed-strip and coincidence argument used there is unchanged. Hence every allocation directly contradicts the original Hall balance. $\square$

Lemma 6.77 (Second root-inactive correlation). *Let a high valuation-one pivot have root count 512, and split the other pivot according as its root count is at most 128 or equals 256. In the first case the cutoff-491 density/endpoint pair remains*

$$(I, \epsilon) < (0.111634279, 244029),$$

but the root sum is only 640. In the second case one has

$$(I, \epsilon) < (0.111612165, 231221),$$

with root sum 768.

Proof. Only the second case is new. For a valuation-one pivot, root count greater than 128 requires at least eight distinct odd prime factors in the effective modulus. At most one is the modulus prime 5, so at least seven non-5 odd primes divide the pivot. At least three of them are at most 47, since otherwise the least possible product is

$$2(3)(7)(53)(59)(61)(67)(71) = 38110106118 > 2.6260 \cdot 10^{10}.$$

Thus at least three events in

$$\{3, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47\}$$

are inactive for the root-256 form.

At the upper endpoint there are 6143 feasible seven/eight-inactive patterns for the root-512 form. Reactivate every event of the root-256 form except for an arbitrary three-element subset of the displayed thirteen primes. Exact degree-3/2 evaluation of all

$$6143 \binom{13}{3} = 1756898$$

relaxed pattern pairs is maximized by the high inactive set

$$\{7, 11, 13, 17, 23, 29, 31, 37\}$$

and the low inactive set $\{41, 43, 47\}$, with the prime-3 residue pair coincident. This gives the second density/endpoint pair. Since all other forced low-form inactive events were reactivated, it is a uniform upper bound. $\square$

Lemma 6.78 (Second-correlation endpoint block). *On*

$$2.6222 \cdot 10^{10} \leq N \leq 2.6260 \cdot 10^{10},$$

the root-sum-640 and root-sum-768 prime-root payments are respectively below

$$0.002538247876, \quad 0.002681279433,$$

and

$$\frac{\mathcal{S}(\lfloor r_N + 1 \rfloor)}{N} < 0.000681458623.$$

The two high-low alternatives are below 0.039859701 and 0.039999987.

Proof. Both unit Y -block maxima occur at

$$Y = 216710779, \quad \pi(Y) - \pi(491) = 11951269.$$

Substitution with root sums 640 and 768 gives the first two numbers. The unit diagonal maximum is

$$\frac{\mathcal{S}(21183049)}{26222000000} = \frac{17869208}{26222000000} < 0.000681458623,$$

which gives $\Delta_{25} < 0.003168950774$. Apply Lemma 6.77, the cutoff-491 tail, and $C_{\text{quad}} < 0.027346508$ to obtain $0.03985970054 \dots$ and $0.03999998609 \dots$. $\square$

Theorem 6.79 (Second root-inactive descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 2.6222 \cdot 10^{10}$.*

Proof. Theorem 6.76 leaves the interval in the preceding lemma. The low-root split closes the largest high–low branch in both alternatives. Every other allocation retains larger slack. At the new endpoint the mixed strip is $> 0.000481N$; after removing an even part of size at most $10^{-5}N$, the final odd class has more than $0.000471N$ pivots and supplies a pair within $2124 < 4(3 \cdot 7 \cdot 11)^2$. Thus its at-most-two-coincidence argument is unchanged, and all six allocations close the original Hall balance. $\square$

Lemma 6.80 (Four-event low-root refinement). *In the root-256 alternative of Lemma 6.77, one may replace its density/endpoint pair by*

$$(I, \epsilon) < (0.111609892, 227133).$$

Proof. The seven non-5 odd prime divisors of the low pivot include at least four primes at most 79. Indeed, if at most three did, their least possible product would give

$$2(3)(7)(11)(83)(89)(97)(101) = 33435142818 > 2.6222 \cdot 10^{10}.$$

They still include at least three primes at most 47. Reactivate every other low-form event and enumerate the 2717 four-subsets of the primes through 79 having at least three members at most 47. At the upper endpoint there are 6125 feasible high-form patterns, so the enumeration has 16641625 pairs. Its maximum uses high inactive set

$$\{7, 11, 13, 17, 23, 29, 31, 37\}$$

and low inactive set $\{41, 43, 47, 79\}$, with the prime-3 pair coincident. Exact degree-3/2 evaluation gives the displayed bound. $\square$

Lemma 6.81 (Four-event endpoint block). *On*

$$2.6216 \cdot 10^{10} \leq N \leq 2.6222 \cdot 10^{10},$$

the root-sum-768 prime–root payment is below 0.002681504757, and

$$\frac{\mathcal{S}(\lfloor r_N + 1 \rfloor)}{N} < 0.000681507463.$$

The root-256 high–low envelope is below 0.039999944.

Proof. The unit prime–root maximum occurs at

$$Y = 216661157, \quad \pi(Y) - \pi(491) = 11948721,$$

and the diagonal maximum is

$$\frac{\mathcal{S}(21179821)}{26216006421} = \frac{17866404}{26216006421} < 0.000681507463.$$

The latter gives $\Delta_{25} < 0.003169173036$. Substitution of Lemma 6.80 gives $0.03999994397 \dots$; the root-128 alternative remains far smaller. $\square$

Theorem 6.82 (Four-event descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 2.6216 \cdot 10^{10}$.*

Proof. The preceding theorem and lemma leave only the displayed interval and close its controlling root-256 branch. At the new endpoint the mixed strip is $> 0.000480N$; after the even part is removed, the last odd class supplies a pair within $2128 < 4(3 \cdot 7 \cdot 11)^2$. Every other allocation retains larger slack, so the original Hall balance closes directly. $\square$

Lemma 6.83 (Five-event low-root refinement). *In the root-256 alternative, one may use*

$$(I, \epsilon) < (0.111609312, 223134).$$

Proof. At least five of the seven non-5 odd prime divisors are at most 157: if at most four were, the minimal product is already greater than

$$2(3)(7)(11)(13)(163)(167)(173) = 28283653398 > 2.6216 \cdot 10^{10}.$$

Combine this with the previously proved requirements of at least four divisors at most 79 and at least three at most 47.

For fixed high-form data, write $x = p^{-2}$ for a low-form event that is declared inactive. Direct differentiation of the degree-3 individual polynomial and degree-2 joint polynomial shows that the resulting intersection upper bound is nonincreasing in x for $0 \leq x \leq 1/9$; the same check covers whether the high event at p is active and whether the least active common event changes. Thus the componentwise largest admissible relaxed five-set is the worst one. It is

$$\{41, 43, 47, 79, 157\}.$$

Evaluating this set against all 6121 feasible high patterns is maximized by the high inactive set $\{7, 11, 13, 17, 23, 29, 31, 37\}$ with the prime-3 pair coincident. The exact polynomials give the displayed density and endpoint count. $\square$

Lemma 6.84 (Five-event endpoint block). *On*

$$2.6211 \cdot 10^{10} \leq N \leq 2.6216 \cdot 10^{10},$$

the root-sum-768 prime-root payment is below 0.002681688359, and

$$\frac{\mathcal{S}(\lfloor r_N + 1 \rfloor)}{N} < 0.000681546710.$$

The controlling envelope is below 0.039999961.

Proof. The unit prime-root maximum occurs at

$$Y = 216620017, \quad \pi(Y) - \pi(491) = 11946587,$$

and the unit diagonal maximum is

$$\frac{\mathcal{S}(21177127)}{26211004702} = \frac{17864024}{26211004702} < 0.000681546710.$$

Thus $\Delta_{25} < 0.003169353914$. With Lemma 6.83, the envelope is

$$0.03999996020 \dots$$

$\square$

Theorem 6.85 (Five-event descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 2.6211 \cdot 10^{10}$.*

Proof. The four-event theorem leaves the interval of the preceding lemma. The five-event bound closes its controlling root-256 alternative; every other allocation retains larger slack. The mixed strip remains $> 0.000480N$, so the final odd-class coincidence argument also remains unchanged. This closes the original Hall balance directly. $\square$

Lemma 6.86 (Nested six-event refinement). *At cutoff 503, the root-256 high-low alternative admits*

$$(I, \epsilon) < (0.111560667, 234703),$$

while the root-128 alternative admits

$$(I, \epsilon) < (0.111634809, 260615)$$

with root sum 640.

Proof. The seven non-5 odd prime divisors of a root-256 pivot obey the nested order constraints

$$d_1 \leq 17, \quad d_2 \leq 29, \quad d_3 \leq 47, \quad d_4 \leq 79, \quad d_5 \leq 157, \quad d_6 \leq 503.$$

For example, the six successive contrary minimal products are

$$51253692706, 30223218306, 38110106118, 33435142818, 28283653398, 27076327278,$$

each larger than $2.6211 \cdot 10^{10}$. The last number is $2(3)(7)(11)(13)(17)(509)(521)$.

The monotonicity calculation in Lemma 6.83 therefore reduces the relaxed low inactive set to

$$\{17, 29, 47, 79, 157, 503\}.$$

At the upper endpoint, exact enumeration gives 6123 feasible high patterns. Evaluation against all of them is maximized by the high inactive set $\{7, 11, 13, 17, 23, 29, 31, 37\}$ with the prime-3 pair coincident, and gives the first density/endpoint pair. Reactivating every low event gives the second pair. A scan of every safe prime cutoff from 109 through 1009, with its corresponding forced nested event count, is uniquely minimized at 503; beyond 1009 the explicit endpoint increment exceeds the possible tail decrement. $\square$

Lemma 6.87 (Nested six-event endpoint block). *On*

$$2.6153 \cdot 10^{10} \leq N \leq 2.6211 \cdot 10^{10},$$

the root-sum-640 and root-sum-768 prime-root payments are below

$$0.002540414524, \quad 0.002683823447,$$

and

$$\frac{\mathcal{S}(\lfloor r_N + 1 \rfloor)}{N} < 0.000682004015.$$

The root-128/root-sum-640 and root-256/root-sum-768 high-low envelopes are, respectively, below 0.039864461 and 0.039999957.

Proof. Exact summation and the negative boundary term give

$$\sum_{p>503} \frac{1}{p^2} < 0.000269280.$$

Both unit prime-root maxima occur at

$$Y = 216140527, \quad \pi(Y) - \pi(503) = 11921650.$$

The unit diagonal maximum is

$$\frac{S(21145872)}{26153000000} = \frac{17836451}{26153000000} < 0.000682004015,$$

and gives $\Delta_{25} < 0.003171459351$. Substitution in the root-128 and root-256 cases of Lemma 6.86 gives, respectively,

$$0.03986446071\dots \quad \text{and} \quad 0.03999995674\dots$$

□

Theorem 6.88 (Nested six-event descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 2.6153 \cdot 10^{10}$.*

Proof. The five-event theorem leaves the interval of the preceding lemma, whose two root alternatives close the controlling branch. At the new endpoint the mixed strip is $> 0.000472N$; after removing the even part, the final odd class still supplies a pair within $2165 < 4(3 \cdot 7 \cdot 11)^2$. All other allocations retain larger slack, so the original Hall balance closes directly. □

Lemma 6.89 (Joint gap divisibility). *Let $b, c \in E_1$ be distinct, put $d = |b - c| \leq 834$, and use the cutoff-503 inactive sets H, L for the two forms. If*

$$C = \prod_{p \in H \cap L} p$$

and an active common prime q has coincident residue classes, then

$$4Cq^2 \mid d.$$

In particular $4C \leq 834$, and a coincident prime must be one of 3, 7, 11, 13 and must satisfy $4Cq^2 \leq 834$.

Proof. Both valuation-one pivots are $2 \pmod{4}$, so $4 \mid d$. Every prime in $H \cap L$ divides both pivots and hence divides d . A coincident active residue pair at q means $b \equiv c \pmod{q^2}$, so $q^2 \mid d$. The three factors $4, C, q^2$ are pairwise coprime: the cutoff primes are odd, and an event cannot be simultaneously active and inactive. Their product therefore divides d . The numerical conclusions follow from $d \leq 834$. □

Lemma 6.90 (Exact joint-compatible cutoff-503 patterns). *In the valuation-one root-512/root-256 alternative on*

$$2.3775 \cdot 10^{10} \leq N \leq 2.6153 \cdot 10^{10},$$

the degree-3/2 intersection density and endpoint discrepancy satisfy the combined bound

$$\frac{2I_{\text{joint}}^{HL}}{25} + \frac{2\epsilon_{\text{joint}}^{HL}}{N} < 0.001674991.$$

The same maximizing pattern applies when the normalized endpoint term is evaluated at either end of the interval.

Proof. At the upper endpoint there are 94 cutoff primes. Enumerate the root-512 patterns as the eight-subsets H with $2 \prod_{p \in H} p \leq N$, together with the seven-subsets having room for one prime at least 509. There are 6093 such patterns. For the root-256 pivot, enumerate the analogous seven- and six-subsets; there are 560701.

For fixed high data, Lemma 6.83 and the nested six-event argument give the relaxed low set

$$\{17, 29, 47, 79, 157, 503\}.$$

Evaluating this relaxation first excludes 5782 high patterns whose upper bound is already below the incumbent. The remaining 311 patterns require

$$311(560701) = 174378011$$

exact high–low evaluations. Lemma 6.89 leaves 31260793 joint-compatible pairs. For each one, the individual degree-3 and joint degree-2 polynomials are evaluated together with their actual nonconstant CRT count. The maximum is

$$\begin{aligned} H &= \{3, 13, 17, 19, 23, 29, 37, 41\}, \\ L &= \{3, 11, 43, 47, 53, 59, 61\}, \end{aligned}$$

with $C = 3$ and coincident prime $q = 7$; indeed $4Cq^2 = 588 \leq 834$. Exact rational evaluation gives

$$I = 0.020694788706384609\dots, \quad \epsilon = 230703.$$

Repeating the combined density-plus-endpoint ordering at the lower endpoint gives the same maximum. The runner-up normalized payment is smaller by more than $3.17 \cdot 10^{-7}$, while the first screened upper bound is smaller by more than $8.2 \cdot 10^{-4}$, so the strict rounded bounds follow. $\square$

Lemma 6.91 (Joint-compatible endpoint block). *On*

$$2.3976 \cdot 10^{10} \leq N \leq 2.6153 \cdot 10^{10},$$

the six exhaustive Hall allocations are bounded, respectively, by

$$0.039985, \quad 0.0396, \quad 0.0389, \quad 0.0359, \quad 0.039999946, \quad 0.039921.$$

The controlling fifth allocation is the one-odd-class case with $e > 10^{-5}N$, three allowed coincidences, and two low-root even pivots.

Proof. At the lower endpoint the extended half-root degree satisfies

$$\mathcal{M}_{1/2}(N) > \frac{N}{50} + 2.07 \cdot 10^6, \quad \mathcal{R}(N) > 0.00017314N.$$

The two root–Euler comparisons used in Proposition 5.4 are, respectively, below 0.001364 and 0.005245, still strictly below $\kappa_5/200$ and $\kappa_5/75$. Thus the matching and the six-way allocation remain valid on the whole interval.

For the dense E_1 allocation, the low–low, exact joint high–low, and high–high alternatives are below

$$0.039984900, \quad 0.032902569, \quad 0.032184709.$$

The three intervening allocation envelopes increase by less than 0.000322 from their values in Theorem 6.70; this gives the second, third, and fourth displayed bounds.

If the even part in the final one-odd-class allocation exceeds $10^{-5}N$, one valuation class supplies a pair within 300000, so at most three residue pairs coincide. The low-low, high-low, and high-high root alternatives are below

$$0.039999945687, \quad 0.038901921, \quad 0.039425069.$$

For the controlling low-low case use cutoff 509, $(I_3^{LL}, \epsilon_3^{LL}) < (0.137108297, 303436)$, root sum 4096, and $k = 67$. Unit blocks give

$$\begin{aligned} \text{prime-root payment} &< 0.004739167218, \\ \frac{\mathcal{S}(19955490)}{23976000000} &= \frac{16787505}{23976000000} < 0.000700179555, \\ \Delta_W &< 0.005355245344. \end{aligned}$$

Substitution gives 0.0399999456866949...

Finally suppose $e \leq 10^{-5}N$. The single odd class contains more than $0.00016314N$ pivots and hence supplies a pair within 6130; since $6130 < 4(3 \cdot 7 \cdot 11)^2$, at most two residue pairs coincide. Its three root alternatives are below

$$0.039920971, \quad 0.039306063, \quad 0.039573819.$$

Every displayed bound is uniform and strictly below $1/25 - 7/(25N)$. $\square$

Theorem 6.92 (Joint-compatible descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 2.3976 \cdot 10^{10}$.*

Proof. Theorem 6.88 leaves precisely the interval of Lemma 6.91. That lemma closes all six mass allocations directly in the original Hall inequality. No residual cut is introduced. $\square$

Lemma 6.93 (Valuation-coincidence correlation). *In the final one-odd-class allocation with $e > 10^{-5}N$, select two pivots from one common 2-adic valuation class at distance at most 300000. If that class is E_1 , the three-coincidence low-low, high-low, and high-high alternatives are below*

$$0.037712106, \quad 0.036293211, \quad 0.036561762.$$

If it is E_2 or E_3 , at most two residue pairs coincide, and the three corresponding alternatives are below

$$0.039482093, \quad 0.038881874, \quad 0.039406733.$$

These bounds hold on $2.3784 \cdot 10^{10} \leq N \leq 2.3976 \cdot 10^{10}$.

Proof. For E_1 , three coincidences are arithmetically possible, but the root split is the valuation-one split at 256: the three root sums are 512, 768, 1024, not 4096, 6144, 8192. Substitute those sums in the $r = 3$ rows of Lemma 6.68.

For E_2 and E_3 , the two selected pivots are congruent modulo 8. Three coincident odd primes would therefore force

$$8(3 \cdot 7 \cdot 11)^2 = 426888 \mid |b - c|,$$

contrary to the gap bound. Hence the $r = 2$ rows apply. Using the conservative root sums 4096, 6144, 8192 gives the second triple. Direct endpoint substitution gives the displayed values, and unit-block monotonicity makes them uniform. The value

$$0.039999945686 \dots$$

combined mutually incompatible data: three coincidences from E_1 and root sum 4096 from a higher valuation class. $\square$

Lemma 6.94 (Valuation-correlated endpoint block). *On*

$$2.3784 \cdot 10^{10} \leq N \leq 2.3976 \cdot 10^{10},$$

the six exhaustive Hall allocations are below, respectively,

$$0.0399999979, \quad 0.0397, \quad 0.0389, \quad 0.0359, \quad 0.039483, \quad 0.039937.$$

Proof. At the lower endpoint,

$$\mathcal{M}_{1/2}(N) > \frac{N}{50} + 1.72 \cdot 10^6, \quad \mathcal{R}(N) > 0.00014499N,$$

and the two root–Euler differences are below 0.001367 and 0.005259. Thus the matching and allocation structure remain valid.

After Lemma 6.93, the dense E_1 low–low branch is the only close alternative. Use cutoff 509, $(I_1^{LL}, \epsilon_1^{LL}) < (0.113400471, 303811)$, root sum 512, and $k = 135$. Exact unit blocks give

$$\begin{aligned} \text{prime-root payment} &< 0.002438141663, \\ \frac{\mathcal{S}(19848823)}{23784018997} &= \frac{16693717}{23784018997} < 0.000701887978, \\ \Delta_{25} &< 0.003262997405. \end{aligned}$$

The maximum prime–root block occurs at $N = 23784015555$, $Y = 176177893$, and $\pi(Y) = 9828932$. Substitution gives

$$0.0399999788173336 \dots < \frac{1}{25} - \frac{7}{25N}.$$

The exact joint-compatible high–low maximum of Lemma 6.90 is unchanged at the new lower endpoint. The three middle allocations retain their stated slack, Lemma 6.93 gives the fifth, and the three $r = 2$ final-odd root alternatives are below

$$0.039936452, \quad 0.039322465, \quad 0.039591016.$$

Here $e \leq 10^{-5}N$ leaves more than $0.00013499N$ pivots in the final odd class, hence a pair within $7408 < 4(3 \cdot 7 \cdot 11)^2$. This completes all six allocations. $\square$

Theorem 6.95 (Valuation-correlated descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 2.3784 \cdot 10^{10}$.*

Proof. Theorem 6.92 leaves precisely the interval of Lemma 6.94, whose six alternatives all close in the original Hall balance. $\square$

Lemma 6.96 (Re-optimized low–low window). *On*

$$2.3775 \cdot 10^{10} \leq N \leq 2.3784 \cdot 10^{10},$$

the dense E_1 low–low branch is below

$$0.0399999941.$$

The optimal finite certificate uses cutoff 467, individual degree 3, joint degree 2, and split point $k = 135$.

Proof. For an arbitrary prime cutoff Q , evaluate the odd individual lower Bonferroni degrees 1, 3, 5 and even joint upper degrees 0, 2, 4, keeping the zero- and one-coincidence endpoint counts separate. Subtract the exact finite prime-square sum from the boundary-corrected cutoff-509 tail and retain $Y = \lfloor N/k \rfloor$ in the prime-root term.

At the lower endpoint, exact enumeration of every prime cutoff through 4999, every one of the nine degree pairs, and every $2 \leq k \leq 1000$ is uniquely minimized at

$$Q = 467, \quad (d_{\text{ind}}, d_{\text{joint}}) = (3, 2), \quad k = 135.$$

The zero/one-coincidence densities and endpoint counts there are

	0	1
I	0.013585721441	0.113397645522
ϵ	250980	250803.

The one-coincidence alternative controls. Exact subtraction gives

$$\sum_{p>467} \frac{1}{p^2} < 0.000289972.$$

Higher Bonferroni degrees have larger endpoint counts and were already worse at the larger endpoint in Lemma 6.62; the comparison only strengthens as N decreases. Beyond cutoff 4999, the same one-prime increment comparison used in that lemma has an increasing positive endpoint term and a decreasing possible density-tail saving, so no omitted cutoff can improve the bound.

Unit blocks on the stated interval give

$$\begin{aligned} \text{prime-root payment} &< 0.002438481401, \\ \frac{\mathcal{S}(19843811)}{23775011058} &= \frac{16689295}{23775011058} < 0.000701967918, \\ \Delta_{25} &< 0.003263366199. \end{aligned}$$

The first maximum occurs at $N = 23775001605$, $Y = 176111123$, and $\pi(Y) = 9825451$. Substitution gives

$$0.0399999402040402 \dots < \frac{1}{25} - \frac{7}{25N}.$$

All other Hall allocations retain the strict margins of Lemma 6.94. $\square$

Theorem 6.97 (Cutoff-467 descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 2.3775 \cdot 10^{10}$.*

Proof. Theorem 6.95 leaves only the interval of Lemma 6.96, whose controlling branch and all five complementary allocations close directly. $\square$

Lemma 6.98 (Integer negative-Pell count). *In every exact aggregate diagonal formula, the last factor*

$$1 + \log_{2+\sqrt{3}} N$$

may be replaced by

$$P(N) := \left\lfloor 1 + \log_{2+\sqrt{3}} N \right\rfloor.$$

In particular $P(N) = 19$ throughout

$$2.2827 \cdot 10^{10} \leq N \leq 2.3775 \cdot 10^{10}.$$

Proof. The proof of Lemma 4.2 bounds, for each fixed d , the number of positive solutions of the negative Pell equation by $1 + \log_{2+\sqrt{3}} N$. That number is an integer, so it is at most the floor of the displayed real bound. Finally direct squaring in $\mathbb{Q}(\sqrt{3})$ gives

$$(2 + \sqrt{3})^{18} < 2.2827 \cdot 10^{10} < 2.3775 \cdot 10^{10} < (2 + \sqrt{3})^{19},$$

which makes the floor equal to 19. □

Lemma 6.99 (Integer-Pell endpoint block). *On*

$$2.3669 \cdot 10^{10} \leq N \leq 2.3775 \cdot 10^{10},$$

the dense E_1 low-low branch is below 0.039999960 and all five complementary Hall allocations retain strict slack.

Proof. Use Lemma 6.98 in the exact diagonal term and repeat the cutoff/order/split enumeration. At the lower endpoint the unique optimum is cutoff 463, degrees 3/2, and $k = 135$. Its one-coincidence data are

$$I = 0.113397118126\dots, \quad \epsilon = 242617, \quad \sum_{p>463} \frac{1}{p^2} < 0.000294557.$$

With this fixed certificate, unit blocks give

$$\begin{aligned} \text{prime-root payment} &< 0.002442402990, \\ \frac{\mathcal{S}(19784791)}{23669021538} &= \frac{16637351}{23669021538} < 0.000702916721, \\ \Delta_{25}^{\text{Pell}} &< 0.003259369807. \end{aligned}$$

The prime-root maximum occurs at $N = 23669006085$, $Y = 175325971$, and $\pi(Y) = 9784038$. Thus the controlling envelope is

$$0.0399999596197116\dots < \frac{1}{25} - \frac{7}{25N}.$$

At the same endpoint the half-root matching margin exceeds $1.51 \cdot 10^6$, the two root-Euler differences are below 0.001369 and 0.005268, and $\mathcal{R}(N) > 0.00012798N$. If the final even part is at most $10^{-5}N$, the remaining odd class therefore supplies a pair within $8476 < 4(3 \cdot 7 \cdot 11)^2$. Every structural and complementary allocation used above remains valid; the integer Pell replacement only lowers their diagonal payments. □

Theorem 6.100 (Integer-Pell descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 2.3669 \cdot 10^{10}$.*

Proof. Theorem 6.97 leaves the interval of Lemma 6.99, which closes every allocation in the original Hall inequality. □

Lemma 6.101 (Restricted small-modulus diagonal sum). *Put*

$$\mathcal{F}(R) = \sum_{\substack{s \leq R \\ p|s \Rightarrow p \equiv 1 \pmod{4}, \, p \geq 13}} 2^{\omega(s)}.$$

In every exact aggregate diagonal formula, the occurrence of $\mathcal{S}(R)$ in the small-modulus rounding term may be replaced by $\mathcal{F}(R)$. On

$$2.2827 \cdot 10^{10} \leq N \leq 2.3669 \cdot 10^{10}$$

unit blocks give

$$\frac{\mathcal{F}(19312757)}{22827035690} = \frac{3991879}{22827035690} < 0.000174875050.$$

With the integer Pell factor 19, the corresponding maxima satisfy

$$\Delta_{25}^{\mathcal{F}} < 0.001689240822, \quad \Delta_W^{\mathcal{F}} < 0.002213420548, \quad \Delta_{100}^{\mathcal{F}} < 0.001687899155.$$

On the extended lower interval

$$1.382 \cdot 10^{10} \leq N \leq 2.2827 \cdot 10^{10},$$

the corresponding unit-block maxima are

$$\frac{\mathcal{F}(13821334)}{13820000000} = \frac{2856719}{13820000000} < 0.000206709045$$

and

$$\Delta_{25}^{\mathcal{F}} < 0.001997137693, \quad \Delta_W^{\mathcal{F}} < 0.002616654327, \quad \Delta_{100}^{\mathcal{F}} < 0.001995296367.$$

Proof. In the proof of Lemma 4.2, a modulus s contributes only when it is squarefree and every prime divisor is 1 (mod 4) and at least 13. Its exact root count is $2^{\omega(s)}$. Thus the small- s rounding loss is at most $3\mathcal{F}(R)$ in the q -formulas and $6\mathcal{F}(R)$ in the W -formula. Replacing this by $\mathcal{S}(R)$ would enlarge both the support and the weight.

For completeness, the finite certificate starts with weight one at $s = 1$. From a permitted squarefree product s , append each permitted prime larger than its last prime factor, provided $sp \leq 19784778$, and assign weight $2 \cdot 2^{\omega(s)}$ to sp . Unique increasing prime factorization makes this an exact, duplication-free enumeration. There are 1224624 products at the upper end. Prefixing their integer weights and scanning every change of $\lfloor r_N + 1 \rfloor$ gives the displayed maximum. Substitution in the three diagonal formulas, with Lemma 6.98, gives the last line. Repeating the same prefix and unit-block scan on the extended range gives the second set of maxima. The conservative Pell factor 19 remains valid there (and drops to 18 below $(2 + \sqrt{3})^{18}$). $\square$

Lemma 6.102 (Restricted-diagonal endpoint block). *On*

$$2.2827 \cdot 10^{10} \leq N \leq 2.3669 \cdot 10^{10},$$

the six exhaustive Hall allocations are below, respectively,

$$0.038464, \quad 0.0389, \quad 0.0382, \quad 0.0363, \quad 0.038539, \quad 0.039711.$$

Proof. At the lower endpoint, Proposition 5.4 gives

$$\mathcal{M}_{1/2}(N) > \frac{N}{50} + 411, \quad \mathcal{R}(N) > 0.000000036N.$$

The root–Euler differences are below 0.001386 and 0.005332, so the matching and the six-way allocation remain valid.

For the dense E_1 low-low branch retain the cutoff-463, degree-3/2, $k = 135$ certificate of Lemma 6.99, but use Lemma 6.101. Unit Y - and R -blocks give

$$\begin{aligned}\text{prime-root payment} &< 0.002474825011, \\ \Delta_{25}^{\mathcal{F}} &< 0.001689240822,\end{aligned}$$

and the full envelope is

$$0.038463008852 \dots$$

Direct substitution of the same restricted diagonal terms in the next three allocation formulas gives the second, third, and fourth displayed bounds.

It remains to check the one-odd-class allocation at the almost-vanishing strip. Split at $e = 1.5 \cdot 10^{-8}N$. If e is larger, one of the three even valuation classes contains at least 115 pivots and supplies a pair at distance below $2.01 \cdot 10^8$. If e is smaller, the final odd class contains at least 481 pivots and supplies a pair at distance below $4.8 \cdot 10^7$. In both cases the difference is divisible by 4, and five coincident odd primes would force

$$4(3 \cdot 7 \cdot 11 \cdot 13 \cdot 17)^2 = 10424818404$$

to divide it. Hence at most four residue pairs coincide. Lemma 6.68 gives

$$I_4^{LL} < 0.142005658, \quad \epsilon_4^{LL} = 303250.$$

Reactivating every finite event is a valid common upper bound for all root alternatives. For the even pair, use the worst root sum 8192 and fixed $k = 52$; for the odd pair use root sum 1024 and $k = 105$. Unit blocks give

	prime-root	full envelope
even	0.006026552133	0.038538552454
odd	0.003090437113	0.039710563116.

The latter row includes the raw $1.5 \cdot 10^{-8}$ even charge. All six bounds are therefore strictly below $1/25 - 7/(25N)$ throughout the interval. $\square$

Theorem 6.103 (Restricted-diagonal descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 2.2827 \cdot 10^{10}$.*

Proof. Theorem 6.100 leaves precisely the interval of Lemma 6.102, which closes every allocation directly in the original Hall inequality. $\square$

Lemma 6.104 (Eight-prime endpoint block). *On*

$$1.382 \cdot 10^{10} \leq N \leq 2.2827 \cdot 10^{10},$$

all six Hall allocations close directly. The dense E_1 branch is below 0.039366, the worst higher-valuation even branch is below 0.039753, and the final odd branch is below 0.039577.

Proof. At the lower endpoint, Proposition 5.5 and Corollary 5.22 give

$$\mathcal{M}_{1/4}(N) > \frac{N}{50} + 16356, \quad \mathcal{R}(N) > 0.00000236N.$$

Use Lemma 6.101 in all diagonal terms. The preceding computation already covers $N \geq 1.383 \cdot 10^{10}$; scan the new unit interval down to $1.382 \cdot 10^{10}$. For the cutoff-463, $k = 135$ dense certificate, the maxima are

$$\text{prime-root} < 0.003055994166, \quad \Delta_{25}^{\mathcal{F}} < 0.001997137693,$$

and the full envelope is $0.039365928857\dots$. Direct substitution in the three middle allocation formulas leaves larger slack.

For the last two allocations split at $e = 10^{-7}N$. If e is larger, one even valuation class contains more than $e/3$ pivots and supplies a pair at distance below $3.01 \cdot 10^7$. Four coincidences would force $4(3 \cdot 7 \cdot 11 \cdot 13)^2 = 36072036$ to divide the gap, so $r \leq 3$. For E_1 use the worst root sum 1024; for E_2, E_3 use the conservative root sum 8192. Unit blocks give

$$0.036240855052\dots, \quad 0.039752885052\dots$$

If $e \leq 10^{-7}N$, the final odd class contains more than $0.00000226N$ pivots and supplies a pair at distance below $4.43 \cdot 10^5$. Four coincidences would force $4(3 \cdot 7 \cdot 11 \cdot 13)^2 = 36072036$ to divide the gap, so $r \leq 3$. A low-low pair uses $(I_3^{LL}, \epsilon_3^{LL}) < (0.137108297, 303436)$ and root sum 512. If one pivot is high-root, the forced inactive events give $(I_3^{HL}, \epsilon_3^{HL}) < (0.115175656, 268824)$; use root sums 768 and 1024 for the high-low and high-high alternatives. On this new unit interval the exact Pell factor is 18. Optimizing at fixed $k = 113, 99, 89$ gives, respectively,

$$0.039576187839\dots, \quad 0.038656450868\dots, \quad 0.038982482756\dots$$

The higher-valuation even branch is controlling, and it too is strictly below $1/25 - 7/(25N)$. All six therefore close on the stated interval. $\square$

Theorem 6.105 (Eight-prime root-Euler descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 1.382 \cdot 10^{10}$.*

Proof. Theorem 6.103 leaves the interval of Lemma 6.104, whose six allocations all close inside the original Hall inequality. $\square$

Lemma 6.106 (Second eight-prime endpoint block). *On*

$$1.29 \cdot 10^{10} \leq N \leq 1.382 \cdot 10^{10},$$

all six Hall allocations close directly. The dense branch is below 0.039519, the worst even branch is below 0.039926, and the final odd branch is below 0.039706.

Proof. At the lower endpoint, Proposition 5.6 and Corollary 5.22 give

$$\mathcal{M}_{1/8}(N) > \frac{N}{50} + 6728, \quad \mathcal{R}(N) > 0.00000104N.$$

The exact restricted-diagonal scan gives

$$\frac{\mathcal{F}(13200941)}{12900021174} = \frac{2728601}{12900021174} < 0.000211519111,$$

$$\Delta_{25}^{\mathcal{F}} < 0.002043605517, \quad \Delta_W^{\mathcal{F},18} < 0.002603499910.$$

The cutoff-463 dense envelope is $0.039518076041\dots$

Split at $e = 9 \cdot 10^{-8}N$. In the dense-even case a valuation class supplies a pair below $3.34 \cdot 10^7$, so $r \leq 3$. The E_1 alternative is below 0.036401; for E_2, E_3 , use root sum 8192, exact Pell factor 18, and $(I_3^{LL}, \epsilon_3^{LL})$ to obtain

$$0.039925196317 \dots$$

If $e \leq 9 \cdot 10^{-8}N$, the final odd class has more than $0.00000095N$ pivots, hence a pair below $1.06 \cdot 10^6$ and again $r \leq 3$. The low–low, high–low, and high–high root-stratified envelopes are

$$0.039705845760 \dots, \quad 0.038796221886 \dots, \quad 0.039129213946 \dots$$

The first four allocation formulas retain larger slack. Thus every branch is strictly below $1/25 - 7/(25N)$. $\square$

Theorem 6.107 (Second eight-prime root–Euler descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 1.29 \cdot 10^{10}$.*

Proof. Theorem 6.105 leaves the interval of Lemma 6.106, which closes every allocation in the original Hall inequality. $\square$

Lemma 6.108 (Third eight-prime endpoint block). *On*

$$1.279 \cdot 10^{10} \leq N \leq 1.29 \cdot 10^{10},$$

all six Hall allocations close directly. The dense, worst even, and final odd envelopes are below 0.039538, 0.039957, and 0.039723.

Proof. At the lower endpoint,

$$\mathcal{M}_{1/16}(N) > \frac{N}{50} + 5090, \quad \mathcal{R}(N) > 0.00000079N.$$

Unit blocks give

$$\frac{\mathcal{F}(13125781)}{12790008075} = \frac{2713031}{12790008075} < 0.000212121133,$$

and the cutoff-463 dense envelope is $0.039537470760 \dots$. Split at $e = 9 \cdot 10^{-8}N$. The dense-even valuation class supplies a pair below $3.34 \cdot 10^7$, so $r \leq 3$. With exact Pell factor 18, the worst E_2, E_3 envelope is

$$0.039956442803 \dots$$

In the sparse-even case the final odd class contains more than $0.00000070N$ pivots, hence a pair below $1.43 \cdot 10^6$ and again $r \leq 3$. The low–low, high–low, and high–high envelopes are

$$0.039722473493 \dots, \quad 0.038814112743 \dots, \quad 0.039147949978 \dots$$

All other allocations retain larger slack, proving the claim. $\square$

Theorem 6.109 (Third eight-prime root–Euler descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 1.279 \cdot 10^{10}$.*

Proof. Theorem 6.107 leaves the interval of Lemma 6.108, which closes every Hall allocation directly. $\square$

Lemma 6.110 (Retained-modulus endpoint block). *On*

$$1.2643 \cdot 10^{10} \leq N \leq 1.279 \cdot 10^{10},$$

all six Hall allocations close directly. The dense, worst even, and final odd envelopes are below 0.039564, 0.039999135, and 0.039745.

Proof. At the lower endpoint, Proposition 5.10 and Corollary 5.22 give

$$\mathcal{M}_{1/32}(N) > \frac{N}{50} + 1442, \quad \mathcal{R}(N) > 0.000000228N.$$

The exact restricted-diagonal unit scan gives

$$\frac{\mathcal{F}(13025077)}{12643098938} = \frac{2692239}{12643098938} < 0.000212941386,$$

and the cutoff-463 dense envelope is 0.039563962205... Split at $e = 10^{-7}N$. In the dense-even case one valuation class supplies a pair below $3 \cdot 10^7$, so $r \leq 3$. With exact Pell factor 18, the worst E_2, E_3 envelope is

$$0.039999134344... < \frac{1}{25} - \frac{7}{25N}.$$

If $e \leq 10^{-7}N$, the final odd class contains more than $0.000000128N$ pivots and supplies a pair below $7.82 \cdot 10^6$, again with $r \leq 3$. The low-low, high-low, and high-high envelopes are

$$0.039744956074..., \quad 0.038838362547..., \quad 0.039173393787....$$

The remaining allocations have larger slack. Thus every branch is strictly below the exact target. $\square$

Theorem 6.111 (Retained-modulus root-Euler descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 1.2643 \cdot 10^{10}$.*

Proof. Theorem 6.109 leaves precisely the interval of Lemma 6.110, which closes every allocation in the original Hall inequality. $\square$

Lemma 6.112 (Quadratic-residue endpoint block). *On*

$$1.15 \cdot 10^{10} \leq N \leq 1.2643 \cdot 10^{10},$$

all six Hall allocations close directly. The dense, worst even, and final odd envelopes are below 0.039791, 0.039896, and 0.039534.

Proof. At the lower endpoint, Proposition 5.9 and Corollary 5.22 give

$$\mathcal{M}_{\text{QR}}(N) > \frac{N}{50} + 154290, \quad \mathcal{R}(N) > 0.0000268N.$$

Twelve exact blocks of width at most 10^8 cover the interval. The restricted-diagonal scan is controlled by the first block and gives

$$\frac{\mathcal{F}(12227654)}{11500000000} = \frac{2527449}{11500000000} < 0.000219778174,$$

with cutoff-463 dense envelope 0.039790171170...

Split at $e = 1.5 \cdot 10^{-5}N$. In the dense-even case one valuation class has more than $5 \cdot 10^{-6}N$ pivots and supplies a pair below 200001. Three coincident active primes would make

$$4(3 \cdot 7 \cdot 11)^2 = 213444$$

divide their difference, so $r \leq 2$. Exact Pell factor 18, root sum 8192, and $(I_2^{LL}, \epsilon_2^{LL})$ give the worst E_2, E_3 envelope

$$0.039895678510 \dots$$

If $e \leq 1.5 \cdot 10^{-5}N$, the final odd class contains more than $0.0000118N$ pivots, so a pair lies below 84746 and again $r \leq 2$. The low-low, high-low, and high-high envelopes, using the exact root sums 512, 768, 1024, are

$$0.039533065858 \dots, \quad 0.039010120997 \dots, \quad 0.039358389351 \dots$$

All block maxima occur at the lower end, and the other allocations retain larger slack. Hence every branch is strictly below the exact target. $\square$

Theorem 6.113 (Quadratic-residue root-tail descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 1.15 \cdot 10^{10}$.*

Proof. Theorem 6.111 leaves the interval of Lemma 6.112, which closes every allocation directly in the original Hall inequality. $\square$

Lemma 6.114 (QR-hierarchy endpoint block). *On*

$$9.38 \cdot 10^9 \leq N \leq 1.15 \cdot 10^{10},$$

all six Hall allocations close directly. The dense, worst even, and final odd envelopes are below 0.039998505, 0.038930, and 0.039954.

Proof. At the lower endpoint, Proposition 5.8 and Corollary 5.22 give

$$\mathcal{M}_{\text{QR2}}(N) > \frac{N}{50} + 7633040, \quad \mathcal{R}(N) > 0.001627N.$$

For the dense branch optimize the complete finite-window family while retaining the exact restricted diagonal sum. The lower block is uniquely minimized by cutoff 367, Bonferroni degrees $3/2$, and $k = 100$, with

$$\frac{\mathcal{F}(10674474)}{9380000000} = \frac{2206435}{9380000000} < 0.000235227612,$$

and the full dense envelope is

$$0.039998504675 \dots < \frac{1}{25} - \frac{7}{25N}.$$

Split at $e = 1.5 \cdot 10^{-5}N$. A dense even valuation class supplies a pair below 200001, so $r \leq 2$ as before. On the present interval a pivot cannot have more than 2048 transformed roots; consequently the two-pivot root sum in the E_2, E_3 branch is at most 4096. With exact Pell factor 18 its envelope is

$$0.038929045621 \dots$$

In the sparse-even case the final odd class contains more than $0.001612N$ pivots and supplies a pair below 621, again with $r \leq 2$. The low-low, high-low, and high-high envelopes are

$$0.039953388923\dots, \quad 0.039460160705\dots, \quad 0.039834609222\dots$$

All later unit blocks decrease and the remaining allocations have larger slack. Hence every branch closes in the original Hall balance. $\square$

Theorem 6.115 (Quadratic-residue hierarchy descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 9.38 \cdot 10^9$.*

Proof. Theorem 6.113 leaves the interval of Lemma 6.114, which closes every allocation directly. $\square$

Lemma 6.116 (Cutoff-383 root-inactive endpoint block). *On*

$$8.78 \cdot 10^9 \leq N \leq 9.38 \cdot 10^9,$$

all six Hall allocations close directly. The dense, worst even, and final odd envelopes are below 0.039997644, 0.039192926, and 0.039954208.

Proof. At the lower endpoint, Proposition 5.8 and Corollary 5.22 give

$$\mathcal{M}_{\text{QR2}}(N) > \frac{N}{50} + 6261235, \quad \mathcal{R}(N) > 0.001426N.$$

The higher-root lower bound in the proof of the proposition is still more than $9.49 \cdot 10^6$ above $\mathcal{M}_{\text{QR2}}$ there, so no degree stratum has been lost by extending the endpoint.

It remains to strengthen the controlling dense E_1 branch. Its selected pair has even gap at most 834 and at most one coincident finite-prime residue pair. A valuation-one pivot has root count at most 512 on the present interval. First split each root count as at most 128, exactly 256, or exactly 512.

A root-256 pivot has at least seven non-5 odd prime divisors. At most one can exceed 383, since the least product with two such primes already exceeds the upper endpoint. Thus at cutoff 383 it forces either seven inactive events with

$$2 \prod_{p \in D} p \leq 9380000000,$$

or six inactive events with room for one further prime at least 389. There are respectively 129618 and 713 such forced patterns. Exact degree-3/2 evaluation gives the following coordinatewise density/endpoint bounds for a full-active pair and a pair with one root-256 pivot:

$$(I_{128,128}, \epsilon_{128,128}) < (0.113388315, 146003),$$

$$(I_{256,128}, \epsilon_{256,128}) < (0.112531936, 130146).$$

For two root-256 pivots, let C be the product of their common forced inactive primes and let q be the sole coincident active prime. By Lemma 6.89, the exact compatibility condition is $4Cq^2 \leq 834$. Reactivating all events of either pivot gives an upper bound for every pair whose one-form bound is too small to control. Only 6955 of the 130331 patterns remain. Exhausting their gap-compatible Cartesian square gives

$$I_{256,256} < 0.111792015, \quad \epsilon_{256,256} \leq 109217.$$

The density maximum has common product 19, coincidence prime 3, and inactive sets

$$\{7, 17, 19, 31, 37, 41, 43\}, \quad \{11, 13, 19, 23, 29, 47, 53\}.$$

The complete enumeration is reproduced by `scripts/dense_root_inactive_359_scan.py` with `--q 383`.

Now

$$\sum_{p>383} \frac{1}{p^2} < 0.000371164574418.$$

The exact restricted diagonal and prime blocks on the whole interval, with root sums 256, 384, 512 and split points 124, 108, 98, give

$$0.039443871, \quad 0.039759223, \quad 0.039997643212.$$

The controlling block has

$$\frac{\mathcal{F}(10214297)}{8780025882} = \frac{2111247}{8780025882} < 0.000240460226.$$

A root-512 pivot forces eight non-5 odd prime divisors. At cutoff 359 there are only 658 feasible forced patterns; even after reactivating every event of the other pivot one has

$$(I, \epsilon) < (0.021511201, 105714).$$

The worst root-sum-1024 envelope is therefore below 0.033652, so these higher-root alternatives close with large slack.

For the dense higher-valuation even branch, the pair gap remains below 200001, hence $r \leq 2$, and the actual two-pivot root sum is 4096. The exact Pell-18 block is below 0.039192926. If the even mass is at most $1.5 \cdot 10^{-5}N$, the final odd class contains more than $0.001411N$ pivots and supplies a pair below 710, still with $r \leq 2$. At cutoff 359, splitting its two root counts at 128/256 gives the three exact block bounds

$$0.039301217, \quad 0.039647372, \quad 0.039954208.$$

The remaining allocations retain larger slack. Hence every branch closes inside the original Hall balance on the stated interval. $\square$

Theorem 6.117 (Dense root–inactive descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 8.78 \cdot 10^9$.*

Proof. Theorem 6.115 leaves precisely the interval of Lemma 6.116, which closes all six allocations directly. $\square$

Lemma 6.118 (Free integer split endpoint block). *On*

$$8.77 \cdot 10^9 \leq N \leq 8.78 \cdot 10^9,$$

all six Hall allocations close directly. The dense, worst even, and final odd envelopes are below 0.039999903, 0.039197329, and 0.039956604.

Proof. The intermediate-prime/root decomposition is valid for every integer split point Y ; the earlier substitution $Y = \lfloor N/k \rfloor$ was a convenient one-parameter subfamily, not an arithmetic restriction. Use cutoff 379. At the upper pattern endpoint there are 129618 forced root-256 inactive patterns. Applying $4Cq^2 \leq 834$ and exact combined density-plus-endpoint ordering leaves 6954 candidates and gives

$$I_{256,256} < 0.111791625, \quad \epsilon_{256,256} \leq 104529.$$

The maximizing inactive sets are the same two sets displayed in Lemma 6.116; again $C = 19$ and $q = 3$. Moreover

$$\sum_{p>379} \frac{1}{p^2} < 0.000377981718171.$$

Choose the fixed integer threshold

$$Y = 89823400, \quad \pi(Y) = 5207279.$$

The prime-root payment is convex as a function of N on this block, so its maximum occurs at an endpoint; exact evaluation gives

$$\mathfrak{P}_{379,512,Y} < 0.003488984652.$$

The restricted-diagonal block is maximal at

$$N = 8770002913, \quad R = 10206521, \quad \mathcal{F}(R) = 2109663,$$

and is below 0.002324486044. Substitution gives the dense envelope

$$0.039999902986 \dots < \frac{1}{25} - \frac{7}{25N}.$$

At the new lower endpoint the QR degree remains more than 6238718 above half a base class and the mixed strip exceeds $0.001422N$. The exact higher-valuation even root-4096 block is below 0.039197329; the three cutoff-359 final-odd root alternatives are below

$$0.039303347, \quad 0.039649632, \quad 0.039956604.$$

All remaining allocations retain larger slack, proving the claim inside the original Hall balance. $\square$

Theorem 6.119 (Free-split dense descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 8.77 \cdot 10^9$.*

Proof. Theorem 6.117 leaves precisely the short block of Lemma 6.118, which closes every allocation directly. $\square$

Lemma 6.120 (Joint-multiplier and one-coincidence block). *On*

$$7.8 \cdot 10^9 \leq N \leq 8.77 \cdot 10^9,$$

all six Hall allocations close directly. The controlling dense high-low, worst even, and final odd envelopes are below 0.039989608, 0.039667100, and 0.039176885.

Proof. Retain cutoff 379 and recompute the root-256 patterns at the upper endpoint. A seven-event pattern has actual divisor

$$M_D = 2 \prod_{p \in D} p.$$

For every such pattern enumerate all multiples $mM_D \leq 8770000000$. Reactivating the other form first leaves 32775 one-form candidates. Of these, 70 are six-event relaxations with one omitted large prime and are compared against every candidate. The exact seven-event patterns give 53150 actual multiple entries; sorting them leaves only 865 support pairs represented by two values within 834. Applying $4Cq^2 \leq 834$ to every retained pair gives the safe coordinatewise bound

$$I_{256,256} < 0.020715563, \quad \epsilon_{256,256} \leq 109217.$$

Thus the dense high–high bound is strictly below the target. The full-active and one-root-256 bounds at the same cutoff are

$$(I_{128,128}, \epsilon_{128,128}) < (0.113387532, 140307),$$

$$(I_{256,128}, \epsilon_{256,128}) < (0.112523483, 124883).$$

Exact restricted-diagonal blocks give the root-256 low–low envelope below 0.039658794. For the high–low root sum 384, use the five split blocks

interval for N	k	envelope
$[8.50, 8.77] \cdot 10^9$	108	0.039817900
$[8.30, 8.50] \cdot 10^9$	107	0.039864973
$[8.10, 8.30] \cdot 10^9$	106	0.039913645
$[7.95, 8.10] \cdot 10^9$	105	0.039951123
$[7.80, 7.95] \cdot 10^9$	104	0.039989608.

The joint-multiplier high–high envelope is below 0.034 throughout.

At the lower endpoint the QR degree is more than 4113070 above half a base class and the mixed strip exceeds $0.001054N$. In the sparse-even allocation the final odd class therefore contains more than $0.001039N$ pivots and supplies a pair below 963. Two coincident active primes would force

$$4(3 \cdot 7)^2 = 1764$$

to divide its difference, which is impossible. Hence this branch has $r \leq 1$. The cutoff-359 root-128/root-256 alternatives are respectively below

$$0.038513847, \quad 0.038856177, \quad 0.039176885.$$

The actual higher-valuation even root-4096 block is below 0.039667100, and all remaining allocations have larger slack. Every branch therefore closes in the original Hall balance. $\square$

Theorem 6.121 (Joint-multiplier descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 7.8 \cdot 10^9$.*

Proof. Theorem 6.119 leaves precisely the interval of Lemma 6.120, which closes every allocation directly. $\square$

Lemma 6.122 (Exact finite high-low block). *On*

$$7.74855 \cdot 10^9 \leq N \leq 7.8 \cdot 10^9,$$

all six Hall allocations close directly. The controlling dense high-low, worst even, and final-odd envelopes are below 0.039999999884, 0.039694959, and 0.039191886.

Proof. Use cutoff 353 and retain the actual floor terms in the degree-3/2 Bonferroni count. For $t \in \{7, 18\}$ write

$$X_t(N) = \left\lfloor \frac{N-t}{25} \right\rfloor + 1.$$

If p is active for a pivot b , the event $p^2 \mid (t + 25n)b + 1$ is the single class

$$n \equiv r_{t,b,p} := -(tb + 1)(25b)^{-1} \pmod{p^2}.$$

Let $C_X(r, m)$ denote the number of integers $0 \leq n < X$ in the class $r \pmod{m}$. Every finite intersection of events is therefore counted exactly by CRT and the floor

$$C_X(r, m) = \begin{cases} 0, & r \geq X, \\ 1 + \lfloor (X - 1 - r)/m \rfloor, & r < X. \end{cases}$$

For one pivot let $L_b = X - S_1(b) + S_2(b) - S_3(b)$ be the odd Bonferroni lower bound for its survivor set, with every S_j evaluated by the preceding CRT floor. For the union of the two event systems, merging coincident classes at the same prime, let $U_{b,c} = X - \Sigma_1(b, c) + \Sigma_2(b, c)$ be the even Bonferroni upper bound for the joint survivor set. Hence the number of common bad values in this progression is at most

$$X - L_b - L_c + U_{b,c}. \tag{77}$$

This is an integer identity in signed CRT floors, so its maximum on a finite N -interval occurs at the left endpoint or at one of the explicitly enumerated floor jumps.

At the upper support endpoint there are 98208 feasible six/seven-event supports. Ranking the uniform one-coincidence bounds gives first

$$D_1 = \{13, 17, 19, 23, 29, 31, 43\}, \quad b_1 = 2 \prod_{p \in D_1} p = 7466754178,$$

then

$$D_2 = \{13, 17, 19, 23, 29, 31, 41\}, \quad b_2 = 2 \prod_{p \in D_2} p = 7119463286.$$

Every other support is bounded by the third uniform value

$$(I, \epsilon) < (0.112497017, 103036),$$

and hence contributes at most

$$\frac{2I}{25} + \frac{2\epsilon}{7.74855 \cdot 10^9} < 0.009026356261. \tag{78}$$

Both displayed bases exceed half the upper endpoint, so they are the actual high pivots. Since there is at most one active coincidence, its prime is one of $3, 7, 11, 13$ and $4q^2 \mid b_i - c$ with

$0 < |b_i - c| \leq 834$. There are 56 admissible offsets for each base. Expanding (77) and scanning every jump on the whole interval gives

$$\max \frac{1}{N} \sum_{t=7,18} (X_t - L_{b_1} - L_c + U_{b_1,c}) < 0.009000586,$$

$$\max \frac{1}{N} \sum_{t=7,18} (X_t - L_{b_2} - L_c + U_{b_2,c}) < 0.009000350.$$

The witnesses occur at $N = 7773532793$, $c = b_1 + 108$, and at $N = 7781651968$, $c = b_2 + 288$, respectively. Thus (78) is the valid uniform pair payment. The complete support ranking, CRT expansion, and jump scan are reproduced by `scripts/exact_dense_high_low_bonferroni.py`.

Now

$$\sum_{p>353} p^{-2} < 0.000407314700.$$

On $7.75 \cdot 10^9 \leq N \leq 7.8 \cdot 10^9$, choose the fixed large-prime split $Y = 74680000$, for which $\pi(Y) = 4376612$. The exact restricted-diagonal scan with root sum 384 gives

$$0.039999616073 \dots < \frac{1}{25} - \frac{7}{25N}.$$

On the bottom block choose instead $Y = 74751552$, for which $\pi(Y) = 4380561$. On every interval with fixed restricted-diagonal parameter R , the full envelope minus the Hall target is a sum of an affine function and positive multiples of $1/N$, and is therefore convex. Scanning both endpoints of every such interval on $7.74855 \cdot 10^9 \leq N \leq 7.75 \cdot 10^9$ gives its maximum at the lower endpoint:

$$0.039999999883517563 < 0.039999999963864207.$$

The full-active low-low block is below 0.039676122; the retained actual multiplier enumeration keeps the high-high block below 0.034, and every root-512 alternative remains below 0.033652.

At the lower endpoint

$$\mathcal{M}_{\text{QR2}}(N) > \frac{N}{50} + 4003743, \quad \mathcal{R}(N) > 0.001033N.$$

The higher-root direct progression lower bound is in fact greater than $N/50 + 10258173$, so the hierarchy itself retains strict slack. The exact higher-valuation even block with actual two-pivot root sum 4096 is below 0.039694959. If the even mass is at most $1.5 \cdot 10^{-5}N$, the final odd class has more than $0.001018N$ pivots and supplies a pair at distance below 982. Two coincidences would force $4(3 \cdot 7)^2 = 1764$ to divide that nonzero difference, so again $r \leq 1$. The three cutoff-359 root alternatives are below

$$0.038527089, \quad 0.038870308, \quad 0.039191886.$$

All six allocations therefore contradict the Hall inequality. $\square$

Theorem 6.123 (Exact finite-pair descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 7.74855 \cdot 10^9$.*

Proof. Theorem 6.121 leaves precisely the interval of Lemma 6.122, which closes every allocation inside the original Hall cut. $\square$

Lemma 6.124 (Twenty-exception exact finite block). *On*

$$7.69835 \cdot 10^9 \leq N \leq 7.74855 \cdot 10^9,$$

all six Hall allocations close directly. The controlling dense high-low, worst even, and final-odd envelopes are below 0.039999992524, 0.039722475, and 0.039206719.

Proof. Keep cutoff 353 and the exact CRT/floor expression (77). At the upper support endpoint there are 97283 feasible six/seven-event supports. Exhaustive ranking shows that only the following twenty actual bases exceed the twenty-first uniform record:

7466754178	7119463286	6424881502	7685998826
7540865882	7190127934	6905745814	6318022766
6024161242	5436438194	7410796822	7466049734
7705963694	7597780762	7601061182	6830641246
7325070038	6737654638	6984369106	6954162358.

Every base exceeds half the support endpoint and hence has no nontrivial multiplier. For each base enumerate its at most 56 offsets satisfying $0 < |b - c| \leq 834$ and the exact one-coincidence divisibility. Scanning every signed-floor jump on the full interval (starting the base 7705963694 only at its first feasible value) gives

$$\max_{b \text{ in the displayed list}} \frac{1}{N} \sum_{t=7,18} (X_t - L_b - L_c + U_{b,c}) < 0.009000587.$$

The maximum is attained for $b = 7466754178$, $c = b + 108$, at $N = 7703499518$.

Every remaining support is bounded by the twenty-first record

$$D_{21} = \{11, 13, 19, 23, 29, 37, 53\}, \quad (I_{21}, \epsilon_{21}) < (0.112331560, 103036).$$

Consequently the true pair payment on this interval is

$$\frac{2I_{21}}{25} + \frac{2\epsilon_{21}}{7698350000} < 0.009013292578,$$

strictly larger than every exact exception and therefore valid uniformly. These support rankings and interval jump scans are reproduced by `scripts/exact_dense_high_low_bonferroni.py`.

Choose the integer split

$$Y = 74567262, \quad \pi(Y) = 4370376.$$

As in the preceding lemma, total envelope minus target is convex on every fixed- R interval. A joint endpoint scan of the whole block, rather than a sum of unrelated component maxima, has its maximum at the lower endpoint and gives

$$0.039999992523402584 < 0.039999999963628570.$$

The full-active low-low block is below 0.039688991; actual multiplier compatibility keeps the high-high block below 0.034, and every root-512 alternative remains below 0.033652.

At the lower endpoint

$$\mathcal{M}_{\text{QR2}}(N) > \frac{N}{50} + 3897425, \quad \mathcal{R}(N) > 0.001012N,$$

while the higher-root direct bound exceeds $N/50 + 9990592$. The exact higher-valuation even root-4096 block is below 0.039722475. In the sparse-even case the final odd class has more than $0.000997N$ pivots and therefore supplies a pair at distance below 1003. Since $1003 < 1764$, two coincidences are again impossible. The three one-coincidence root alternatives are below

$$0.038540147, \quad 0.038884298, \quad 0.039206719.$$

Thus every allocation closes in the unchanged Hall balance. $\square$

Theorem 6.125 (Twenty-exception finite descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 7.69835 \cdot 10^9$.*

Proof. Theorem 6.123 leaves precisely the interval of Lemma 6.124, which closes all six allocations. $\square$

Lemma 6.126 (Fifty-exception exact finite block). *On*

$$7.68768 \cdot 10^9 \leq N \leq 7.69835 \cdot 10^9,$$

all six Hall allocations close directly. The controlling dense high-low, worst even, and final-odd envelopes are below 0.039999997172, 0.039728326, and 0.039209886.

Proof. Keep cutoff 353 and the exact CRT/floor expression (77). At the upper support endpoint there are 96431 feasible supports. The leading fifty bases are obtained from the twenty-base list in Lemma 6.124 by deleting the now-infeasible base 7705963694 and adjoining

$$\begin{array}{llll} 7107601358 & 6630712946 & 6164237222 & 6389954714 \\ 5877528514 & 6302967242 & 7528040806 & 5766544498 \\ 6853887898 & 7294521806 & 6629170262 & 5498333126 \\ 5955017354 & 7532934838 & 7006010726 & 5280864446 \\ 7567600898 & 7390692166 & 6680149762 & 6798014366 \\ 7046939042 & 4831429174 & 6111626378 & 6554010034 \\ 7319347178 & 4606711538 & 7079368582 & 6028427834 \\ 6249172358 & 6359432794 & 5515370146 & \end{array}$$

Denote the resulting fifty-element set by $\mathcal{E}_{50}$. Every one exceeds half the support endpoint. Exhausting all at most 56 one-coincidence offsets and every signed-floor jump for each of the fifty actual bases on the whole interval gives

$$\max_{b \in \mathcal{E}_{50}} \frac{1}{N} \sum_{t=7,18} (X_t - L_b - L_c + U_{b,c}) < 0.009000587.$$

The maximum again uses $b = 7466754178$, $c = b + 108$, now at $N = 7694521343$.

Every remaining support is bounded by the fifty-first record

$$D_{51} = \{11, 13, 17, 23, 29, 41, 43\}, \quad (I_{51}, \epsilon_{51}) < (0.112296072, 103036).$$

Thus the valid uniform pair payment is

$$\frac{2I_{51}}{25} + \frac{2\epsilon_{51}}{7687680000} < 0.009010491293,$$

which strictly dominates every exact exception. The exhaustive ranking and jump scans are reproduced by `scripts/exact_dense_high_low_bonferroni.py`.

The integer split remains

$$Y = 74567262, \quad \pi(Y) = 4370376.$$

The joint convex endpoint scan on the whole block gives

$$0.039999997171827493 < 0.039999999963578089.$$

The full-active low-low block is below 0.039691740; actual multiplier compatibility keeps the high-high block below 0.034, and root-512 alternatives remain below 0.033652.

At the lower endpoint

$$\mathcal{M}_{\text{QR2}}(N) > \frac{N}{50} + 3874872, \quad \mathcal{R}(N) > 0.001008N,$$

and the higher-root direct bound exceeds $N/50 + 9933718$. The exact higher-valuation even root-4096 block is below 0.039728326. In the sparse-even case the final odd class has more than $0.000993N$ pivots and supplies a pair below $1007 < 1764$, so two coincidences remain impossible. The three one-coincidence alternatives are below

$$0.038542937, \quad 0.038887268, \quad 0.039209886.$$

Every allocation therefore closes in the unchanged Hall balance. $\square$

Theorem 6.127 (Fifty-exception finite descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 7.68768 \cdot 10^9$.*

Proof. Theorem 6.125 leaves precisely the interval of Lemma 6.126, which closes all six allocations. $\square$

Lemma 6.128 (Ninety-five-exception exact finite block). *On*

$$7.67769 \cdot 10^9 \leq N \leq 7.68768 \cdot 10^9,$$

all six Hall allocations close directly. The controlling dense high-low, worst even, and final-odd envelopes are below 0.03999999924, 0.039733922, and 0.039212887.

Proof. Use cutoff 353 and the exact signed-floor expansion (77). At the upper support endpoint there are 96249 feasible six/seven-event supports. Adjoin to the fifty bases in Lemma 6.126 the following next forty-five:

5717327902	4157276266	5258841302	5639497006
7338804902	7654533458	7137741754	6735615458
5159539814	6132426014	7427283578	5931362866
4919561218	6907749706	5328173422	6586459022
7133521538	7511532314	4724983978	7265252566
6025909318	7162158718	7680584054	6927333842
7428761626	6526413322	4322857682	7522987186
6222859214	7099156922	6673294342	7026917326
4121794534	7359108614	6796526594	5787574078
6463411526	6700083962	6251496394	6480409078
6105354398	5518384586	7235876362	7037633174
5615750998			

Call the resulting set $\mathcal{E}_{95}$. Every displayed base, and every base already in $\mathcal{E}_{50}$, exceeds half the upper support endpoint. Thus all ninety-five are actual pivots with no multiplier ambiguity. For a base or its paired offset that first appears inside the interval, start its floor scan at $\max\{7.67769 \cdot 10^9, b, c\}$; this is the exact first feasible endpoint, not an extrapolation to the left.

Exhausting every admissible one-coincidence offset and every signed-floor jump on the whole interval gives

$$\max_{b \in \mathcal{E}_{95}} \frac{1}{N} \sum_{t=7,18} (X_t - L_b - L_c + U_{b,c}) < 0.009000589.$$

The maximum is attained for $b = 7466754178$, $c = b + 108$, at $N = 7683469568$, where the exact normalized count is 0.009000588001027402.

The ninety-sixth record is

$$D_{96} = \{11, 13, 17, 23, 29, 31, 37\}, \quad (I_{96}, \epsilon_{96}) < (0.112262740, 103036).$$

Its minimal base is 3719668238, below half the support endpoint, so it does not satisfy the actual-base condition and remains the uniform controller. On the present interval its payment is

$$\frac{2I_{96}}{25} + \frac{2\epsilon_{96}}{7677690000} < 0.009007859565,$$

strictly above every exact exception. The exhaustive ranking and the first-feasible signed-floor scans are reproduced by `scripts/exact_dense_high_low_bonferroni.py`.

Choose the fixed integer split

$$Y = 74315802, \quad \pi(Y) = 4356538.$$

Using the rounded upper bound $I_{96} < 0.112262740$, the joint convex endpoint scan on the entire block has its maximum at the lower endpoint and gives

$$0.039999999923493333 < 0.039999999963530697.$$

The full-active low-low block is below 0.039694322; actual multiplier compatibility keeps high-high below 0.034, and every root-512 alternative remains below 0.033652.

At the lower endpoint,

$$\mathcal{M}_{\text{QR2}}(N) > \frac{N}{50} + 3853771, \quad \mathcal{R}(N) > 0.0010038N,$$

while the higher-root direct progression bound exceeds $N/50 + 9880469$. The exact higher-valuation even root-4096 block is below 0.039733922. After charging an even mass of at most $1.5 \cdot 10^{-5}N$, the final odd class retains more than $0.0009888N$ pivots and hence has a pair at distance below $1012 < 1764$. Thus two finite-prime coincidences are impossible. The three one-coincidence root alternatives are below

$$0.038545557, \quad 0.038890115, \quad 0.039212887.$$

Every allocation therefore closes directly in the unchanged Hall balance. $\square$

Theorem 6.129 (Ninety-five-exception finite descent). *The exact bound (1), and equivalently the Hall inequality (2), holds for every $N \geq 7.67769 \cdot 10^9$.*

Proof. Theorem 6.127 leaves precisely the interval of Lemma 6.128, which closes all six allocations. $\square$

Theorem 6.130 (Erdős Problem 848: terminal assembly). *For every integer $N \geq 1$, the maximum cardinality of a set $A \subseteq [1, N]$ for which $ab + 1$ is nonsquarefree for every $a, b \in A$ is*

$$|\{n \leq N : n \equiv 7 \pmod{25}\}|.$$

Equivalently, the original Hall inequality in the Hall inequality (2) holds for every N .

Proof. Theorem 3.1 gives the upper bound for $N \leq 1.5 \cdot 10^6$. Theorems 6.26 and 6.29 give it on $[1.5 \cdot 10^6, 3 \cdot 10^6)$ and $[3 \cdot 10^6, 5 \cdot 10^6)$, respectively. Theorem 6.21 gives it directly for every $N \geq 5 \cdot 10^6$. These paper ranges cover all positive integers. Proposition 1.1 supplies the matching lower bound, and Theorem 2.2 gives the equivalent Hall formulation. $\square$

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